Why simulate cities and how to simulate cities?

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Why simulate cities and how to simulate cities?

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5. **Google scholar:** https://scholar.google.com.hk/citations?user=sIZG1mkAAAAJ&hl=zh-CN&oi=sra

00 Simulaiton is widely used in many fields

Why simulate

Ecolocy& Climate

Simulation model is a powerful tool to reveal the change processes, spatial patterns, drivers and trends of land use change (Pendulum Wanqi & Zhao Shidong 1997). It has widely applied in various fields such as biodiversity and ecosystem assessment (Bouwman A F et al. 2006) and can be used as input to various climate process models such as InVEST, WRF (Nelson E et al. 2010).

Regional planning

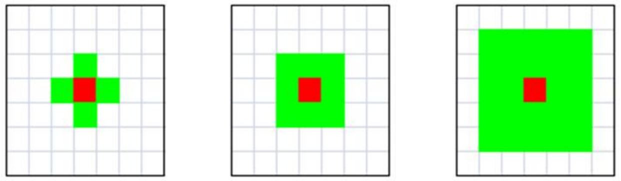
It can be used to study the driving factors of regional development and provide effective decision support for governments at all levels. For example, the model is used to simulate urban development and loss of farmland under the influence of different planning policies (basic farmland, key development zones), to test whether regional development meets planners' expectations, and to assist decision makers in making relatively reasonable decision adjustments before policy implementation.

Landscape & pollution

The simulation methods can combine a variety of environmental, policy, and other elements to provide a scientific and quantitative basis for regional landscape design, landscape connectivity changes, illegal land use monitoring, food security early warning, extreme environmental prediction, soil loss, surface runoff pollution, and flood risk early warning prediction.

00 Geographic simulation and cellular automata

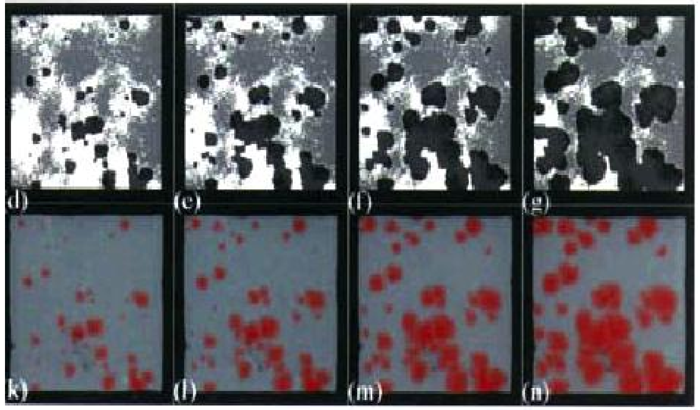
Basic model	Cellular automata is a model with discrete time, space and state, which has powerful spatial modeling and computing ability and can simulate complex dynamic systems with spatio-temporal characteristics.
Principle	By defining transition rules on the cellular neighborhood of the grid space to simulate the interactions of each group of elements, the whole system evolves gradually to generate the final pattern.
Previous studies	In recent decades, many geographers have used CA to simulate complex land use change, and many meaningful results have been obtained. A series of simulation models, such as Patch-CA model, the SLEUTH model; GEOMOD2 model; LTM model; Dyna - CLUE model; have provided important reference and basis for helping urban planning and decision-making(Liu X. 2008; Pijanowski. 2014)。

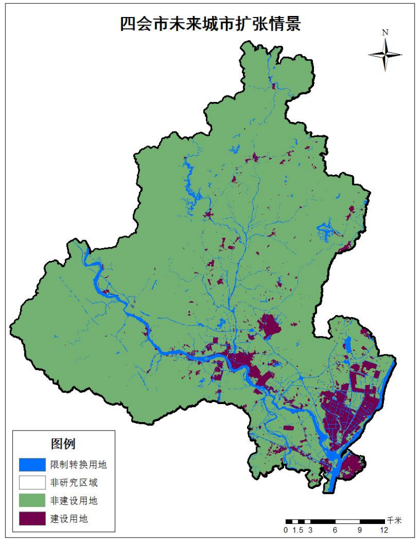


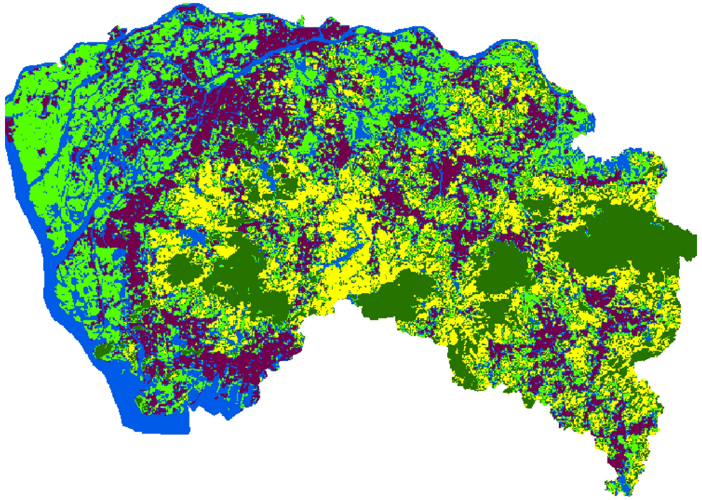
Von. Neumann　　Moore　　Extended Moore

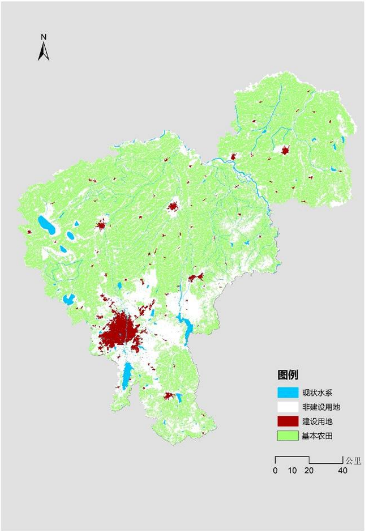
Transition rules f: $s_i^{t+1}=f(s_i^t, s_N^t)$

s_N^t indicates the combination of all the neighbor state at time t









The models involved in this slide are all based on the theory of geographic cellular automata.

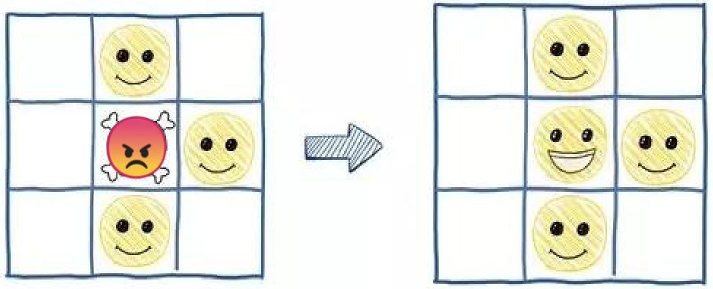
00 Philosophical thought of CA

English proverbs: if you lie down with dogs, you will get up with fleas.

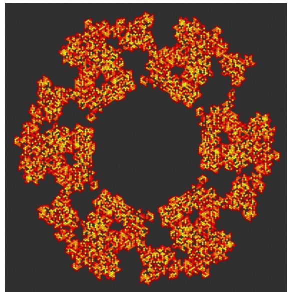
Geographic first law: Everything is related to everything else, but near things are more related to each other.

Complexity Science: We can describs complex global continuous systems with simple local rules and discrete methods

A simple example of CA：



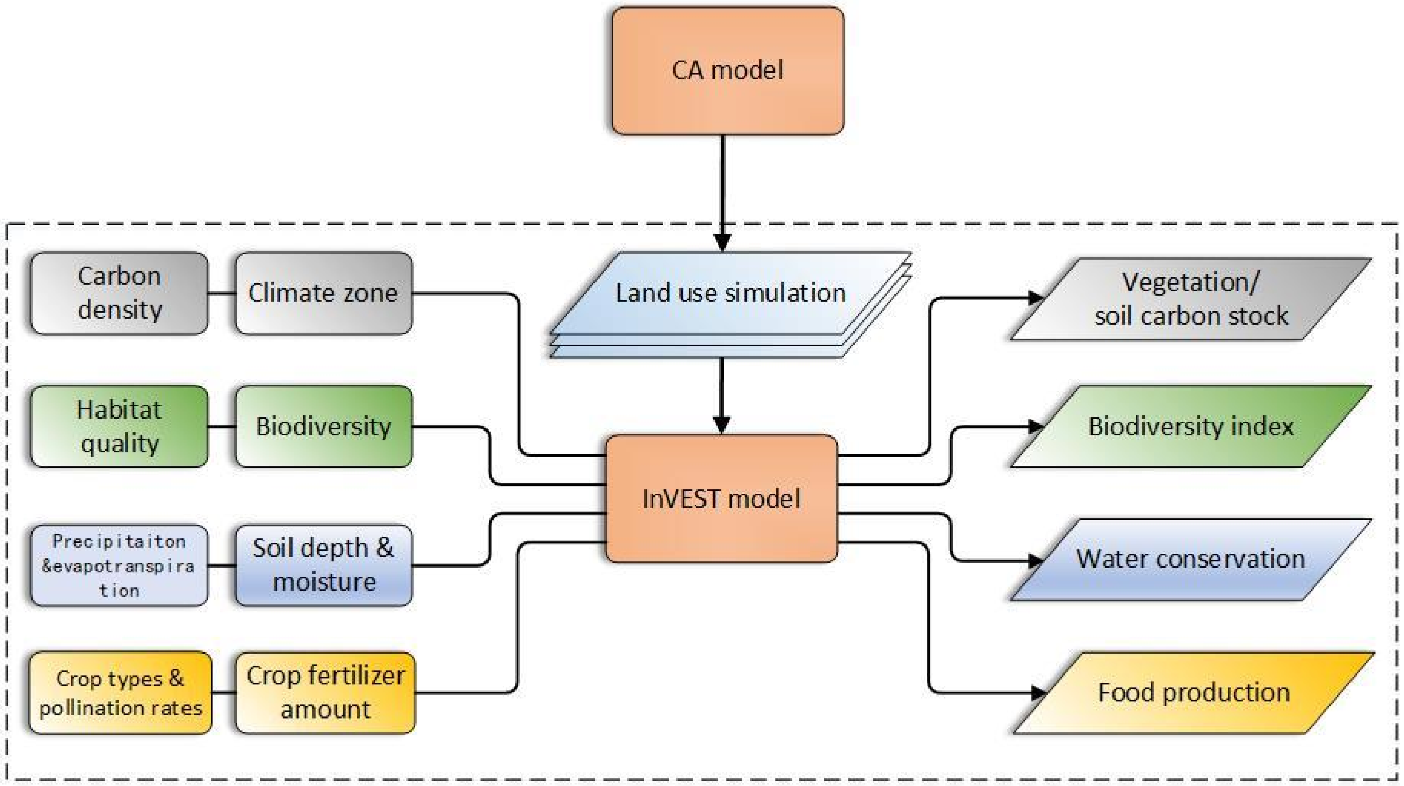
local rules 1： the angery face will change to smile face if there more than two smile faces around it + **Other rules**



Complex future pattern

Impacts of land use on ecological services and climate change

The combination of land use change model (CA) and the InVEST model (developted by Stanford university) can be used to simulate the influence of different land use/cover change on the change of ecosystem service function under different scenarios, which has been widely used all over the world.(Schroter et al., 2005； He et al., 2016； Goldstein et al., 2012)。



agriculture MDPI

Article

Cropland Expansion Mitigates the Supply and Demand Deficit for Carbon Sequestration Service under Different Scenarios in the Future—The Case of Xinjiang

Mingjie Shi [1,2,3], Hongqi Wu [1,2,*], Pingan Jiang [1,2,*], Wenjiao Shi [3,4], Mo Zhang [3,4], Lina Zhang [1,2], Haoyu Zhang [5], Xin Fan [6,7], Zhuo Liu [1,2], Kai Zheng [1,2], Tong Dong [8] and Muhammad Fahad Baqa [4,9]

Journal of Hydrology 610 (2022) 127811

Contents lists available at ScienceDirect

Journal of Hydrology

journal homepage: www.elsevier.com/locate/jhydrol

Research papers

Evaluating potential impacts of land use changes on water supply–demand under multiple development scenarios in dryland region

Xueqi Liu [a,b], Yansui Liu [a,*], Yongsheng Wang [a], Zhengjia Liu [a,*]

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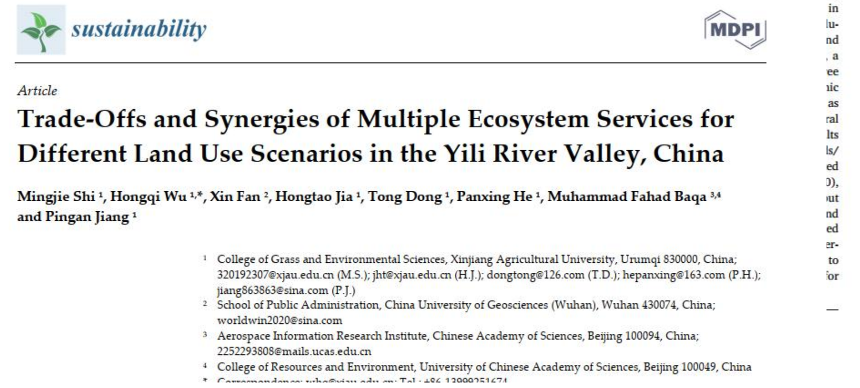
[b] *Faculty of Geographical Science, Beijing Normal University, Beijing 100875, China*



sustainability MDPI

Article

Trade-Offs and Synergies of Multiple Ecosystem Services for Different Land Use Scenarios in the Yili River Valley, China

Mingjie Shi [1], Hongqi Wu [1,*], Xin Fan [2], Hongtao Jia [1], Tong Dong [1], Panxing He [1], Muhammad Fahad Baqa [3,4] and Pingan Jiang [1]

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[4] College of Resources and Environment, University of Chinese Academy of Sciences, Beijing 100049, China

00 Urban planning

Li et al. (2008) used Genetic Algorithm based CA to obtain parameters for reasonable urban morphology

Long et al. (2012) used constrained CA to simulate the urban development of Beijing under different policy constraints, and provide policies to control the urban growth pattern of Beijing.

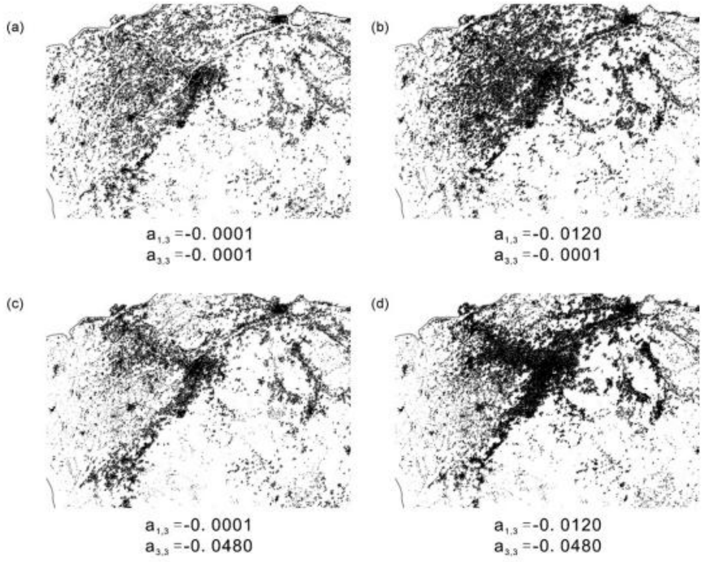
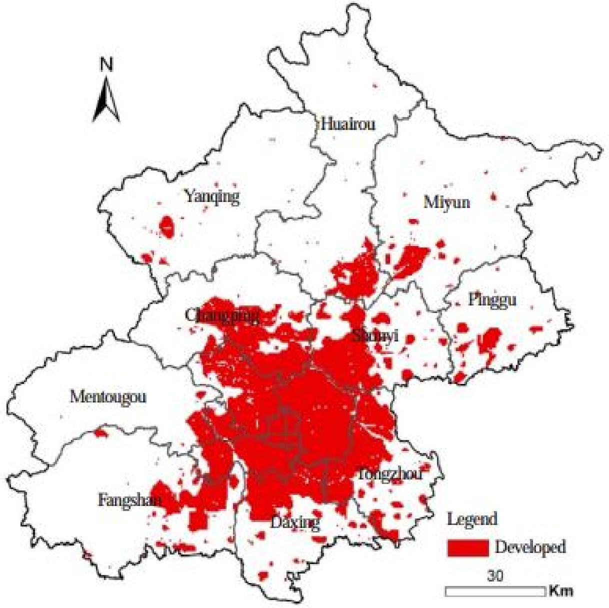


Fig. 5. Obtaining new "genes" by interactive modification of the attractions between city centres and roads.



Landscape and Urban Planning 86 (2008) 177–186

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Landscape and Urban Planning

journal homepage: www.elsevier.com/locate/landurbplan

ELSEVIER

Discovering and evaluating urban signatures for simulating compact development using cellular automata

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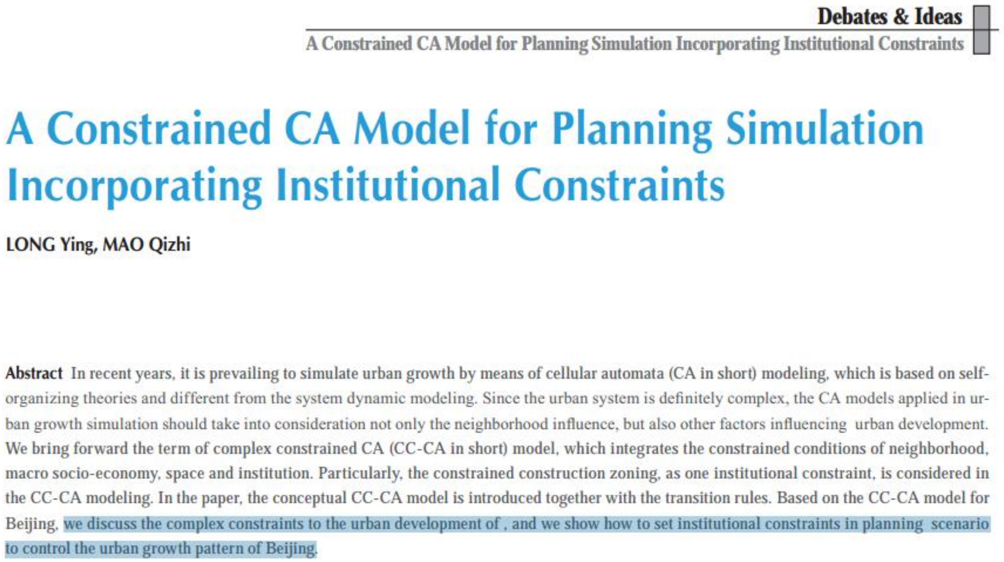
ABSTRACT

This paper provides a new method for retrieving, evaluating and modifying urban signatures for simulating compact development using urban cellular automata (CA). Urban CA usually adopt fixed transition rules for simulating urban dynamics in large complex regions. However, these regions can be segmented into sub-regions so that separate transition rules can be retrieved for generating better simulation results.



A Constrained CA Model for Planning Simulation Incorporating Institutional Constraints

LONG Ying, MAO Qizhi

Abstract In recent years, it is prevailing to simulate urban growth by means of cellular automata (CA in short) modeling, which is based on self-organizing theories and different from the system dynamic modeling. Since the urban system is definitely complex, the CA models applied in urban growth simulation should take into consideration not only the neighborhood influence, but also other factors influencing urban development. We bring forward the term of complex constrained CA (CC-CA in short) model, which integrates the constrained conditions of neighborhood, macro socio-economy, space and institution. Particularly, the constrained construction zoning, as one institutional constraint, is considered in the CC-CA modeling. In the paper, the conceptual CC-CA model is introduced together with the transition rules. Based on the CC-CA model for Beijing, we discuss the complex constraints to the urban development of , and we show how to set institutional constraints in planning scenario to control the urban growth pattern of Beijing.

Landscape dynamic & risk warning

00

Chen et al. (2020) used the CA model to simulate the development of global urban agglomeration and the impact of global expansion on global grain output under multiple SSP scenarios in the future.

Lin et al. (2020) used FLUS to simulate the future urban development and combined it with the inundation analysis model to discuss the possible future coastal flood exposure risk caused by the future land use change.

Xu et al. (2012) used the CLUE-S model to predict the urban development of an urban agglomeration in China in 2020, and combined it with the PM10 measurement data, and predicted the PM10 concentration distribution of future cities.

nature COMMUNICATIONS

ARTICLE

https://doi.org/10.1038/s41467-020-14386-x OPEN

Global projections of future urban land expansion under shared socioeconomic pathways

Guangzhao Chen[1,9], Xia Li[2,9*], Xiaoping Liu[1,3*], Yimin Chen[1*], Xun Liang[4], Jiye Leng[1,5], Xiaocong Xu[1], Weilin Liao[1], Yue'an Qiu[1,6], Qianlian Wu[1,7] & Kangning Huang[8]

Contents lists available at ScienceDirect

Science of the Total Environment

journal homepage: www.elsevier.com/locate/scitotenv

ELSEVIER

Scenario-based flood risk assessment for urbanizing deltas using future land-use simulation (FLUS): Guangzhou Metropolitan Area as a case study

Weibin Lin[a,b], Yimin Sun[a,*], Steffen Nijhuis[b], Zhaoli Wang[c]

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[b] Faculty of Architecture and the Built Environment, Department of Urbanism, Delft University of Technology (TU Delft), Julianalaan …

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Contents lists available at ScienceDirect

Atmospheric Environment

journal homepage: www.elsevier.com/locate/atmosenv

ELSEVIER

A hybrid Grey-Markov/ LUR model for PM_{10} concentration prediction under future urban scenarios

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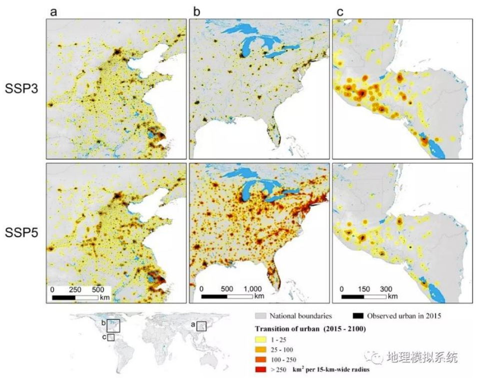

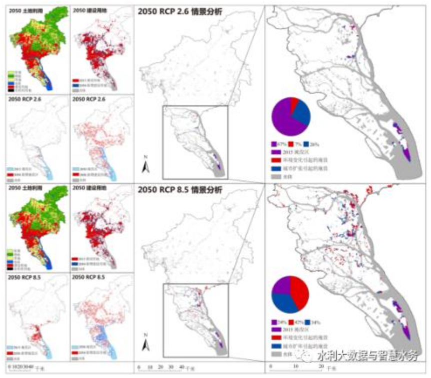

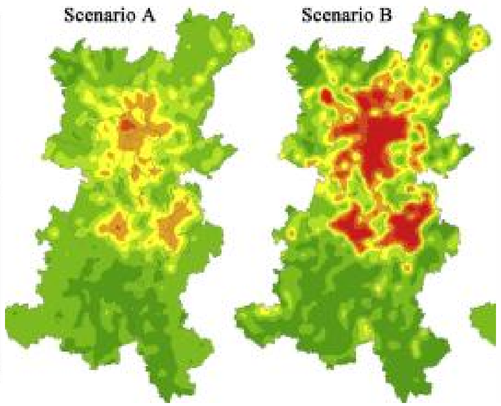

00

Water pollutions & consumption

Zhu et al. (2020) used the improved CLUE-S model to projected the impacts of smart decline on urban runoff contamination levels

Xu et al. (2022) simulated the impacts of land use/land cover patterns on groundwater quality in the Guanzhong Basin of northwest China

Bao et al. (2022) predict the land change trends and water consumption in typical arid regions with CA models

Zhu and Newman *Computational Urban Science* (2021) 1:2
https://doi.org/10.1007/s43762-021-00002-1

Computational Urban Science

ORIGINAL PAPER Open Access

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The projected impacts of smart decline on urban runoff contamination levels

Rui Zhu* and Galen Newman

Impacts of land use/land cover patterns on groundwater quality in the Guanzhong Basin of northwest China

Fei Xu[a,b], Peiyue Li[a,b], Wenqian Chen[a,b], Song He[a,b], Fan Li[a,b], Dawei Mu[a,b] and Vetrimurugan Elumalai[c]

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Ecological Indicators 141 (2022) 109110

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Ecological Indicators

journal homepage: www.elsevier.com/locate/ecolind

ELSEVIER

Predicting land change trends and water consumption in typical arid regions using multi-models and multiple perspectives

Qingling Bao[a,b], Jianli Ding[a,b,*], Lijing Han[a,b], Jiang Li[c], Xiangyu Ge[a,b]

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Keywords:
Patch-generating land use simulation model

ABSTRACT

Water scarcity has emerged as a major impediment to the long-term development of inland arid region basins. Increased human activity has produced land degradation in arid areas and water stress in recent years. The

00 Agricultural application

The Seto team at Yale University, in 2016, used a CA model to simulate future urban land expansion and its impact on global cropland. By 2030, urban expansion will result in a 1.8-2.4% reduction in global arable land area.

Wang et al. (2020) used CA model to conclude that urbanization will facilitate large-scale production in Chinese agriculture. The study shows that if China's urbanization level increases from 56% in 2015 to 80% in 2050, 5.8 million hectare of rural land will actually be released for agricultural production.

PNAS
SPECIAL FEATURE
CrossMark

Future urban land expansion and implications for global croplands

Christopher Bren d'Amour[a,b], Femke Reitsma[c], Giovanni Baiocchi[d], Stephan Barthel[e,f], Burak Güneralp[g], Karl-Heinz Erb[h], Helmut Haberl[h], Felix Creutzig[a,b,1], and Karen C. Seto[i]

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Urban expansion often occurs on croplands. However, there is little scientific understanding of how global patterns of future urban expansion will affect the world's cultivated areas. Here, we combine spatially explicit projections of urban expansion with datasets on global croplands and crop yields. Our results show that urban ex-

India, and other countries (7–9). Although cropland loss has become a significant concern in terms of food production and livelihoods (10) for many countries, there is very little scientific understanding of how future urban expansion and especially growth of MURs will affect croplands. However, this knowledge

nature food
ARTICLES
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Check for updates

Urbanization can benefit agricultural production with large-scale farming in China

Sitong Wang[1,2,10], Xuemei Bai[3,10], Xiaoling Zhang[4,5,10], Stefan Reis[6,7], Deli Chen[8], Jianming Xu[1,9] and Baojing Gu[1,9]

Urbanization has often been considered a threat to food security since it is likely to reduce the availability of croplands. Using spatial statistics and scenario analysis, we show that an increase in China's urbanization level from 56% in 2015 to 80% in 2050 would actually release 5.8 million hectares of rural land for agricultural production—equivalent to 4.1% of China's total cropland area in 2015. Even considering the relatively lower land fertility of these new croplands, crop production in 2050 would still be 3.1–4.2% higher than in 2015. In addition, cropland fragmentation could be reduced with rural land release and a decrease in rural population, benefiting large-scale farming and environmental protection. To ensure this, it is necessary to adopt an integrated urban–rural development model, with reclamation of lands previously used as residential lots. These insights into the urbanization and food security debate have important policy implications for global regions undergoing rapid urbanization.

Ecologial land loss predictions

Ke et al. (2016) and other researchers used CA models to simulate the direct loss of natural habitats such as wetlands and grasslands, and indirect loss of natural habitats, under the policy requirement of " balance of occupancy and fill " of farmland.

Land Use Policy xxx (xxxx) xxx–xxx

Contents lists available at ScienceDirect

Land Use Policy

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Direct and indirect loss of natural habitat due to built-up area expansion: A model-based analysis for the city of Wuhan, China

Xinli Ke[a], Jasper van Vliet[b], Ting Zhou[c], Peter H. Verburg[b], Weiwei Zheng[a], Xiaoping Liu[d,*]

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[b] Environmental Geography Group, Faculty of Earth and Life Sciences, Vrije Universiteit Amsterdam, The Netherlands
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ARTICLE INFO

Keywords:
Land use change
Land use policy
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Built-up area expansion
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Land displacement

ABSTRACT

Urbanization has been responsible for the loss of cropland worldwide, especially in China. To guarantee national food security, China has implemented a series of policies to protect cropland. One of these policies requires that one-hectare cropland should be reclaimed when urban expansion occupies one-hectare cropland. Since most cropland reclamation leads to a conversion of natural habitat, such as wetland and grassland, urban expansion may lead to (indirect) natural habitat loss in addition to direct loss from conversion of into urban area. While several studies assessed the direct habitat loss resulted from built-up area expansion, few studies investigated the indirect losses caused by cropland displacement. In this paper, a model-based approach is applied to explore both direct and indirect impacts of built-up area expansion on natural habitat loss for the city of Wuhan, China.

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Projecting the future impacts of China's cropland balance policy on ecosystem services under the shared socioeconomic pathways

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Shared socioeconomic pathways

ABSTRACT

China has implemented a strict policy of cropland balance to reduce the impacts of built-up land expansion on food production. However, this policy only alleviates the quantitative loss of cropland, while exerting unexpected damages to ecosystem services. Given that China's built-up land will continue to expand in the foreseeable future, the evaluation of potential ecological cost of obtaining a cropland balance is important to help seeking a sustainable strategy for cropland protection. To address this issue, this study develops a cellular automaton model to conduct multiple scenario simulations under the assumptions of shared socioeconomic pathways. The results reveal the inferior physical condition of the reclaimed cropland for compensation. Despite the quantity balance achieved, the lost productivity is compensated by only 28–31%. If the lost productivity is intended to be fully offset, however, the amounts of reclaimed cropland should be more than double, leading to substantial declines in ecosystem services. These findings suggest the failure of the current cropland balance policy, because either the lost cropland or productivity had to be compensated at a high cost of ecological declines. The results also imply that upgrading the quality of remaining cropland is more important than maintaining the quantity balance.

nature sustainability

ANALYSIS

https://doi.org/10.1038/s41893-019-0340-0

Direct and indirect loss of natural area from urban expansion

Jasper van Vliet

Global losses of natural area are primarily attributed to cropland expansion, whereas the role of urban expansion is considered minor. However, urban expansion can induce cropland displacement, potentially leading to a loss of forest elsewhere. The extent of this effect is unknown. This study shows that indirect forest losses, through cropland displacement, far exceed direct losses from urban expansion. On a global scale, urban land increased from 33.2 to 71.3 million hectares (Mha) between 1992 and 2015, leading to a direct loss of 3.3 Mha of forest and an indirect loss of 17.8 to 32.4 Mha. In addition, this urban expansion led to a direct loss of 4.6 Mha of shrubland and an indirect loss of 7.0 to 17.4 Mha. Guiding urban development towards more sustainable trajectories can thus help preserve forest and other natural area at a global scale.

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Urban governance & early warning

Liang et al.(2018) delineated multi-scenario **urban growth boundaries (UGBs)** by coupling a CA-based FLUS model and morphological method

Li et al.(2013) provided an early warning of **illegal development for protected areas** by integrating cellular automata with neural networks

Contents lists available at ScienceDirect

Landscape and Urban Planning

journal homepage: www.elsevier.com/locate/landurbplan

ELSEVIER

Research Paper

Delineating multi-scenario urban growth boundaries with a CA-based FLUS model and morphological method

Xun Liang[a], Xiaoping Liu[a,*], Xia Li[b,a,*], Yimin Chen[b,a], He Tian[a], Yao Yao[c]

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ELSEVIER

Early warning of illegal development for protected areas by integrating cellular automata with neural networks

Xia Li*, Chunhua Lao, Yilun Liu, Xiaoping Liu, Yimin Chen, Shaoying Li, Bing Ai, Zijian He

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ABSTRACT

Ecological security has become a major issue under fast urbanization in China. As the first two cities in this country, Shenzhen and Dongguan issued the ordinance of Eco-designated Line of Control (ELC) to "wire" ecologically important areas for strict protection in 2005 and 2009 respectively. Early warning systems (EWS) are a useful tool for assisting the implementation ELC. In this study, a multi-model approach is proposed for the early warning of illegal development by integrating cellular automata (CA) and artificial neural networks (ANN). The objective is to prevent the ecological risks or catastrophe caused by such development at an early stage. The integrated model is calibrated by using the empirical

Protected area planning and evaluation

He et al. (2005) zones **grassland protection area** using cellular automata modeling in Xilingol steppe grassland in northern China

Mondal et al.(2010) provided an evaluation of **conservation interventions** using a cellular automata-Markov model

Lin et al.(2016) identified the **conflict zones** bewteen future urban development and protected area in Guangzhou, China using a CA model

ELSEVIER
Journal of Arid Environments
Volume 63, Issue 4, December 2005, Pages 814-826

Zoning grassland protection area using remote sensing and cellular automata modeling—A case study in Xilingol steppe grassland in northern China

C. He [a, b], Q. Zhang [c], Y. Li [a, b], X. Li [a, b], P. Shi [a, b]

ELSEVIER
Forest Ecology and Management
Volume 260, Issue 10, 15 October 2010, Pages 1716-1725

Evaluation of conservation interventions using a cellular automata-Markov model

Pinki Mondal, Jane Southworth

Journal of Environmental Management 170 (2016) 177–185
Contents lists available at ScienceDirect
Journal of Environmental Management
journal homepage: www.elsevier.com/locate/jenvman
ELSEVIER

Research article

Conflict resolution in the zoning of eco-protected areas in fast-growing regions based on game theory

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ABSTRACT

Zoning eco-protected areas is important for ecological conservation and environmental management. Rapid and continuous urban expansion, however, may exert negative effects on the performance of practical zoning designs. Various methods have been developed for protected area zoning, but most of them failed to consider the conflicts between urban development (for the benefit of land developers) and ecological protection (local government). Some real-world zoning schemes even have to be modified occasionally after the lengthy negotiations between the government and land developers. Therefore, our

Solution

Do we have a free, practical and easy-used software which can be used as the basis to carry out all the researches I mentioned above?

Of course!

Patch-generating Land Use Simulation Model(PLUS)

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Understanding the drivers of sustainable land expansion using a patch-generating land use simulation (PLUS) model: A case study in Wuhan, China

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Can be used to fine-resolution/large regions!

Can input or output image files (tiff file) with coordinate and projection!

Can simulate urban development and change of multiple land use types!

Can spatio-temperally display the simulation process !

Can reveal the underlying drivers of land use change and simulate the patches change of land uses!

Policy factors can be easily added into the simulation process!

With the newest simulation algorithm, which can obtain higher simulation accuracy than any other software!

With complete user's manual and test data!

With complete modules (data pre-processing, simulation, prediction, evaluation, analysis modules)!

Has user-friendly and beautiful graphical interface !

High running speed!

Totally free!

All data can be freely obtained!

Has been used by users from different countries, and successfully simulate cities in different countries!

CONTENT

01 The Rational of The PLUS Model

02 Tutorial on Using The PLUS Software

03 Application of land use change models

01

The algorithm and innovation of the PLUS Model

01 Introduction

Ideally, cellular automata(CA) models are developed and applied to make land use policy and land management decisions(Sohl & Claggett, 2013). However, most CA modeling research has focused on the improvement of technical modeling procedures or on model calibration and rules, and little attention has been paid to the need for a more conceptual understanding of the underlying causes of land use and land cover(LULC) (Cao, Tang, Shen, & Wang, 2015;Engelen, White, Maarten, & Bernhard, 2002).

For decision-makers, many of the existing CA models are still: (1) weak at revealing the underlying drivers of land use change (Sohl & Claggett, 2013); and (2) unable to spatio-temporally capture the evolution of patches of multiple land uses, especially for the patch evolution of nature land use types (Meentemeyer et al., 2013; Yang, Gong, Tang, & Liu, 2020).

In summary, the existing CA models based on transiton rule mining strategy and landscape dynamic change simulation strategy have some shortcomings, which leads to insufficient progress in the research of these two aspects in recent years.

01

Existing main transforming rule mining strategies

Transition analysis strategy(TAS)：It must extract sample cells with land use states that are changed during the time period between two dates of land use data to examine the neighboring states, or distances and weights of the contributing factor layers. It includes logistic-CA (Wu & Webster, 1998), ANN-CA(Omrani el al., 2019).However, when simulating the transitions of multiple land use types, the number of transition types theoretically increases exponentially with the number of land use types($K \rightarrow (K^2 - K)$). It increases the computational complexity of the model structure of CA and reduces the model's flexibility and general applicability.

Pattern analysis strategy(PAS)：CA models based on PAS calculate the probability of occurrences of land use types in each cell with just one data for the land use data. Such as CLUE-s (Verburg et al., 2002), Fore-SCE（Sohl & Sayler, 2008）and FLUS（Liu et al., 2017）. However, CA models based PAS inherently lack the ability to reveal how the driving factors cause land use change because they are unable to analyze the evolution of land use from the perspective of class transitions.

Future land use types

Original land use types

Convert to	Type 1	Type 2	...	Type K-1	Type K
Type 1	1 persist	1 to 2	...	1 to K-1	1 to K
Type 2	2 to 1	2 persist	...	2 to K-1	2 to K
...	...	...	...	...	...
Type K-1	K-1 to 1	K-1 to 2	...	K-1 persist	K-1 to K
Type K	K to 1	K to 2	...	K to K-1	K persist

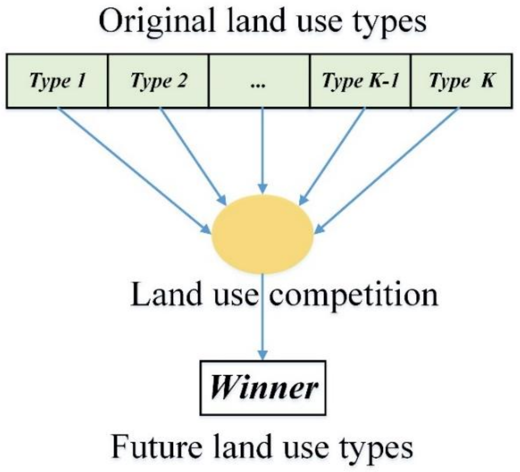
ALL K^2

−

K persistence

=

K^2-K transitions



Factors restricting patch-level landscape simulation

Simulating the succession of land use patches can be used to detect ecologically significant changes and to assess, design, and plan ecosystem management activities in advance (Keane, Parsons, & Hessburg, 2002)。Although researchers have developed a set of CA models to simulate patch changes based on raster or vector data. (Yang, Gong, Tang, & Liu, 2020; Yao et al., 2017)。However, these models generally focus on urban land use and rarely on patch changes of natural land use types.

Sohl, el al., (2007) promoted a patch growth land use simulation by proposing the raster-based Fore-SCE model. However, the Fore-SCE model separately processes the urban land with a simple density slicing technique because it cannot address the cases where there are very large numbers of very small patches. In short, previous studies lack flexible patch-based strategies for modeling the patch growth of multiple natural land use types at a fine-scale resolution, and the use of these CA models is limited for actual planning, decision-making, or policymaking purposes.

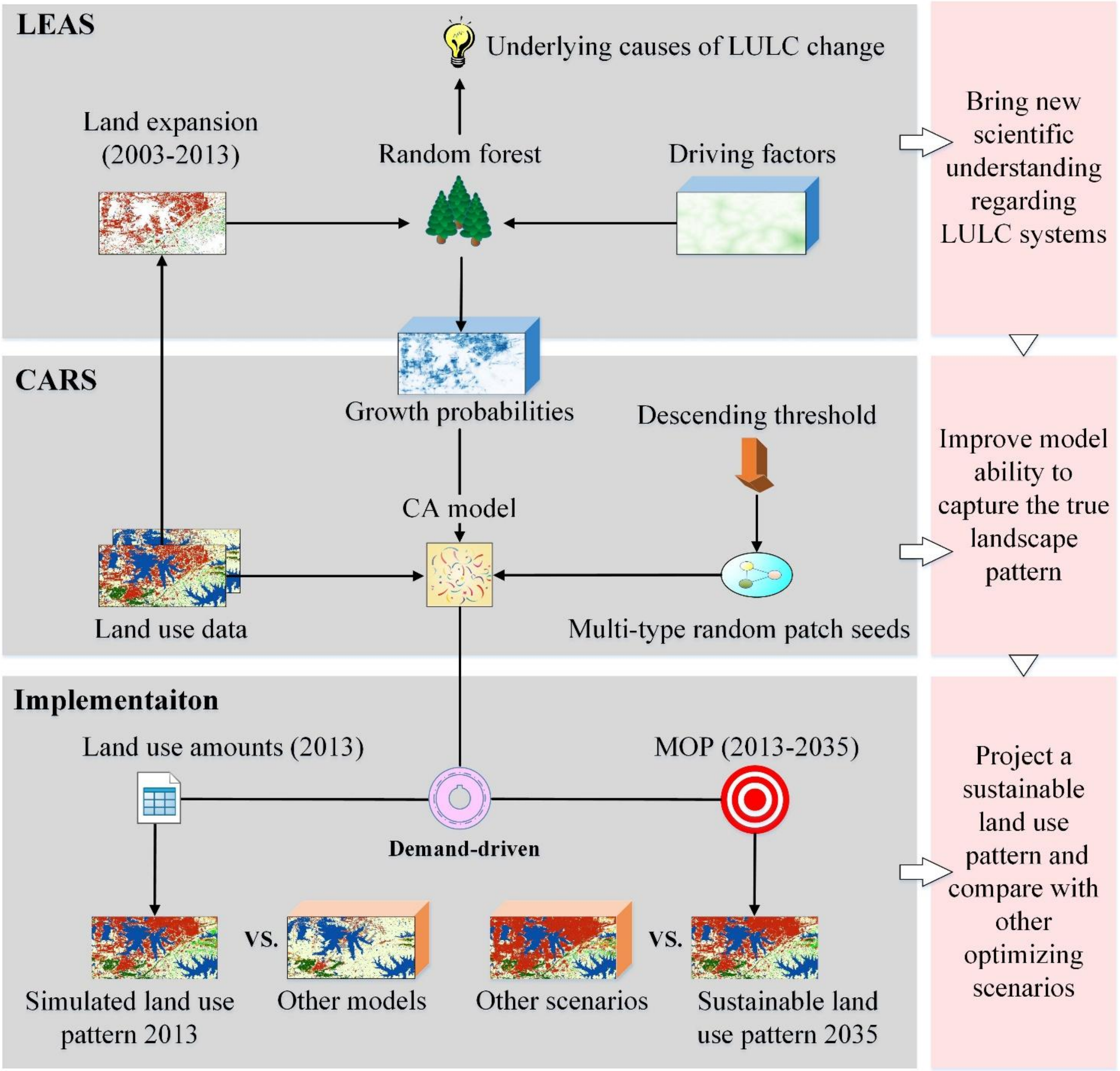
01 Research purpose and significance

Therefore, we developed a raster-based patch-generating land use simulation(PLUS) model. The model:

1） Bring a new analysis strategy, which can better explore the causes of various land use changes.

2） It contains a new multi-class growth mechanism, which can better simulate multi-class land use patch-level changes.

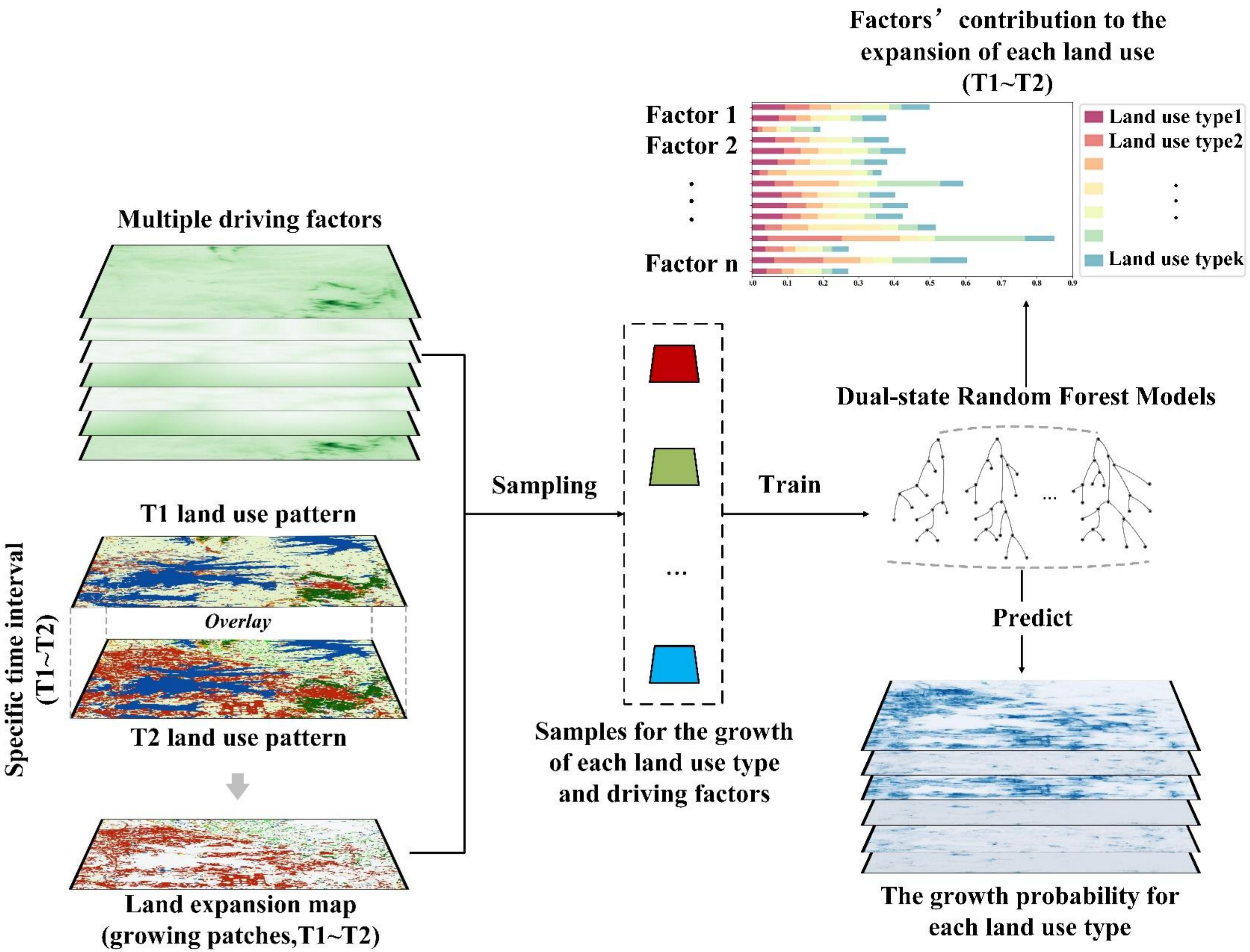
3） Coupled with multiobjective programming, the simulation results can better support the planning policy to achieve sustainable development.



01 Land expansion analysis strategy（LEAS）

The strategy extracted the cells with changed states from the two periods of land use data. And employed a random forest classification(RFC) algorithm to explore the relationship between the growth in each land use type and the multiple driving factors.

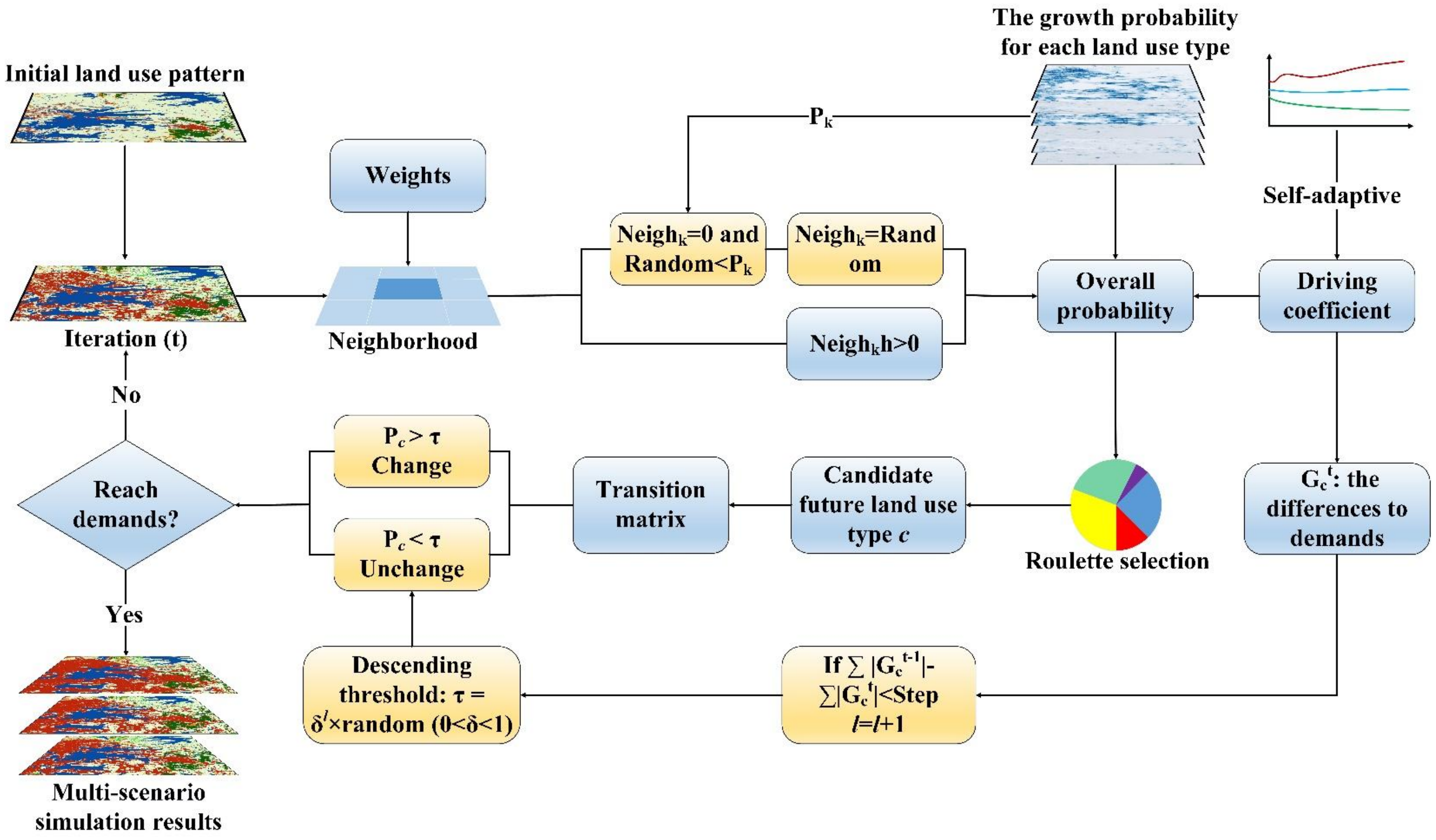
The strategy combines the advantages of existing TAS and PAS, and avoids the analysis of transition types that increase exponentially with the number of categories; Moreover, the mechanism of land use change in a certain period of time, and has better explanatory ability.



01

CA based on multi-type random patch seeds（CARS)

CA models with multi-type random patch seeds and threshold descent rules are spatiotemporally dynamic, which allows the new land use patches to spontaneously grow and freely develop under the constraints of the growth probabilities.



Random land use seeds

$$OP_{i,k}^{1,t} = \begin{cases} P_{i,k}^{1} \times (r \times \mu_k) \times D_k^t & if\ \Omega_{i,k}^t = 0\ and\ r < P_{i,k}^1 \\ P_{i,k}^1 \times \Omega_{i,k}^t \times D_k^t & all\ others \end{cases}$$

$$\Omega_{i,k}^t = \frac{con(c_i^{t-1} = k)}{n \times n - 1} \times w_k$$

$$D_k^t = \begin{cases} D_k^{t-1} & if\ |G_k^{t-1}| \le |G_k^{t-2}| \\ D_k^{t-1} \times \frac{G_k^{t-2}}{G_k^{t-1}} & if\ 0 > G_k^{t-2} > G_k^{t-1} \\ D_k^{t-1} \times \frac{G_k^{t-1}}{G_k^{t-2}} & if\ G_k^{t-1} > G_k^{t-2} > 0 \end{cases}$$

Descending threshold

If $\sum_{k=1}^{N} |G_c^{t-1}| - \sum_{k=1}^{N} |G_c^{t}| < Step$ Then, $d = d + 1$

$$\begin{cases} Change & P_{i,c}^1 > \tau\ and\ TM_{k,c} = 1 \\ Unchange & P_{i,c}^1 \le \tau\ or\ TM_{k,c} = 0 \end{cases}$$

$$\tau = \delta^d \times r$$

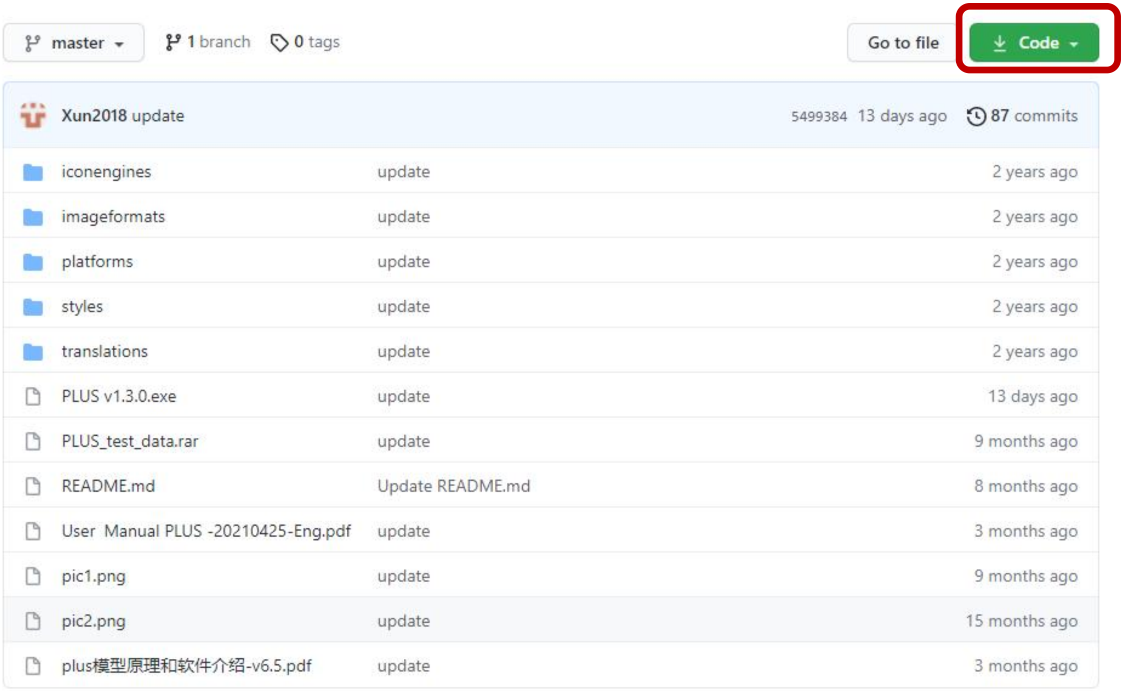
02

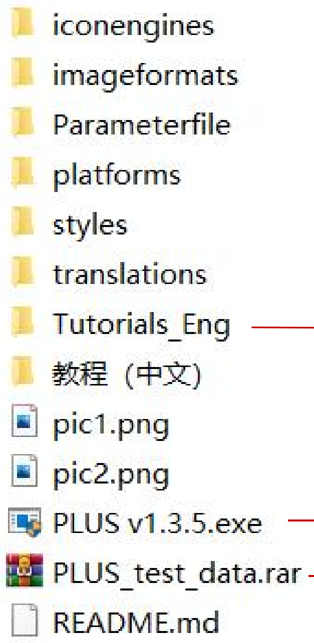
Tutorial on Using The PLUS Software

02 Download PLUS v1.3.5

Link： https://github.com/HPSCIL/Patch-generating_Land_Use_Simulation_Model

Click here to download





User's manual

Click here to start

Example data

PLUS can run in the environment of Windows Vista/7/8/X64 without install process and the support of other software.

Open source library & applications

PLUS was developed purely in the C++ language,

open –source library:

Alglib 3.9.2 (http://www.alglib.net/)

Qt 5 (https://www.qt.io/download/)

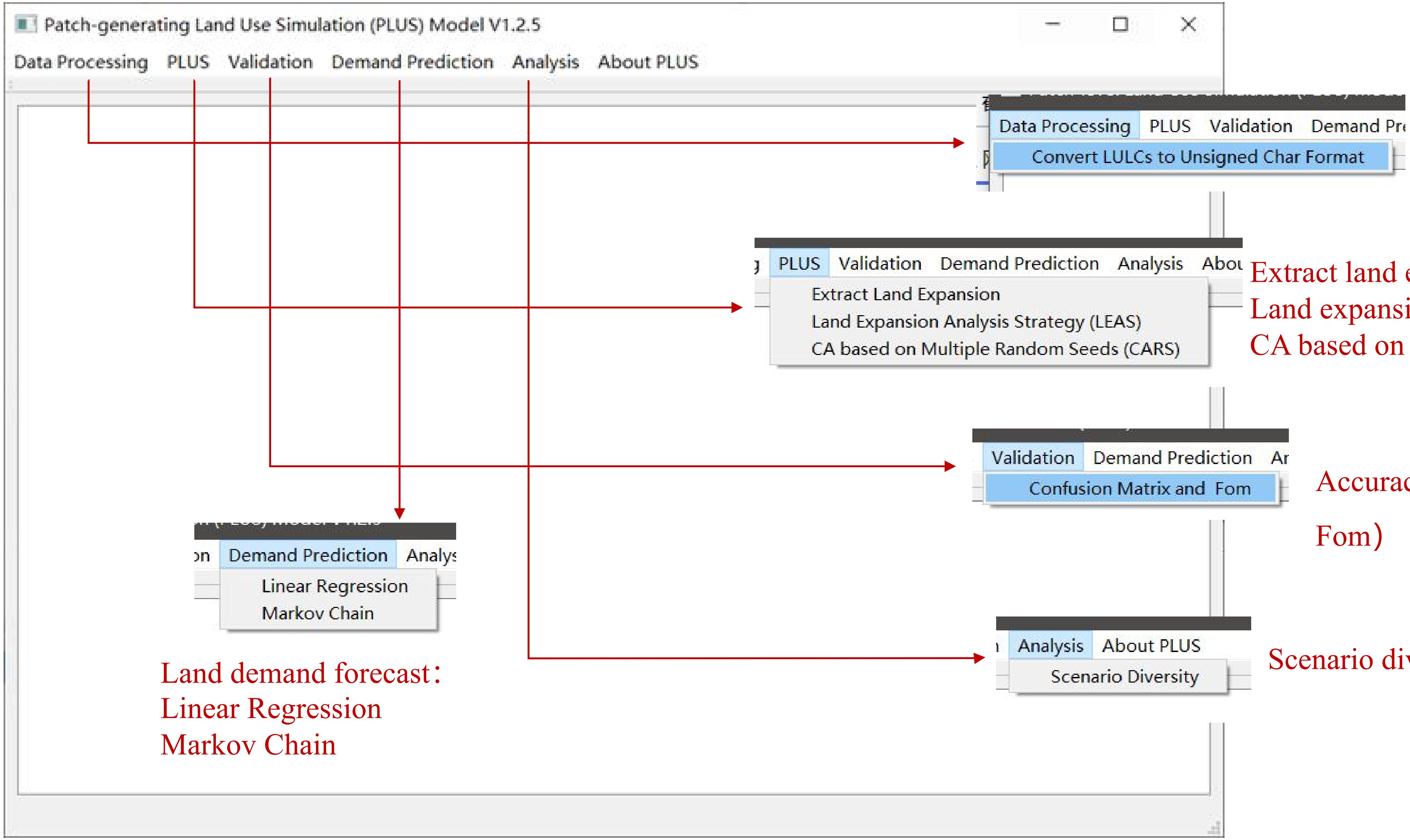
GDAL 2.0.2 (http://www.gdal.org/)

- ✓ Land use/land cover change simulation
- ✓ Policy-making
- ✓ Land use change law excavation
- ✓ Urban planning
- ✓ Ecological security early warning

PLUS can run in the environment of Windows Vista/7/8/X64 without install process and the support of other software.

02

Main interface



Data format conversion

Extract land expansion;
Land expansion analysis strategy;
CA based on multiple random seeds;

Accuracy validation (Kappa and Fom)

Land demand forecast:
Linear Regression
Markov Chain

Scenario diversity analysis

02

Example data description

Category	Data	Filename	Description
Land use data	Land use classification data of Wuhan in 2003	wh2003_refy.tif	1: Grass； 2: Deciduous； 3: Cropland； 4: Urban land； 5: Bare land； 6: Water area； 7: Evergreen forest
	Land use classification data of Wuhan in 2013	wh2013_refy.tif	
Constraint data	Land use constraint	wh_open_water13.tif	
Socioeconomic data	Population	wh_Pop.tif	http://www.geodoi.ac.cn/WebCn/Default.aspx
	GDP	wh_gdp2010.tif	
	Proximity to the arterial road	wh_dist_trunk.tif	OpenStreetMap (https://www.openstreetmap.org/)
	Proximity to the primary road	wh_dist_primary.tif	
	Proximity to the secondary road	wh_dist_secondary.tif	
	Proximity to the tertiary road	wh_dist_teriary.tif	
	Proximity to the railway	wh_dist_railway.tif	
	Proximity to the highway	wh_dist_highway.tif	
	Proximity to the high-speed railway stations	wh_dist_highspdstation.tif	http://lbsyun.baidu.com/
	Proximity to the governments	wh_dist_gov.tif	
Climatic and environmental data	Soil tyoe	wh_soiltype.tif	HWSD v 1.2 (http://westdc.westgis.ac.cn/data/844010ba-d359-4020-bf76-2b58806f9205)
	Annual Mean Temperature	wh_df_tem.tif	WorldClim v2.0 (http://www.worldclim.org/)
	Annual Precipitation	wh_df_pre.tif	
	DEM	wh_df_dem.tif	NASA SRTM1 v3.0
	Slope	wh_df_slope.tif	
	Proximity to the water	wh_dist_openwater.tif	

Other free data source

Global DEM data：
http://www.cgiar-csi.org/data/srtm-90m-digital-elevation-database-v4-1
https://centaurus.caf.dlr.de:8443/eoweb-ng/template/default/welcome/entryPage.vm
http://www.opentopography.org/
Global road traffic data：
http://www.openstreetmap.org
http://www.diva-gis.org/Data
http://sedac.ciesin.columbia.edu/data/set/groads-global-roads-open-access-v1
Global and country level administrative boundaries data：
http://www.naturalearthdata.com/
http://www.diva-gis.org/Data
Global LULC data:
https://globalmaps.github.io/
http://218.244.250.80/GLC30Download/index.aspx
Socio-economic and environmental data:
http://geodata.grid.unep.ch/extras/datasetlist.php
http://geodata.grid.unep.ch/
http://www.worldclim.org/
http://webarchive.iiasa.ac.at/Research/LUC/External-World-soil-database/HTML/
http://cera-www.dkrz.de/WDCC/

Other free data source

https://cgiarcsi.community/2014/11/04/globeland30-glc30/

http://due.esrin.esa.int/page_globcover.php

https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/PRFF8V

https://www.eea.europa.eu/data-and-maps/

https://www.esa-landcover-cci.org/

https://gis.ncdc.noaa.gov/maps/ncei/cdo/daily/

https://sedac.ciesin.columbia.edu/data/collection/gpw-v4

https://luh.umd.edu/

https://freegisdata.rtwilson.com/

https://worldclim.org/data/worldclim21.html

https://www.wcrp-climate.org/

Other free data source

https://www.naturalearthdata.com/

https://globalmaps.github.io/

http://www.diva-gis.org/climate

https://www.fao.org/soils-portal/soil-survey/soil-maps-and-databases/harmonized-world-soil-database-v12/en/

https://catalogue.ceda.ac.uk/uuid/a0913dba885f1b1c0e3eb6dc0c04c188

https://www.fao.org/land-water/land/land-governance/land-resources-planning-toolbox/category/details/en/c/1027491/

https://dataplatform.knmi.nl/?q=PBL

https://www.mrlc.gov/data?f%5B0%5D=category%3Aland%20cover

http://data.ess.tsinghua.edu.cn/

https://lternet.edu/using-lter-data/

https://datadryad.org/stash/dataset/doi:10.5061/dryad.dk1j0

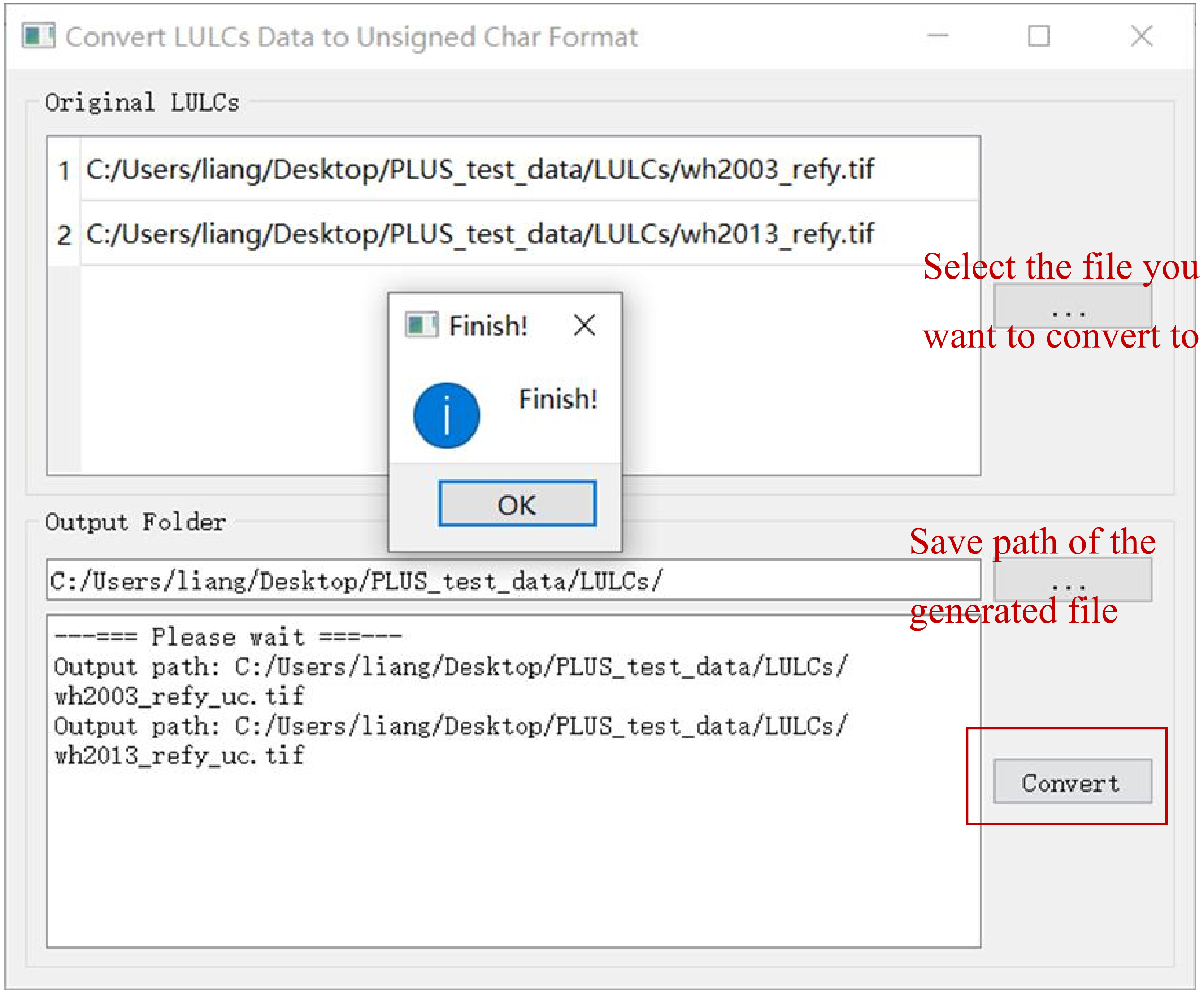
http://earthenginepartners.appspot.com/science-2013-global-forest/download_v1.6.html

http://www.escience.cn/people/feiyunZHU/Dataset_GT.html

02 Data processing-Format conversion

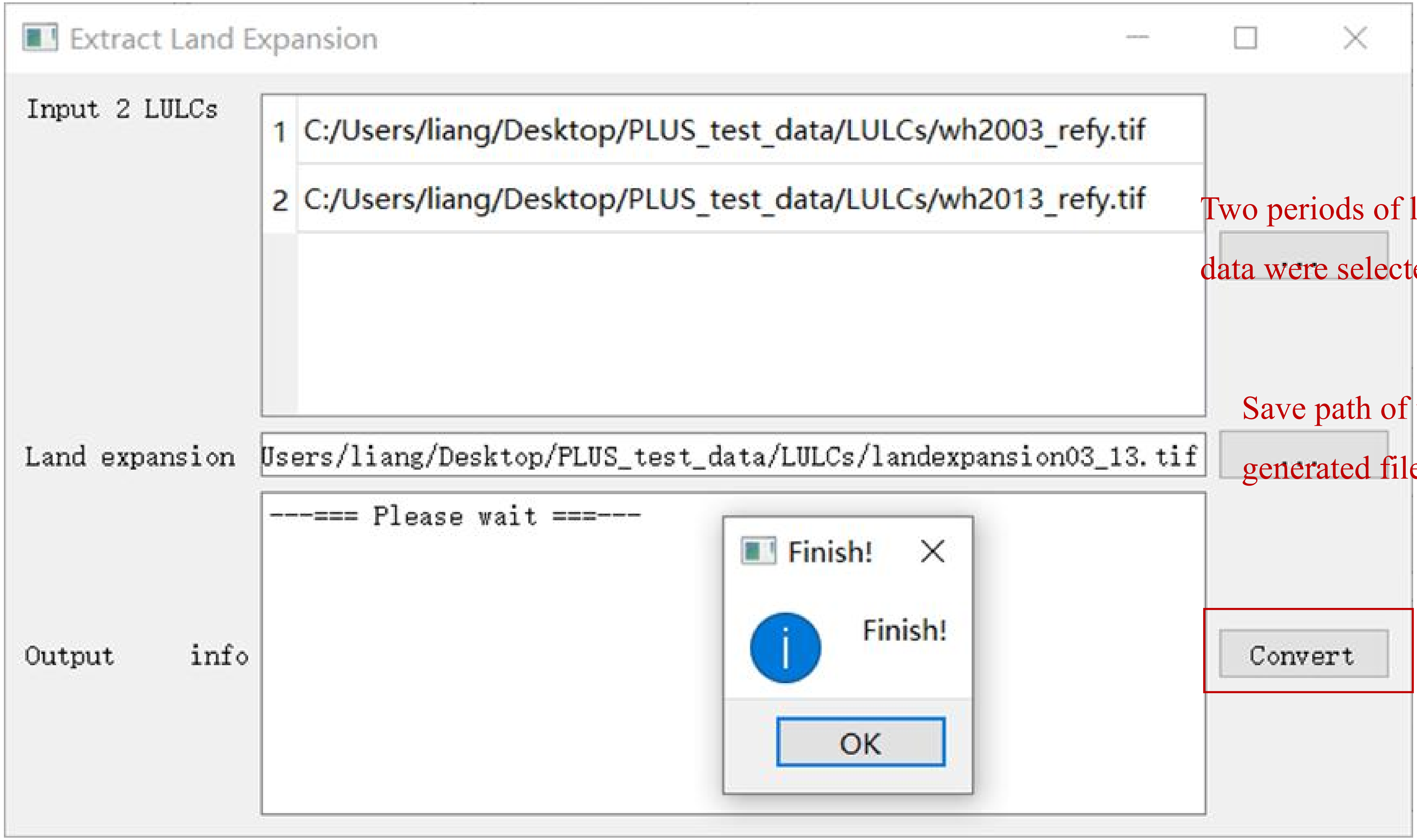
Data format description

- ✓ The PLUS model only supports the ‘unsigned char’ format land use data, and it is recommended that land use data be converted to ‘unsigned char’ format before use. (Only transfer land use data and restricted development area, other driving force data need not go through this transformation, do not carry out redundant transformation) ;
- ✓ The land use data classification number must be a consecutive integer starting from 1;
- ✓ Land use constraint data must be images containing only 0 and 1(0: restricted conversion area; 1: others);
- ✓ All land use data should have the same number of rows and columns.
- ✓ This software all use English path, do not support Chinese,and folder and file do not start with a number, must be a letter!!!



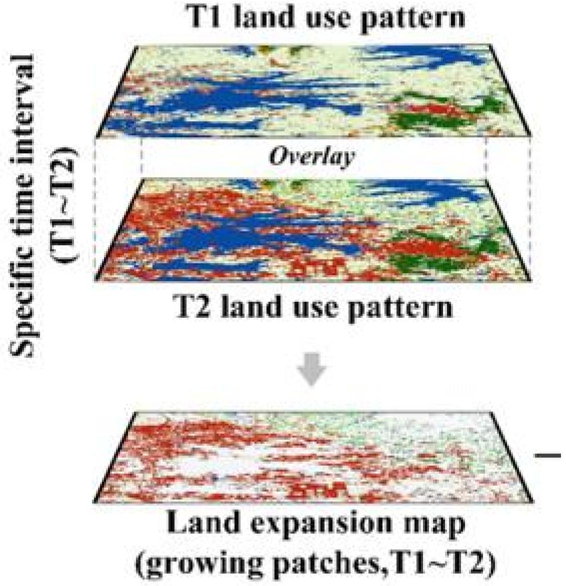
02

PLUS-Extract land expansion



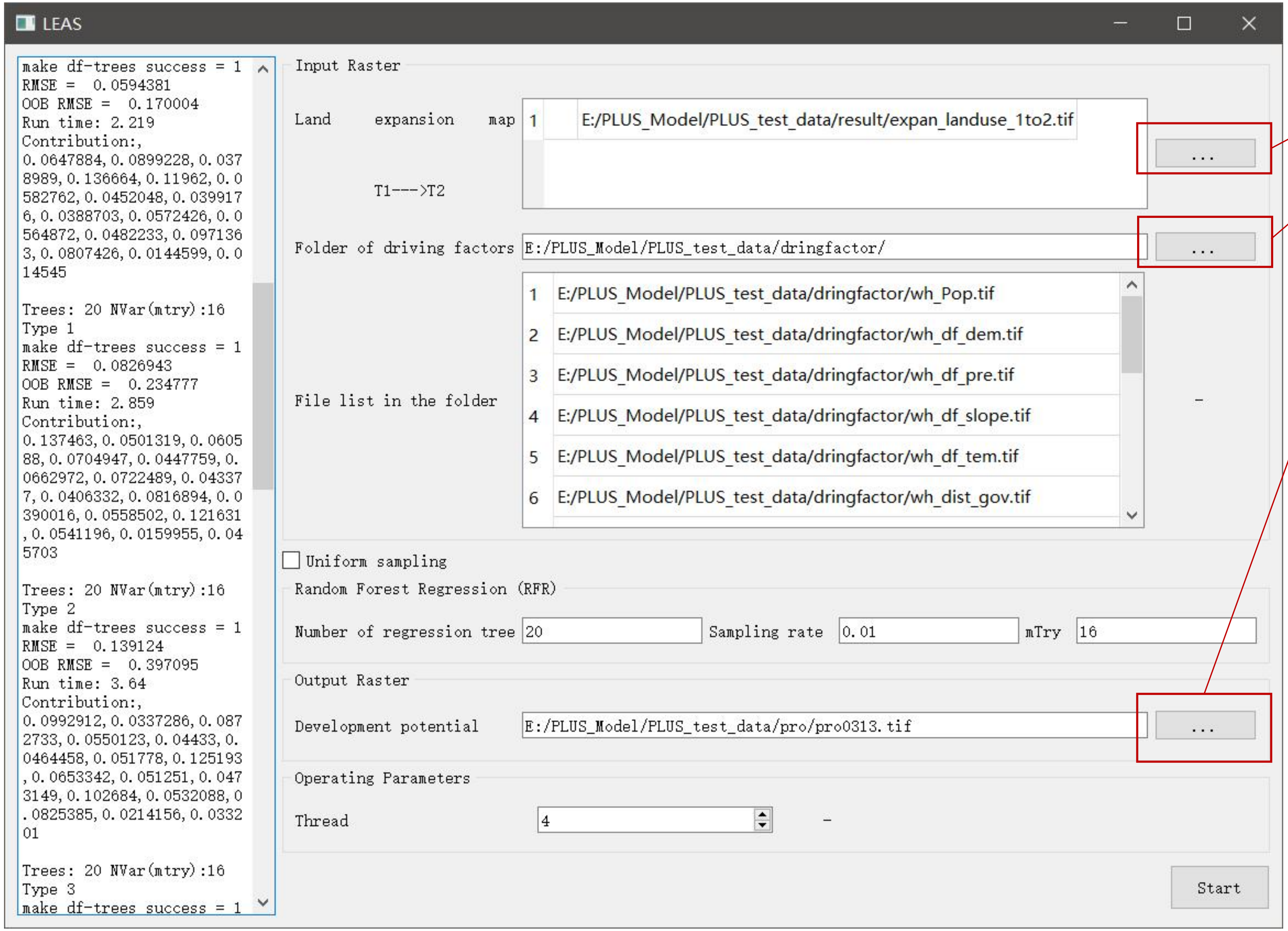
Two periods of land use data were selected

Save path of the generated file



02

PLUS-Land expansion analysis strategy（LEAS）



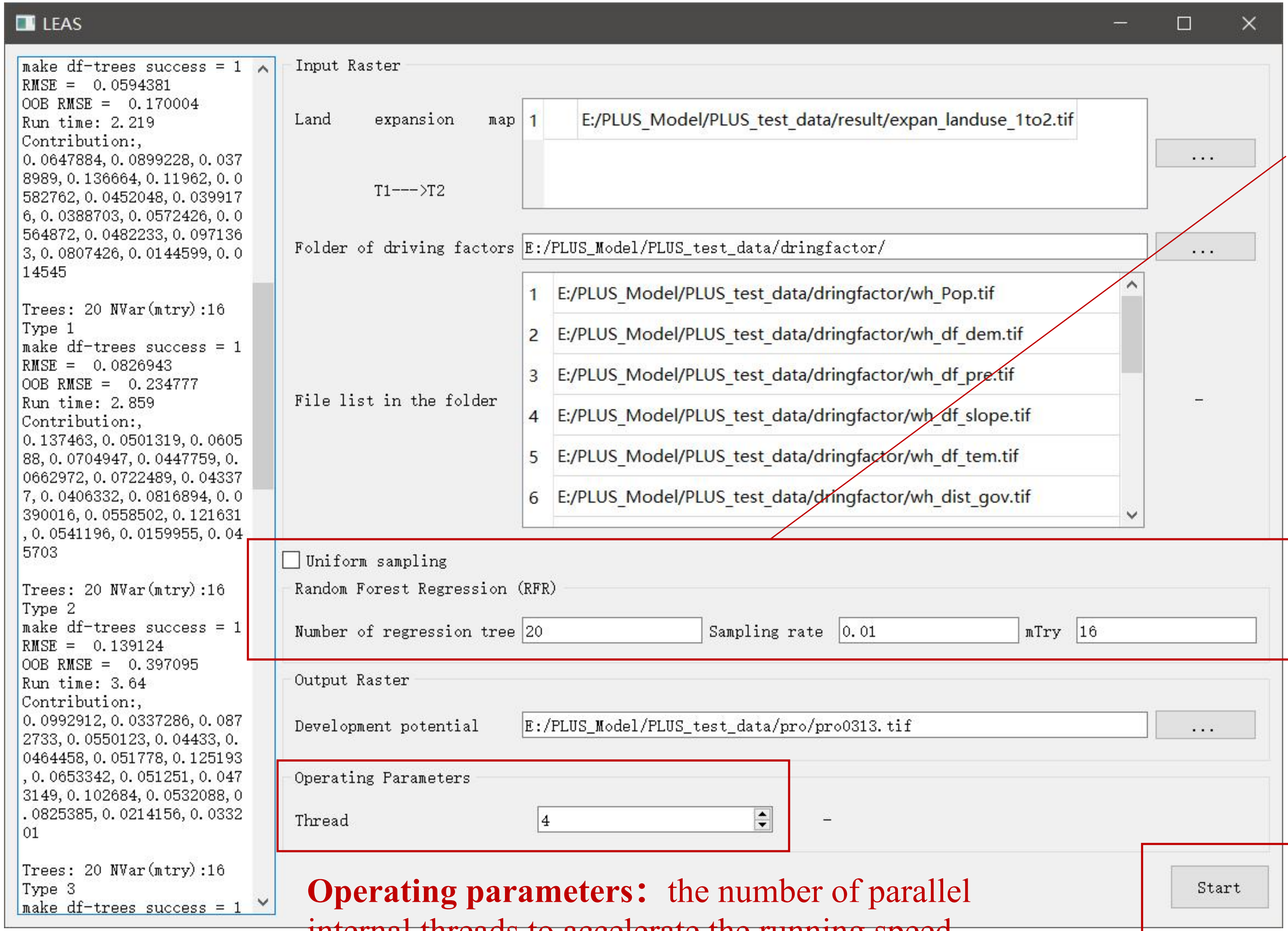
Input and output

- ✓ Land expansion map；
- ✓ Driving factors of LULC；
- ✓ Output path。

LEAS module features

✓ LEAS includes coordinate alignment. the users can import driving factors with common and different coordinate and projection systems to the LEAS sub-model, without unifying them to the same resolution, row count, column count, coordinate, projection, and even map sheet with the ‘land expansion map’ outputted(The LEAS can automatically align the map-areas of those driving factors with the ‘land expansion map’). However, if an error(or null value) is still reported, the users can import only one driving factor into the LEAS module and see if the LEAS can run successfully. If the LEAS is unable to run, the driving factor is problematic. If the LEAS can run with one driving factor, then the users can add new driving factors one by one until the software crashes. Through this way, the users can find out all the problematic driving factors.

02 PLUS-Land expansion analysis strategy（LEAS）



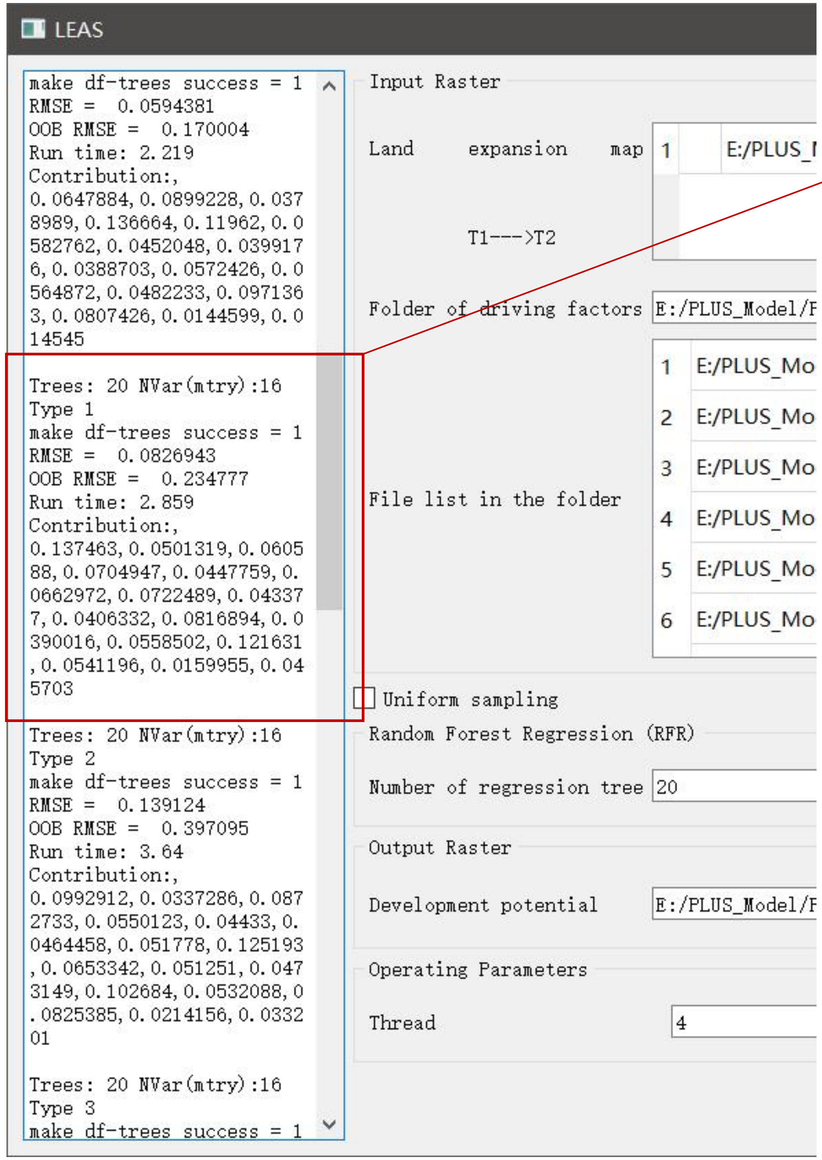
Parameter settings

RFR parameter setting

- ✓ Sampling methods：Uniform sampling（the sampling points for each land use type will be the same）、random sampling（the sampling points will randomly scatter across the study area）；
- ✓ Sampling rate（the default is 0.01）；
- ✓ Number of regression tree；
- ✓ mTry（the number of features for training RF does not exceed the number of drivers）.

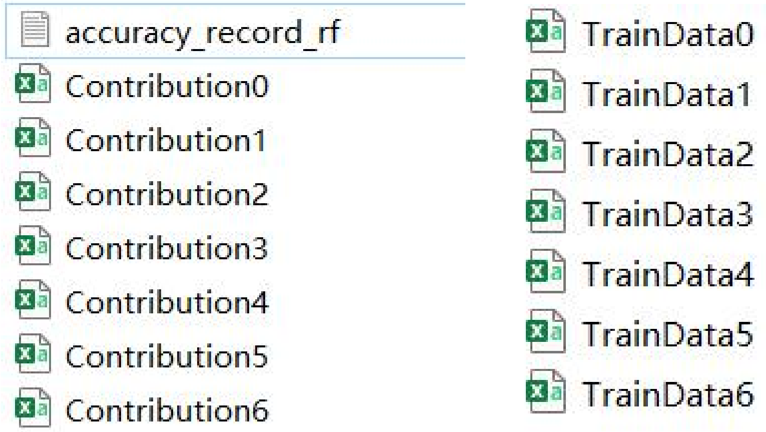
Operating parameters： the number of parallel internal threads to accelerate the running speed.

02 PLUS-Land expansion analysis strategy（LEAS）

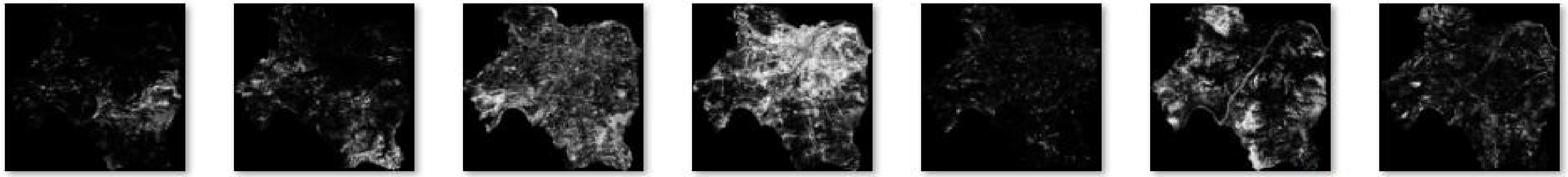


The training results are saved in the “Parameterfile” folder:

- ✓ “accuracy_record_rf.txt”: Random forest training accuracy（RMSE, OOB-RMSE）;
- ✓ ”TrainData.csv” : The sampling data;
- ✓ “Contribution.csv”: Contribution of drivers.



The development probability of various of land

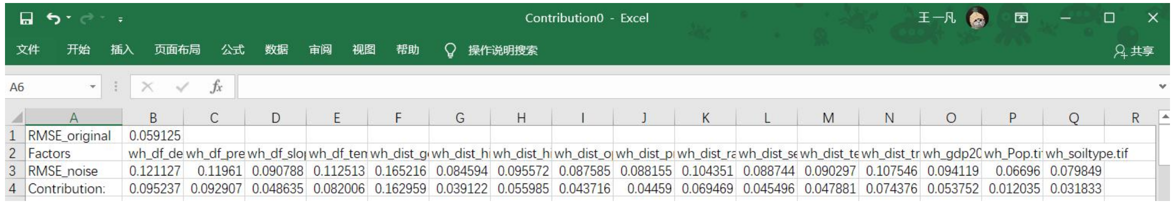


pro0313_band_1 pro0313_band_2 pro0313_band_3 pro0313_band_4 pro0313_band_5 pro0313_band_6 pro0313_band_7

02

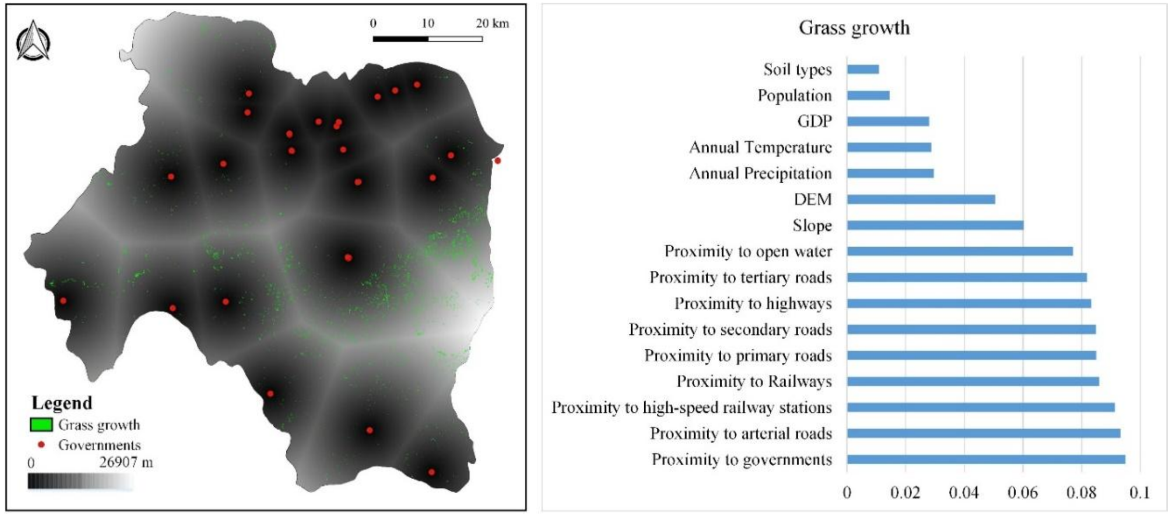
PLUS-Land expansion analysis strategy（LEAS)

Contribution analysis of land use expansion factors

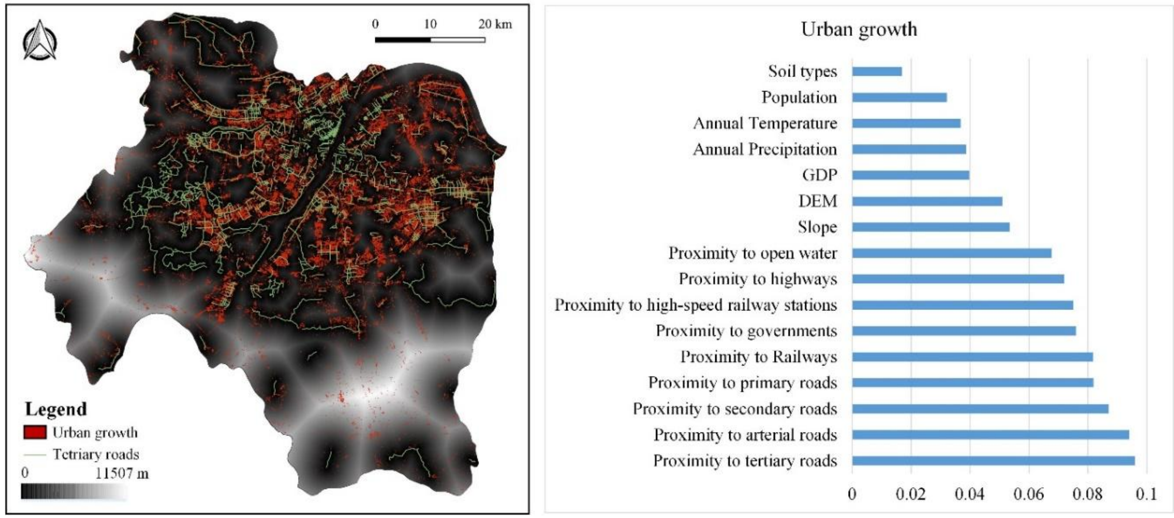


	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	RMSE_original	0.059125																
2	Factors	wh_df_de	wh_df_pre	wh_df_slo	wh_df_ten	wh_dist_g	wh_dist_hi	wh_dist_hi	wh_dist_o	wh_dist_p	wh_dist_ra	wh_dist_se	wh_dist_te	wh_dist_tr	wh_gdp20	wh_Pop.ti	wh_soiltype.tif	
3	RMSE_noise	0.121127	0.11961	0.090788	0.112513	0.165216	0.084594	0.095572	0.087585	0.088155	0.104351	0.088744	0.090297	0.107546	0.094119	0.06696	0.079849	
4	Contribution:	0.095237	0.092907	0.048635	0.082006	0.162959	0.039122	0.055985	0.043716	0.04459	0.069469	0.045496	0.047881	0.074376	0.053752	0.012035	0.031833	

- ✓ According to the Contribution file, the rankingof driving factors for each type of land use can be made to analyze the contribution of land use expansion factors；
- ✓ We found that the new grassland was mainly distributed in areas far away from the governments. The areas around the governments were generally areas where human beings are most active, this result indicates that grass was most likely to grow in areas where it was not strongly impacted by human activities；
- ✓ Urban growth was most influenced by tertiary roads. The distribution of the new urban growth also had a high consistency with the pattern of the tertiary roads.



Ranking of grassland expansion drivers（Contribution0.csv）

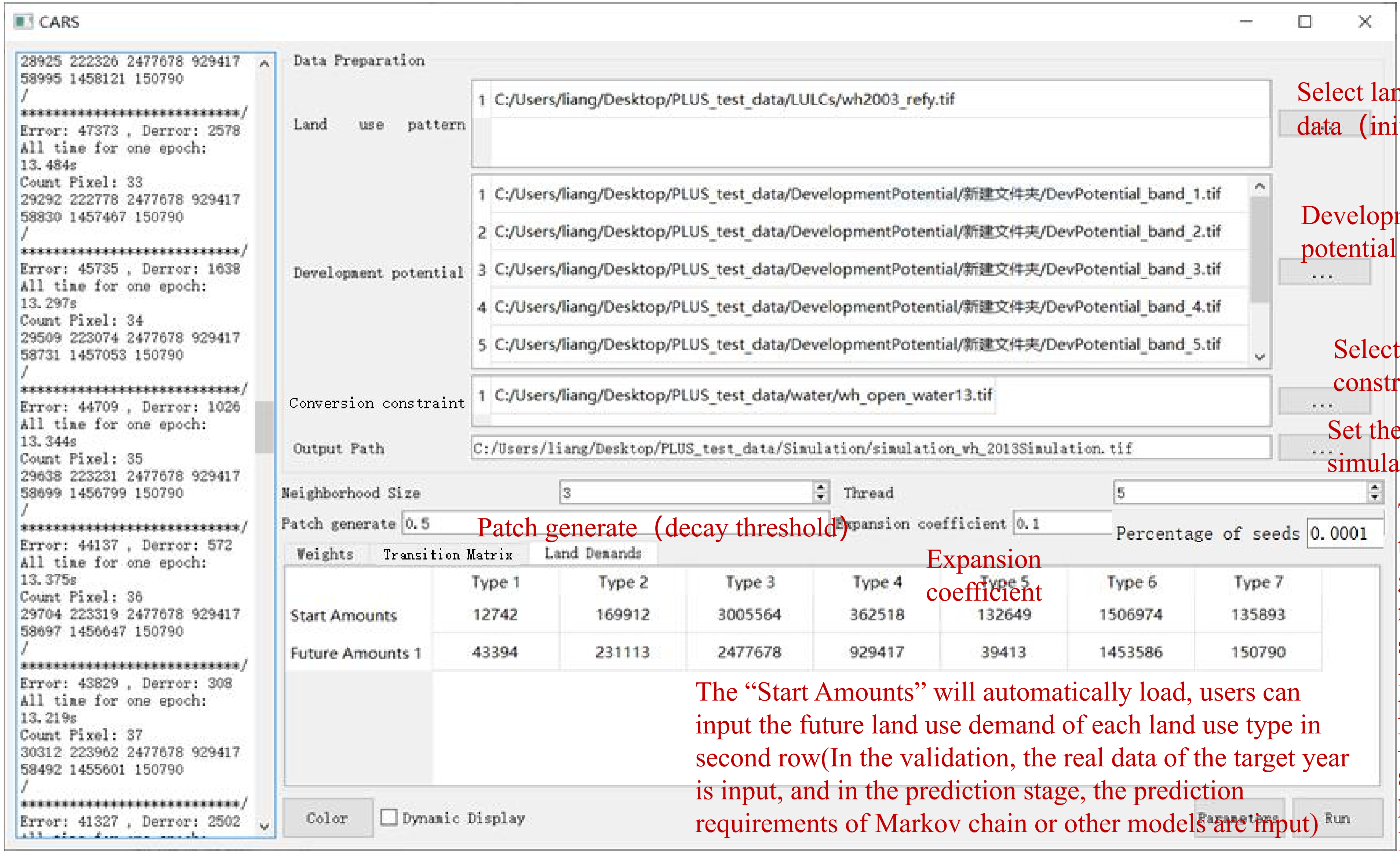


Ranking of urban expansion drivers （Contribution3.csv）

02

PLUS-CA based on multiple random seeds（CARS)

CARS module: It must ensure that all spatial data have the same number of rows and columns



Select land use data（initial year)

Development potential maps

Select the spatial constraint data

Set the path of saving simulation result

Patch generate（decay threshold)

Expansion coefficient

The maximum threshold to generate the number of new seeds, ranging from 0 to 1. A higher Percentage of seeds means a higher dispersive land use pattern.

The “Start Amounts” will automatically load, users can input the future land use demand of each land use type in second row(In the validation, the real data of the target year is input, and in the prediction stage, the prediction requirements of Markov chain or other models are input)

02

PLUS-CA based on multiple random seeds（CARS)

Simulation parameter setting

Neighborhood Size	3	Thread	1
Patch generation	0.5	Expansion coefficient	0.1

- ✓ Neighborhood Size： Neighborhood range， the default value is 3；
- ✓ Thread： Number of threads to improve running speed；
- ✓ Patch generation： The attenuation coefficient of the decreasing threshold ranges from 0 to 1. The higher the value, the less likely the land use type is to be converted；
- ✓ Expansion coefficient： A parameter adjust the ability of model to generate new land use patches, range from 0 to 1. A higher expansion coefficient means a higher ability to generate new patches.

Land use demand

Land Demands | Transtion Matrix | Neighborhood Weights

	Type 1	Type 2	Type 3	Type 4	Type 5	Type 6	Type 7
Start Amounts	12742	169912	3005564	362518	132649	1506974	135893
Future Amounts 1	43394	231113	2477678	929417	39413	1453586	150790

- ✓ Simulation status quo： According to the actual situation of land use in known years or the prediction results of Markov model, etc;

Demand Prediction

- ✓ Predicting the future： Use the “Demand Prediction” function to predict；
- ✓ The “Start Amounts” section will be automatically generated after clicking “Run” , and there is no need to fill in.

02 PLUS-CA based on multiple random seeds（CARS）

Transition matrix

- ✓ A value of 1 means the conversion is allowed, while a value of 0 indicates that the conversion is not possible;
- ✓ It's generally filled according to experience, such as: urban land is not easy to convert, forest land protection is not easy to convert, etc.

Land Demands | Transtion Matrix | Neighborhood Weights

Land use types after simulation

	Type 1	Type 2	Type 3	Type 4	Type 5	Type 6	Type 7
Type 1	1	1	1	1	1	1	1
Type 2	0	1	0	0	0	0	0
Type 3	1	1	1	1	1	1	1
Type 4	0	0	0	1	0	0	0
Type 5	1	1	1	1	1	1	1
Type 6	1	1	1	1	1	1	1
Type 7	0	0	1	0	0	0	1

Current land use types

Neighborhood weights

- ✓ The value ranges from 0 to 1. A larger value indicates a greater influence;
- ✓ Generally, it is filled in according to experience, or it can be calculated according to the proportion of the expanded area of each land type.

Land Demands | Transtion Matrix | Neighborhood Weights

	Type 1	Type 2	Type 3	Type 4	Type 5	Type 6	Type 7
Weight	0.033	0.072	0.214	0.449	0.028	0.131	0.072

- ✓ Click "Parameters" to load all the parameters filled in during the last run.

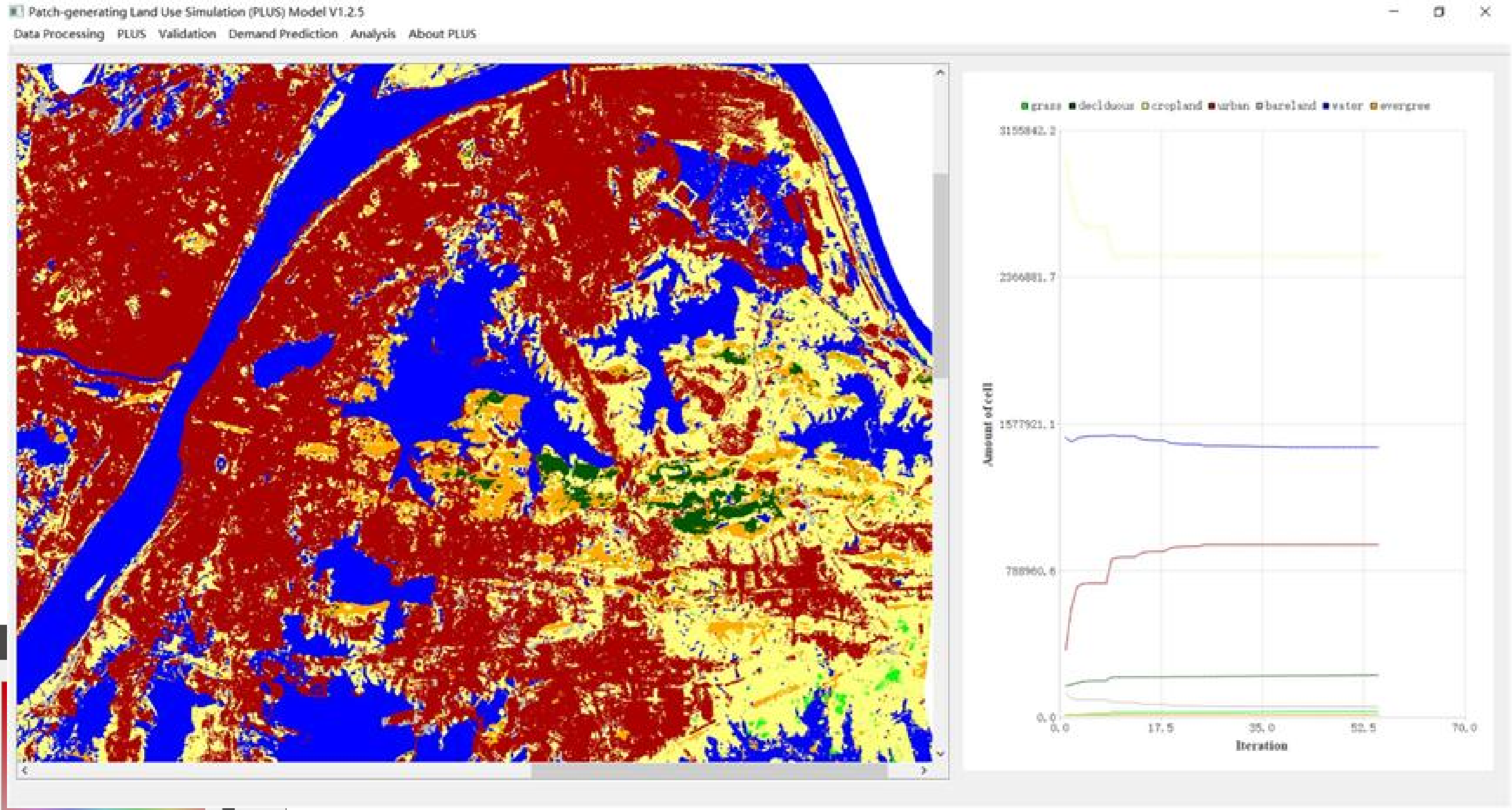
Parameters

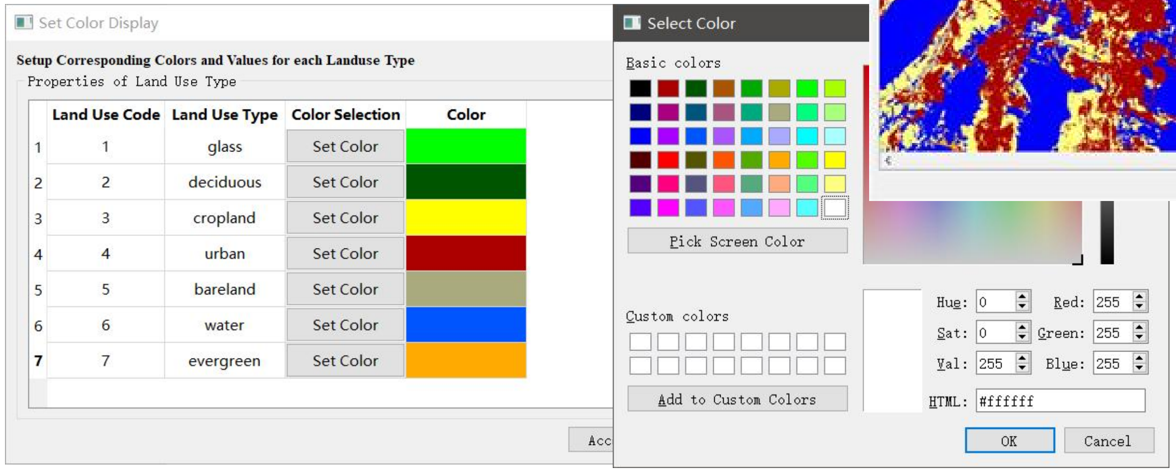
PLUS-CA based on multiple random seeds (CARS)

Start the simulate

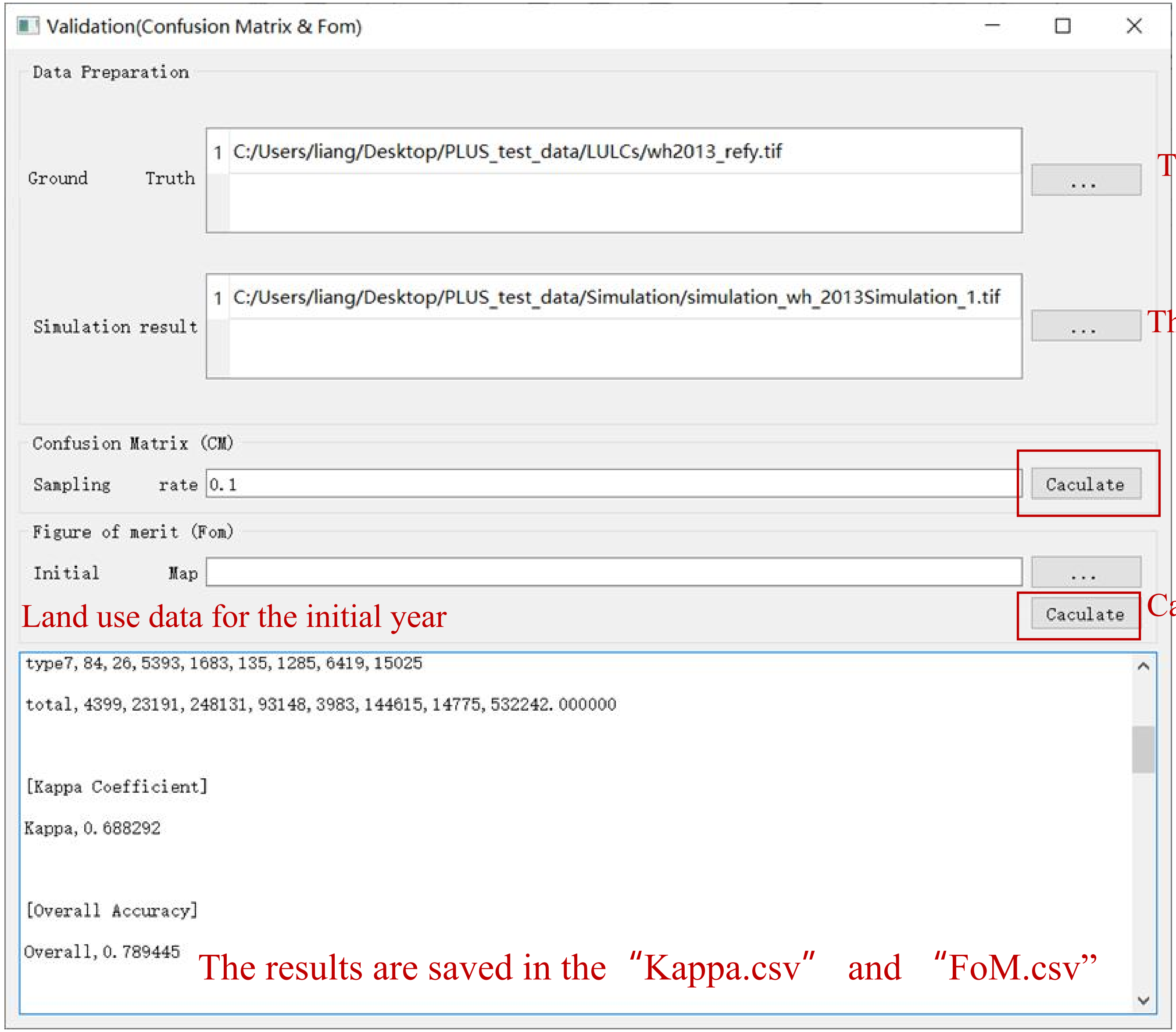
- ✓ Click "Color" button of each row in the list box to set RGB value for all land use types;

- ✓ Check "Dynamic Display" to dynamically display the change of land use status and quantity on the main interface;
- ✓ When all types of land use meet the demand, the simulation is completed.





02 Accuracy validation of simulation-Kappa and FoM



The actual land use data (simulation year)

The simulated land use data

Fill in the sampling rate and calculate the Kappa coefficient

Land use data for the initial year

Calculate the FoM

The results are saved in the "Kappa.csv" and "FoM.csv"

[Kappa Coefficient]

Kappa, 0. 650456

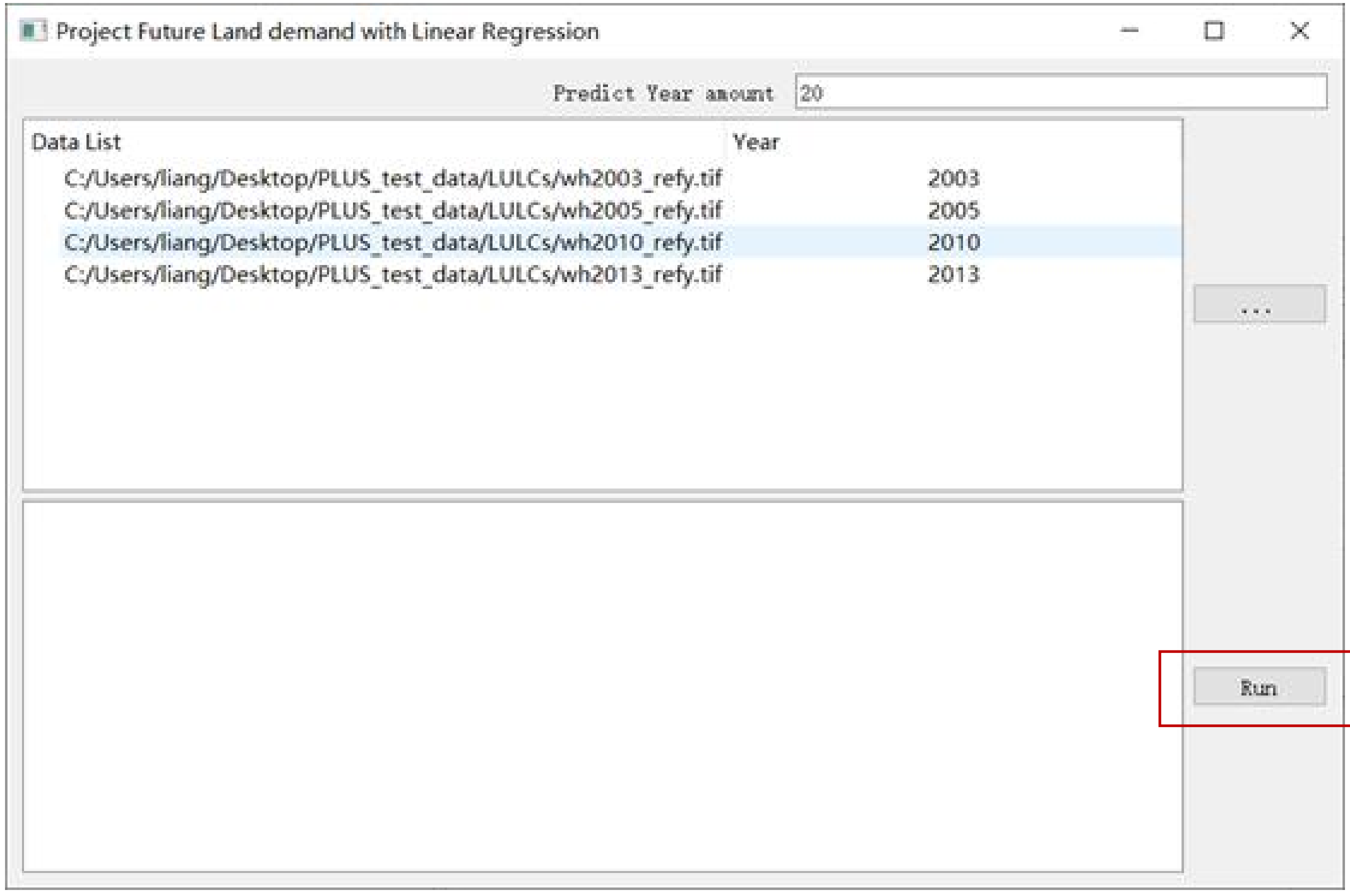
[Overall Accuracy]

Overall, 0. 763807

[Figure of Merit]=B/(A+B+C+D)

FoM=, 0. 227487

Land use demand forecasting-Linear regression



How many years predict

Select the historical land use data

```
2026, 116146, 273124, 1772008, 1581104, 51440, 1382628, 113734
2027, 120655, 278110, 1715672, 1635584, 49764, 1378792, 111590
2028, 125163, 283096, 1659336, 1690056, 48088, 1374956, 109447
2029, 129672, 288082, 1603000, 1744528, 46411, 1371120, 107303
2030, 134180, 293068, 1546664, 1799008, 44735, 1367284, 105160
2031, 138688, 298054, 1490328, 1853480, 43059, 1363448, 103016
2032, 143197, 303040, 1433992, 1907960, 41383, 1359612, 100873
2033, 147705, 308025, 1377648, 1962432, 39707, 1355776, 98729

---=== Finish! ===---
```

Run

Land use in 2033

The results will be saved in a file named "PredictDemand.csv" in the folder 'Parameterfile'.

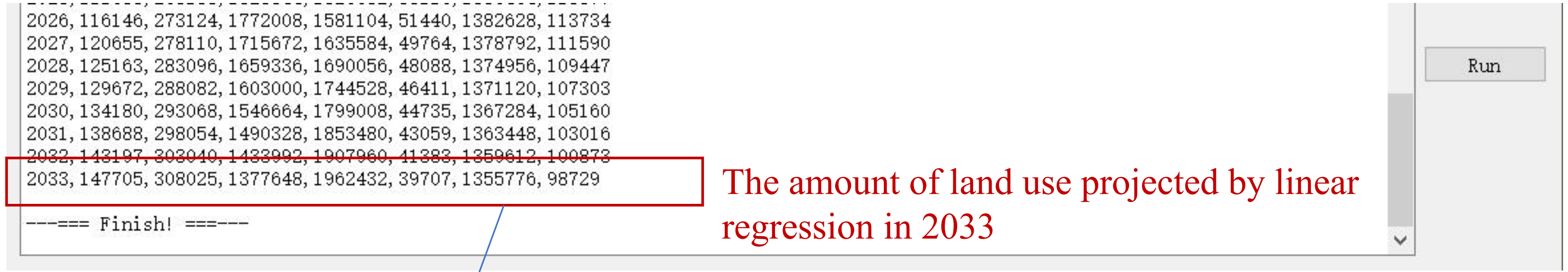
	A	B	C	D	E	F	G	H
1	Category	Type1	Type2	Type3	Type4	Type5	Type6	Type7
2	2003	12742	169912	3005564	362518	132649	1506974	135893
3	2005	13785	166569	3098723	401413	6300	1432942	206618
4	2010	66408	162629	2774437	612505	148823	1433723	127594
5	2013	44394	231113	2477356	929417	39413	1453586	150790
6	2014	62045	213293	2448040	927408	71554	1428659	139455
7	2015	66553	218279	2391704	981880	69878	1424823	137312
8	2016	71062	223265	2335368	1036352	68202	1420987	135168
9	2017	75570	228251	2279032	1090832	66526	1417151	133025
10	2018	80079	233237	2222696	1145304	64850	1413315	130882
11	2019	84587	238223	2166360	1199776	63173	1409479	128738
12	2020	89096	243209	2110024	1254256	61497	1405643	126595
13	2021	93604	248195	2053688	1308728	59821	1401808	124451
14	2022	98112	253180	1997352	1363208	58145	1397972	122308
15	2023	102621	258166	1941016	1417680	56469	1394136	120164
16	2024	107129	263152	1884680	1472152	54792	1390300	118021
17	2025	111638	268138	1828344	1526632	53116	1386464	115877
18	2026	116146	273124	1772008	1581104	51440	1382628	113734
19	2027	120655	278110	1715672	1635584	49764	1378792	111590
20	2028	125163	283096	1659336	1690056	48088	1374956	109447
21	2029	129672	288082	1603000	1744528	46411	1371120	107303
22	2030	134180	293068	1546664	1799008	44735	1367284	105160
23	2031	138688	298054	1490328	1853480	43059	1363448	103016
24	2032	143197	303040	1433992	1907960	41383	1359612	100873
25	2033	147705	308025	1377648	1962432	39707	1355776	98729
26								

02

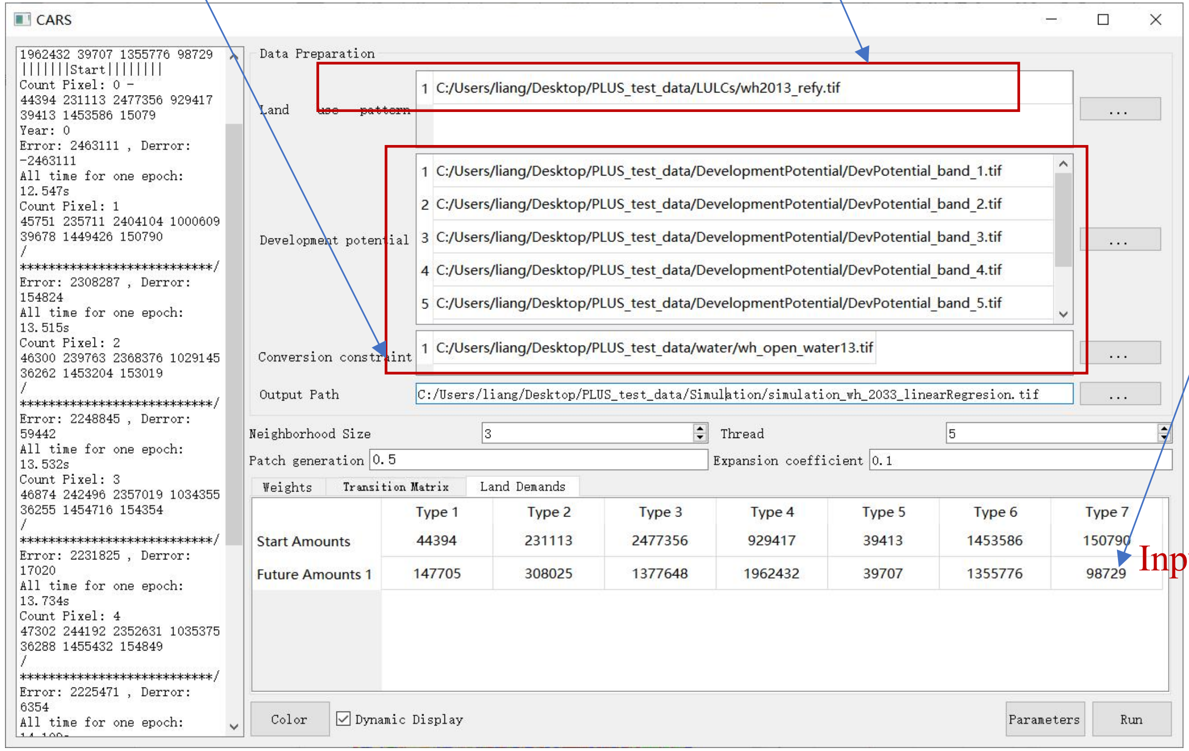
Land use scenario simulation-Linear regression scenario

Switch to initial land use data for 2013

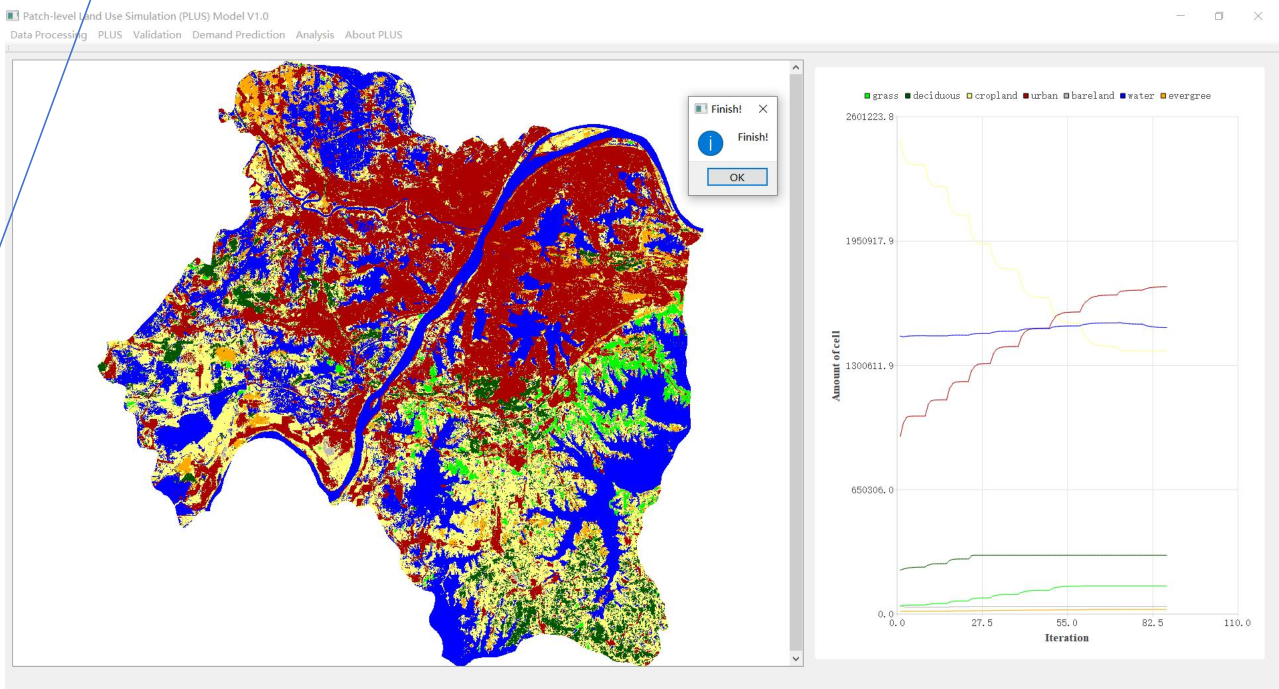
In the future simulation stage, the development probability can follow the result of the previous step, or it can be re-calculated. For example, recent land use data from 2010 to 2013 are used to recalculate the development probability



The amount of land use projected by linear regression in 2033



Input



02

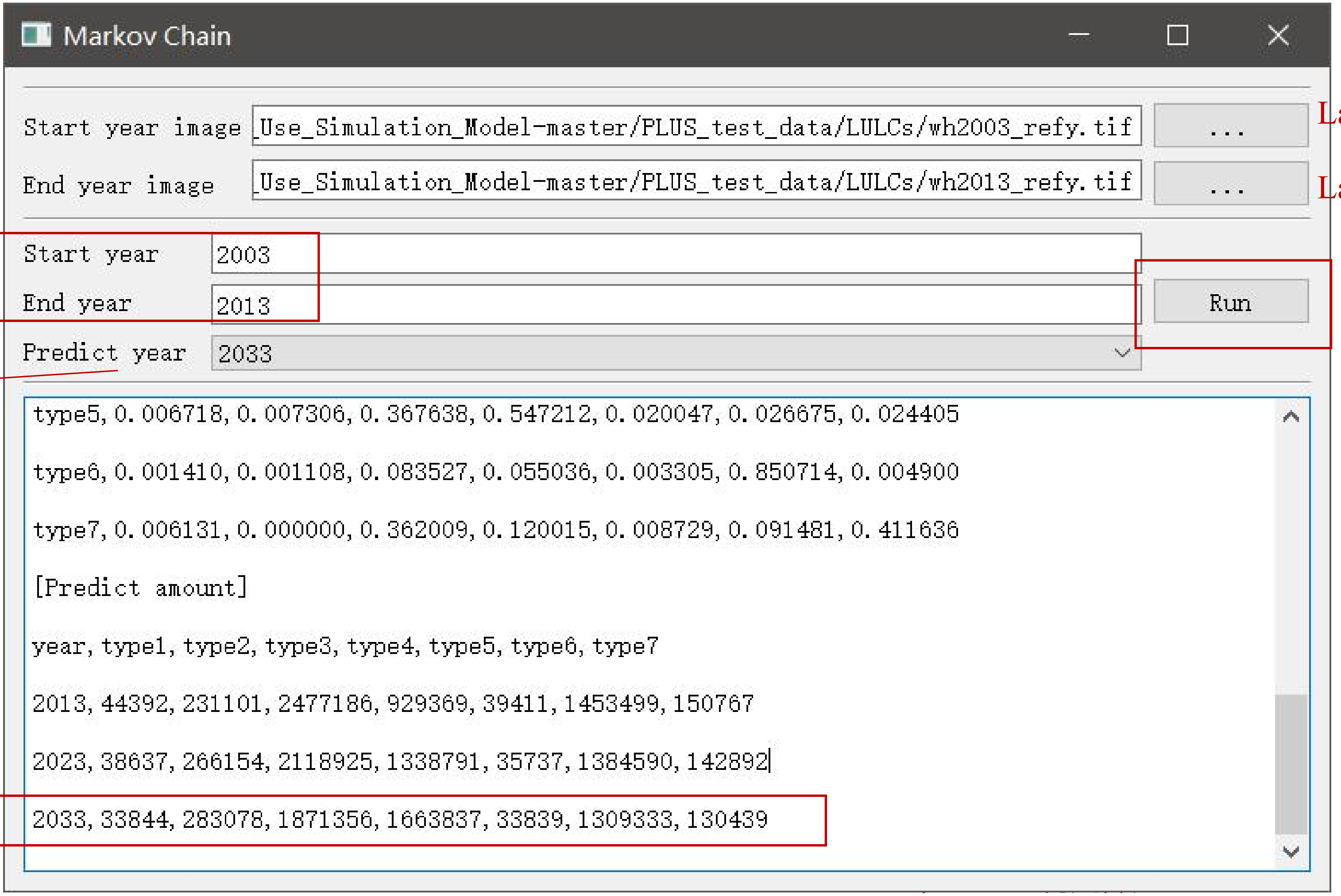
Land use demand forecasting-Markov Chain

Land use data for initial year

Land use data for termination year

Fill in the start and end years

Choose the year of the forecast （equal time interval）

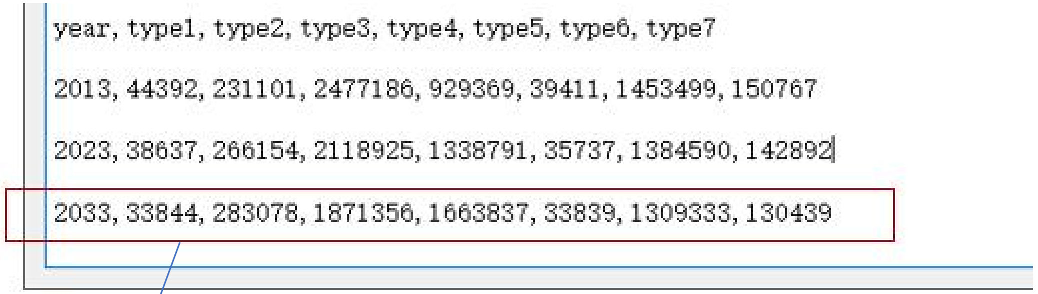


The results are saved in the “MarkovChain.csv” file

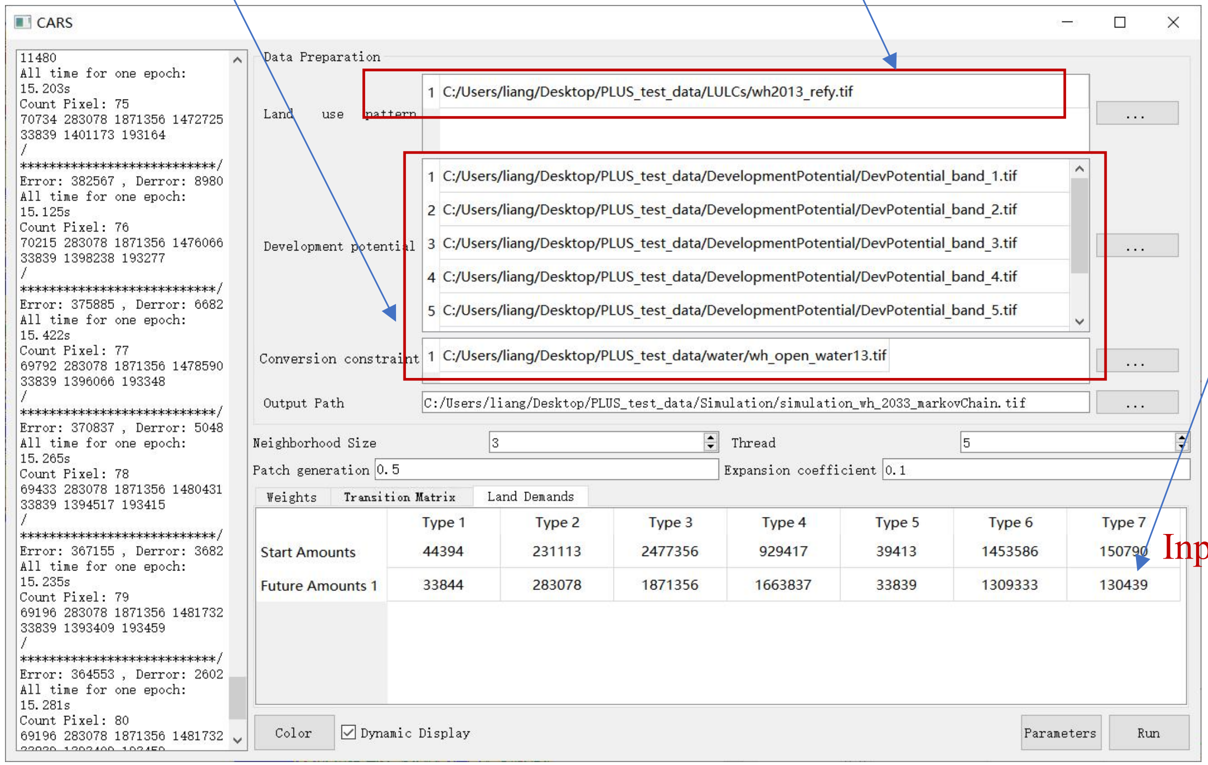
02 Land use scenario simulation-Markov Chain scenario

Switch to initial land use data for 2013

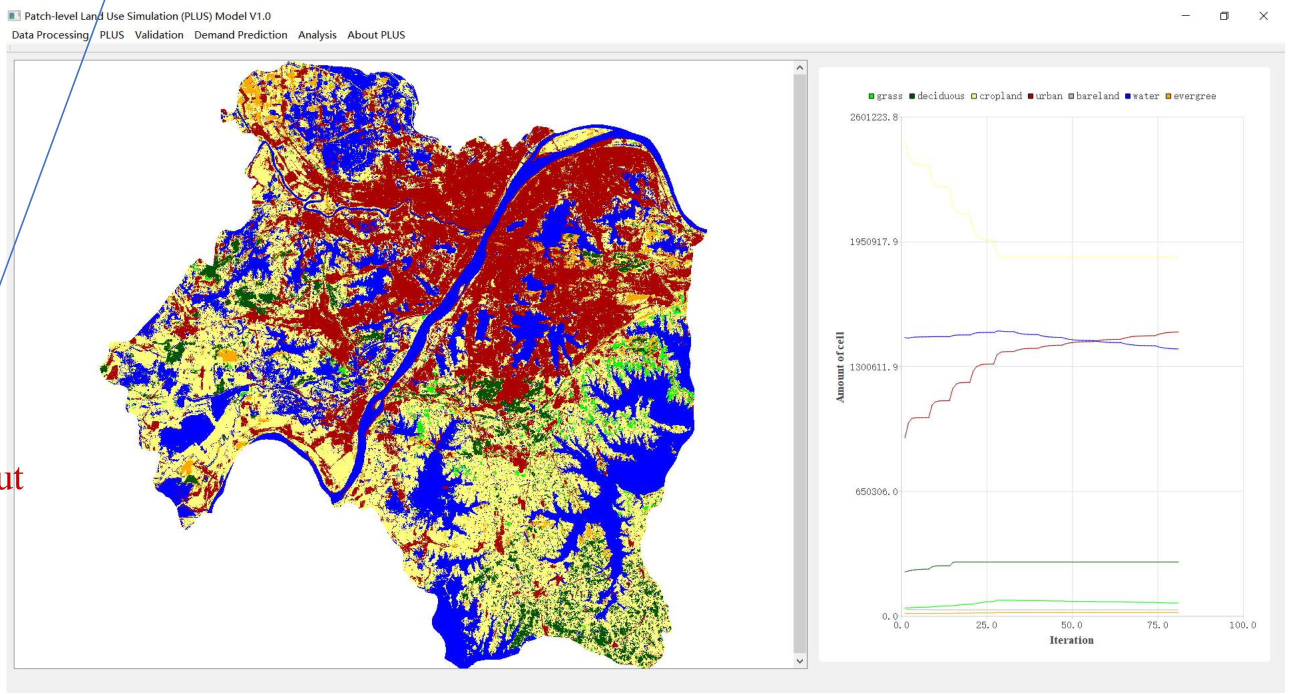
In the future simulation stage, the development probability can follow the result of the previous step, or it can be re-calculated. For example, recent land use data from 2010 to 2013 are used to recalculate the development probability


year, type1, type2, type3, type4, type5, type6, type7
2013, 44392, 231101, 2477186, 929369, 39411, 1453499, 150767
2023, 38637, 266154, 2118925, 1338791, 35737, 1384590, 142892
2033, 33844, 283078, 1871356, 1663837, 33839, 1309333, 130439

The amount of land use projected by Markov Chain in 2033



Input

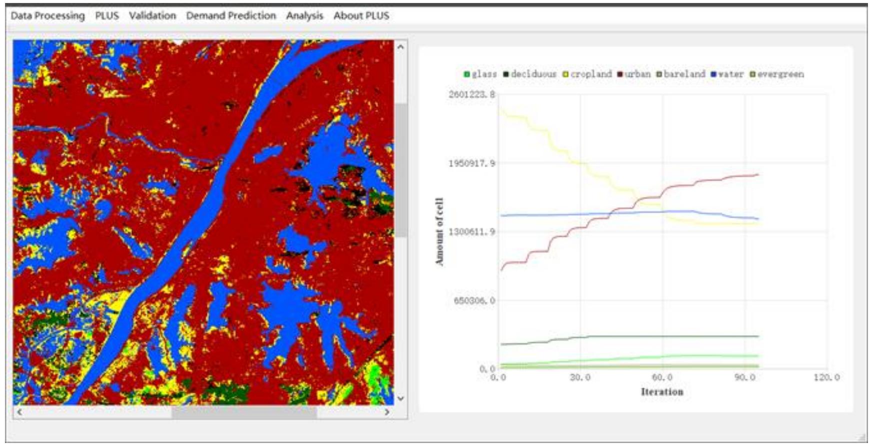


02

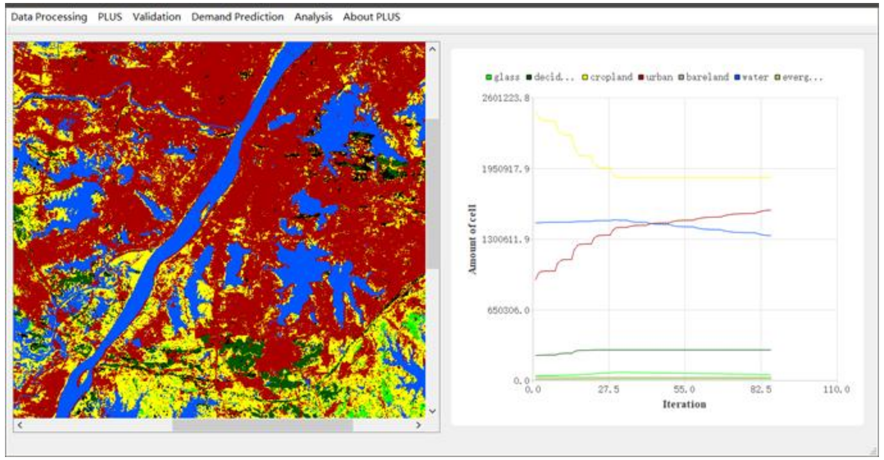
Analysis module-Analysis of scenario diversity

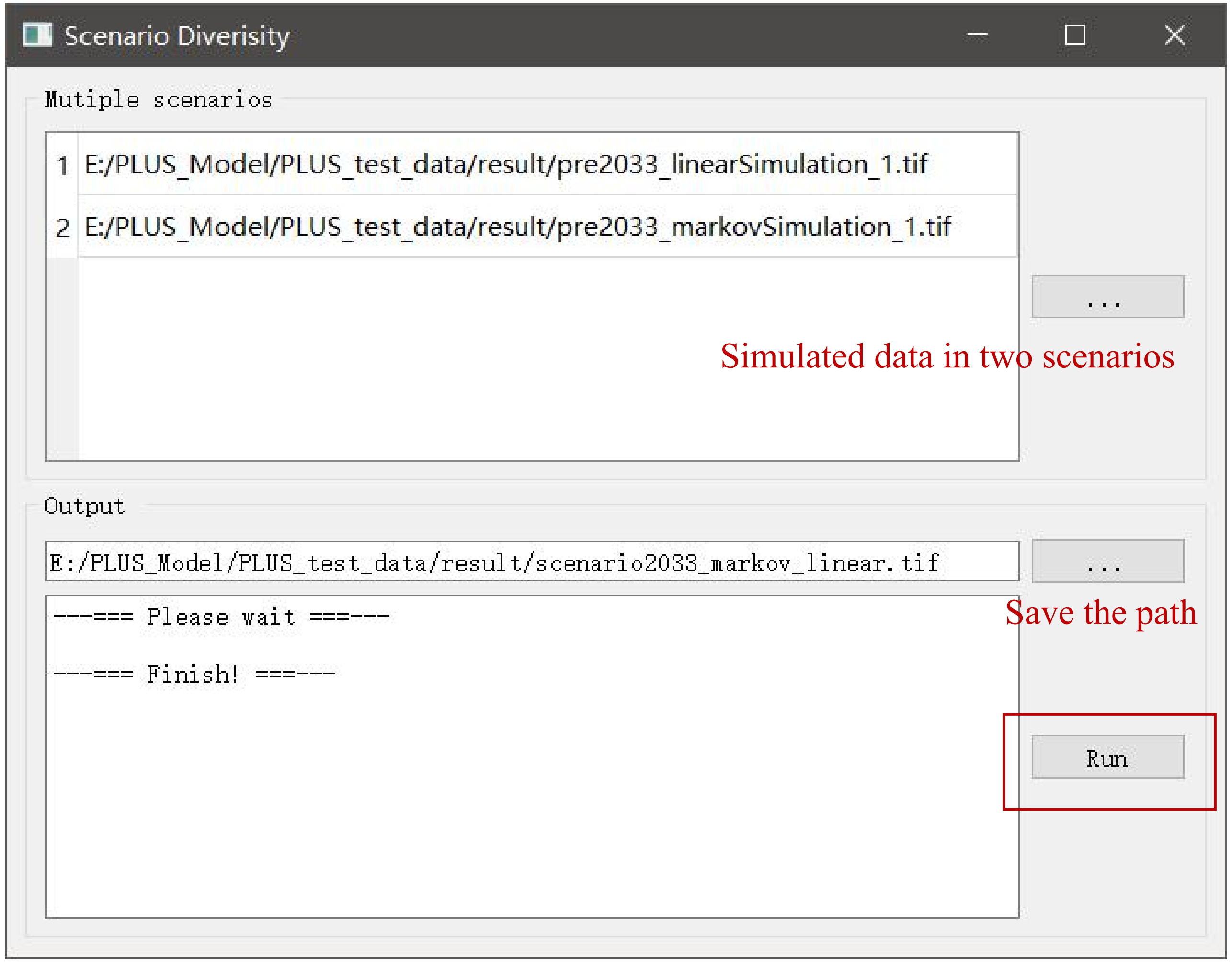
Simulation of land use scenarios under two scenarios

Predicted demand based on **linear regression**



Predicted demand based on **Markov**

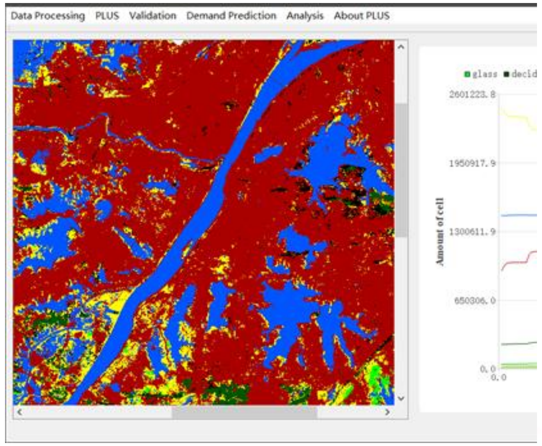




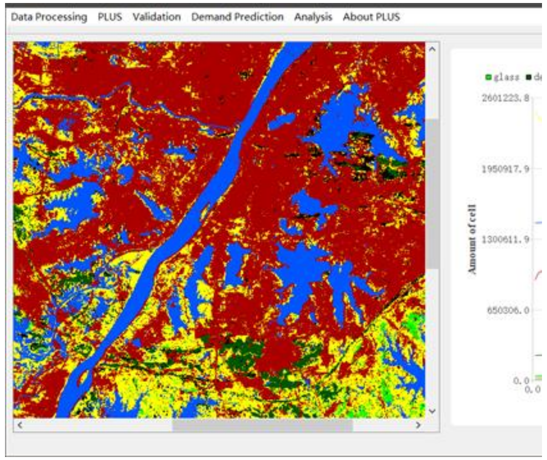
Analysis module-Analysis of scenario diversity

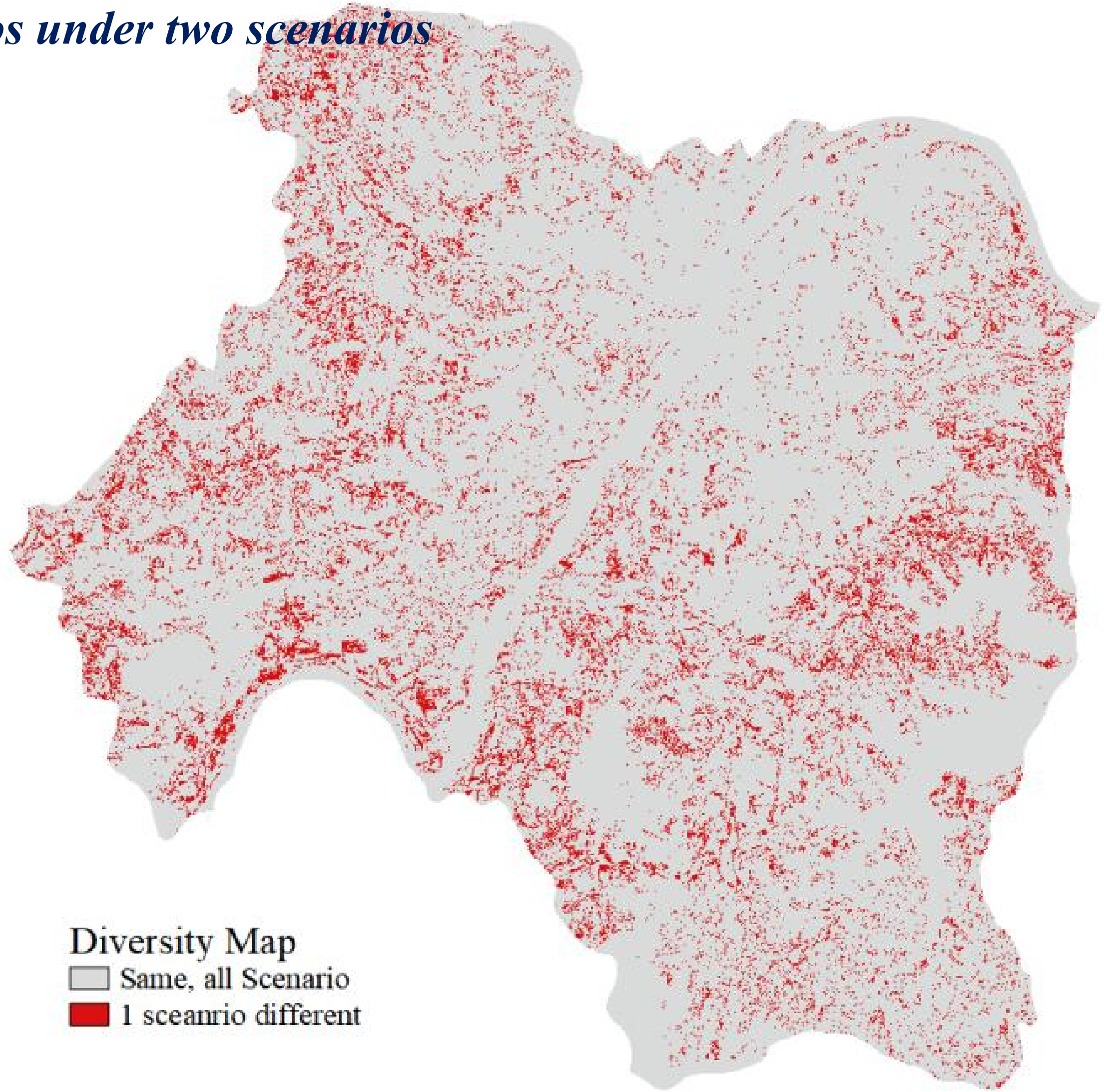
Simulation of land use scenarios under two scenarios

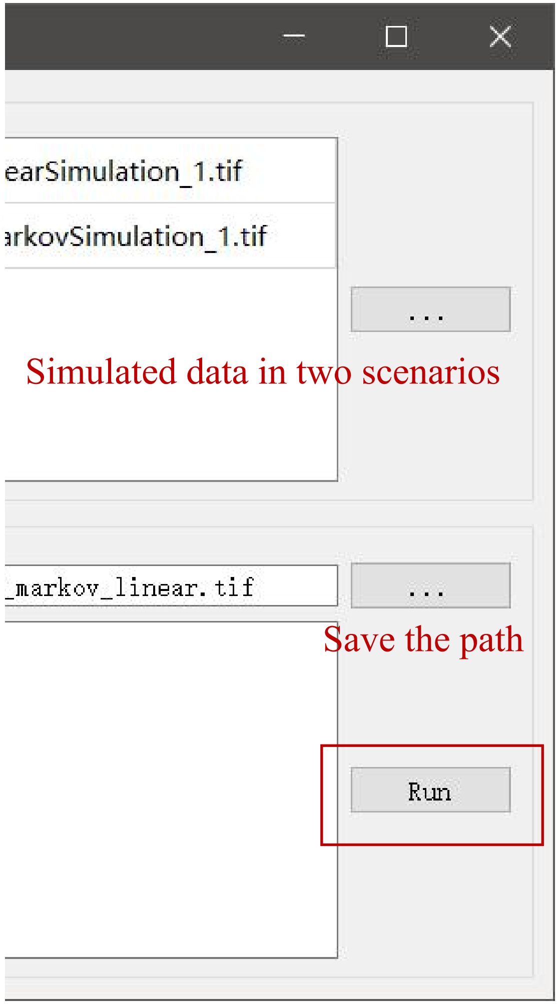
Predicted demand based on **linear regression**



Predicted demand based on **Markov**







The red part is the area where the simulation is inconsistent between the two scenarios, and the hot area of scene change is explored

03

Application of land use change models

03 Current application of PLUS

The paper was published in 2021 and **selected as ESI highly cited and hot papers** in 2022.

1 Understanding the drivers of sustainable land expansion using a patch-generating land use simulation (PLUS) model: A case study in Wuhan, China

Liang, X; Guan, QF; (...); Yao, Y

Jan 2021 | COMPUTERS ENVIRONMENT AND URBAN SYSTEMS 85

Cellular Automata (CA) are widely used to model the dynamics within complex land use and land cover (LULC) systems. Past CA model research has focused on improving the technical modeling procedures, and only a few studies have sought to improve our understanding of the nonlinear relationships that underlie LULC change. Many CA models lack the ability to simulate the detailed patch evolution of multiple land use types. This study introduces a patch-generating land use simulation (PLUS) model that integrates a land expansion analysis strategy and a CA model based on multi-type random patch seeds. These were used to understand the drivers of land expansion and to investigate the landscape dynamics in Wuhan, China. The proposed model achieved a higher simulation accuracy and more similar landscape pattern metrics to the true landscape than other CA models tested. The land expansion analysis strategy also uncovered some underlying transition rules, such as that grassland is most likely to be found where it is not strongly impacted by human activities, and that deciduous forest areas tend to grow adjacent to arterial roads. We also projected the structure of land use under different optimizing scenarios for 2035 by combining the proposed model with multi-objective programming. The results indicate that the proposed model can help policymakers to manage future land use dynamics and so to realize more sustainable land use patterns for future development. Software for PLUS has been made available at https//gitub.com/HPSCIL/patch-generating_Land_Use_Simulation_Model

Based on LULC to evaluate the ESV of 2010 at the grid scale in the ARB, two models, the CA-Markov and PLUS model coupled with multiple linear regression and a Markov chain model, were adopted. By comparison, the PLUS model has higher accuracy than the CA-Markov by IDRISI (Table 8).

Fig. 14 shows that in the oasis region of the ARB, the simulation results of the PLUS are closer to the actual results; however, the simulation results of the CA-Markov model show that construction land is significantly exaggerated, which is obviously different from reality. Overall, PLUS model can better perform a good spatial simulation based on land use data and driving factors.

The PLUS model has been applied not only in more than 100 cities in China, but also in other countries, such as Portugal, Malaysia, Rwanda, Cote d'Ivoire, The aral sea basin.

land MDPI

Article

Understanding Industrial Land Development on Rural-Urban Land Transformation of Jakarta Megacity's Outer Suburb

Adib Ahmad Kurnia [1,2], Ernan Rustiadi [3,4,*], Akhmad Fauzi [1], Andrea Emma Pravitasari [3,4], Izuru Saizen [2] and Jan Ženka [5]

Spatiotemporal changes, trade-offs, and synergistic relationships in ecosystem services provided by the Aral Sea Basin

Chao liang Chen[1,2,3,4], Xi Chen[1,2,4], Jing Qian[1,3,4,5], Zengyun Hu[1,2,3,4], Jun Liu[6], Xiuwei Xing[1,2,3,4], Duman Yimamaidi[2], Zhanar Zhakan[7], Jiayu Sun[3] and Shujie Wei[3]

1 State Key Laboratory of Desert and Oasis Ecology, Xinjiang Institute of Ecology and Geography, Chinese Academy of Sciences, Urumqi, China
2 Research Center for Ecology and Environment of Central Asia, CAS, Urumqi, China
3 Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Shenzhen, China
4 University of Chinese Academy of Sciences, Beijing, China
5 Key Laboratory of GIS & RS Application Xinjiang Uygur Autonomous Region, Urumqi, China
6 TripleSAI Technology, Shenzhen, China
7 Kazakh Agro Technical University named Saken Seifullin, Nur-sultan, Kazakstan

ABSTRACT

Intense human activities in the Aral Sea Basin have changed its natural distributi of land use. Although they provide certain economic benefits, these anthropoge influences have led to the rapid shrinkage of the Aral Sea, severely affecting t region's ecosystem. However, the spatiotemporal variability of the Aral Sea Basi Ecosystem Service Values (ESVs) is not well understood. In this study, we used 3

European Journal of Sustainable Development Research
2022, 6(4), em0193
e-ISSN: 2542-4742

VERITAS

Research Article

Modeling Future Impacts on Land Cover of Rapid Expansion of Hazelnut Orchards: A Case Study on Samsun, Turkey

Ersin Aytaç [1*]

RESEARCH ARTICLE

The role of littoral cliffs in the niche delimitation on a microendemic plant facing climate change

Miguel R. Ferreira[1*], Alice Maria Almeida[1,2], Celestino Quintela-Sabarís[3], Natália Roque[1,4], Paulo Fernandez[1,5], Maria Margarida Ribeiro[1,6,7]

The Fom values of PLUS, FLUS and CLUS-S models at 50m research scale are 0.184, 0.133 and 0.133, respectively. It is easy to know that the PLUS model is better than the FLUS and CLUS-S models in simulation accuracy. The Kappa coefficient values of the three models are all greater than 0.8, and K_{no} is greater than $K_{location}$, indicating that the quantitative prediction ability of the three models is better than the location prediction ability. Three Kappa coefficients of PLUS model are the highest, all greater than 0.95, indicating that the simulation results of the model are basically consistent with the actual situation.

03

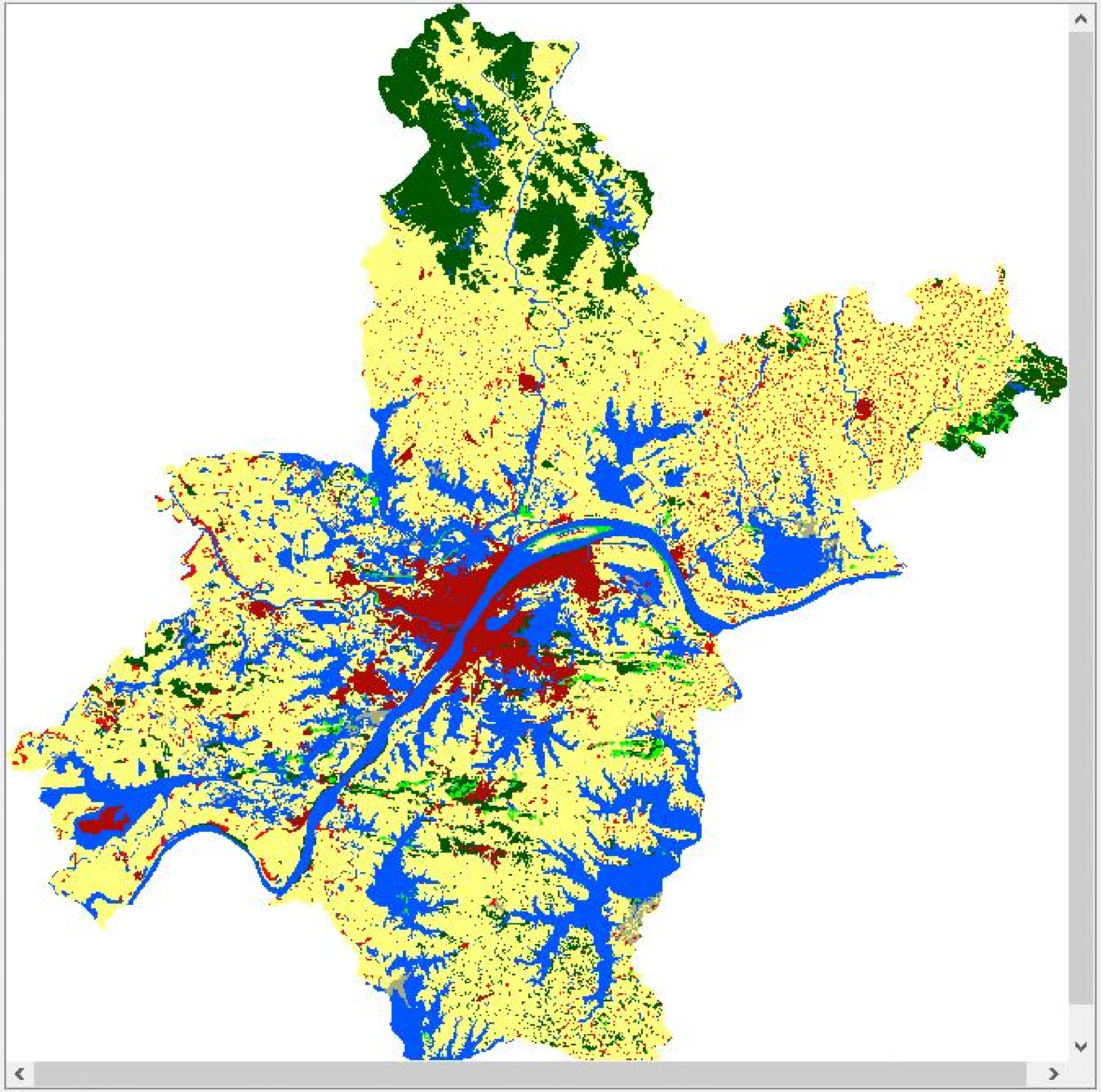
Social benefits of PLUS

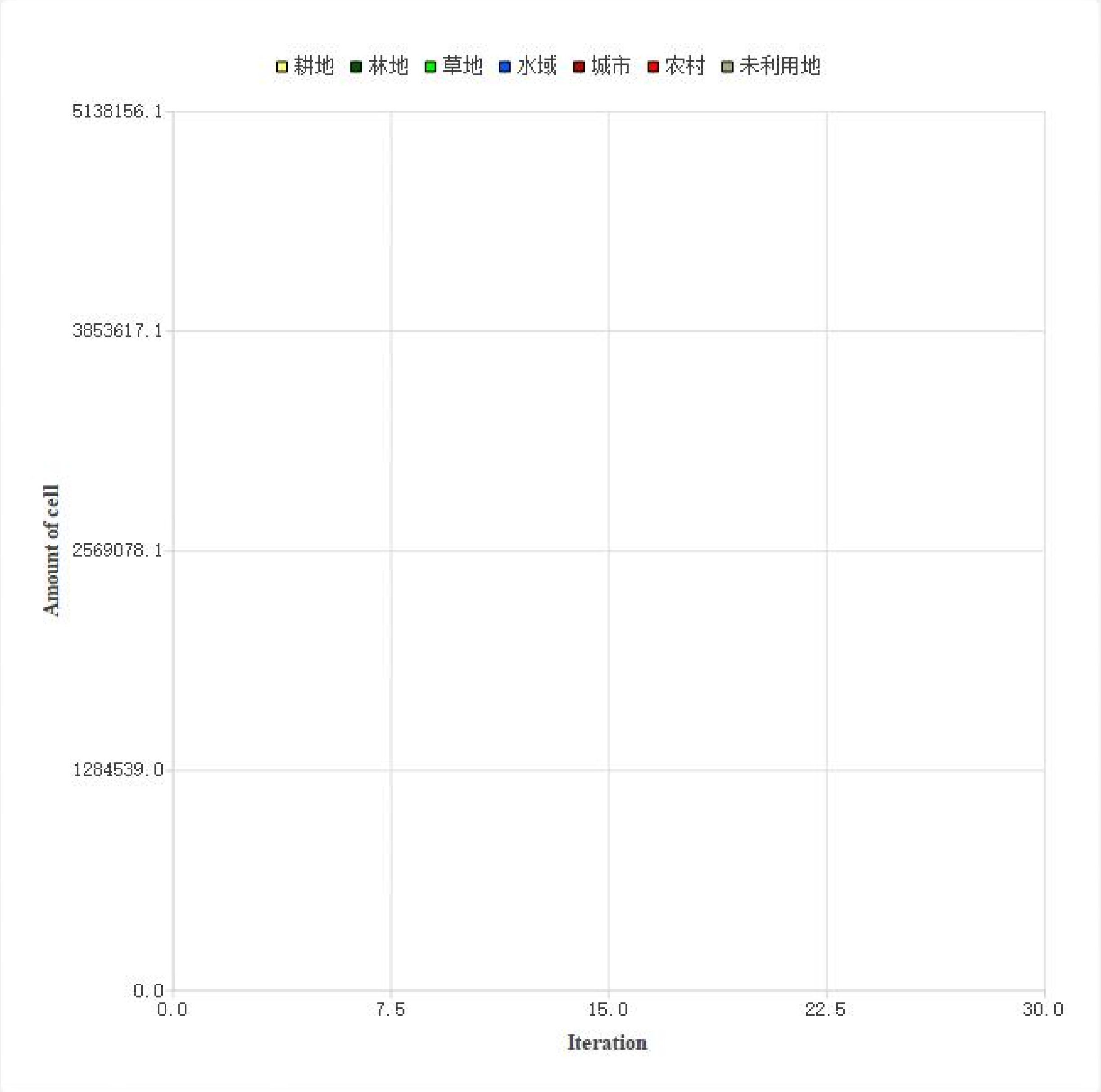
PLUS model has been adopted by many universities and research institutes in the United States, Australia, Japan, Portugal, South Korea, Turkey and China. , Its results are published in ***Journal of Hydrology***, ***Global Ecology and Conservation*** , ***Ecological Indicators*** , ***"China Environmental Science "*** and other well-known journals at home and abroad (a total of 64 articles, including 45 international journal papers and 19 domestic journal papers)

Some articles that used the PLUS model
Multi-scenario simulation of ecosystem service value for optimization of land use in the Sichuan-Yunnan ecological barrier, China
The ecosystem service values simulation and driving force analysis based on land use/land cover: A case study in inland rivers in arid areas of the Aksu River Basin, China
Impacts of land use/land cover changes on ecosystem services in ecologically fragile regions
Cost-benefit analysis of ecological restoration based on land use scenario simulation and ecosystem service on the Qinghai-Tibet Plateau
[1]Impact of urban expansion on carbon storage under multi- scenario simulations in Wuhan, China [2]Understanding the drivers of sustainable land expansion using a patch-generating land use simulation (PLUS) model: A case study in Wuhan, China
Evaluating potential impacts of land use changes on water supply–demand under multiple development scenarios in dryland region
Coupling Coordination Analysis and Prediction of Landscape Ecological Risks and Ecosystem Services in the Min River Basin
Assessment and Prediction of Carbon Storage Based on Land Use/Land Cover Dynamics in the Tropics: A Case Study of Hainan Island, China
Effects of urban growth boundaries on urban spatial structural and ecological functional optimization in the Jining Metropolitan Area, China
Dynamic simulation of land use change and assessment of carbon storage based on climate change scenarios at the city level: A case study of Bortala, China
Trade-Offs and Synergies of Multiple Ecosystem Services for Different Land Use Scenarios in the Yili River Valley, China
Identification and risk prediction of potentially contaminated sites in the Yangtze River Delta
Construction of GI Network Based on MSPA and PLUS Model in the Main Urban Area of Zhengzhou: A Case Study
Forecasting Urban Land Use Change Based on Cellular Automata and the PLUS Model
GAN-Based LUCC Prediction via the Combination of Prior City Planning Information and Land-Use Probability
Forecasting Urban Land Use Change Based on Cellular Automata and the PLUS Model
Urban Spatial Development Based on Multisource Data Analysis: A Case Study of Xianyang City’s Integration into Xi’an International Metropolis
[1]Spatial and Temporal Variation, Simulation and Prediction of Land Use in Ecological Conservation Area of Western Beijing [2]Impacts of Land Use Changes on Net Primary Productivity in Urban Agglomerations under Multi-Scenarios Simulation
How to Account for Changes in Carbon Storage from Coal Mining and Reclamation in Eastern China? Taking Yanzhou Coalfield as an Example to Simulate and Estimate
The role of littoral cliffs in the niche delimitation on a microendemic plant facing climate change
Understanding Industrial Land Development on Rural-Urban Land T ransformation of Jakarta Megacity’s Outer Suburb
How Will Rwandan Land Use/Land Cover Change under High Population Pressure and Changing Climate?
Spatiotemporal changes, trade-offs, and synergistic relationships in ecosystem services provided by the Aral Sea Basin

Patch-level Land Use Simulation (PLUS) Model V1.0

Data Processing PLUS Validation Demand Prediction Analysis About PLUS





Team：High-performance Spatial Computational Intelligence Lab @ CUG

Contact：Dr. Xun Liang（liangxun@cug.edu.cn）

Supervised by：Prof. Qingfeng Guan guanqf@cug.edu.cn）

China University of Geosciences, Wuhan

School of Geography and Information Engineering & National Engineering Research Center of GIS

High-performance Spatial Computational Intelligence Lab （HPSCIL）

Thank you！

Simulating the driving effects of planning policies or future variables on LUCC with the PLUS model

Team: High-performance Spatial Computational Intelligence Lab @ CUG

Contact: Dr. Xun Liang (liangxun@cug.edu.cn)

Supervised by: Prof. Qingfeng Guan (guanqf@cug.edu.cn)

China University of Geosciences

School of Geography and Information Engineering

National Engineering Research Center of GIS

CONTENT

1. Methodology
2. Consider the driving effects of planning transport infrastructure on LUCC
3. Consider the driving effects of development zone on LUCC

Note that these functions are only integrated into PLUS v1.3.5 and later versions. Please learn tutorial A before reading this tutorial. The planning data in this tutorial is the dummy data for the model test. Please don' t regard them as the real planning data.

01

Methodology

01

Method and thought

We used an updated mechanism of planning transport infrastructure based on random forest (RF) and a random seeding mechanism based on planning development zone, which can consider the driving effects of planning policies or future variables on LUCC into the simulation.

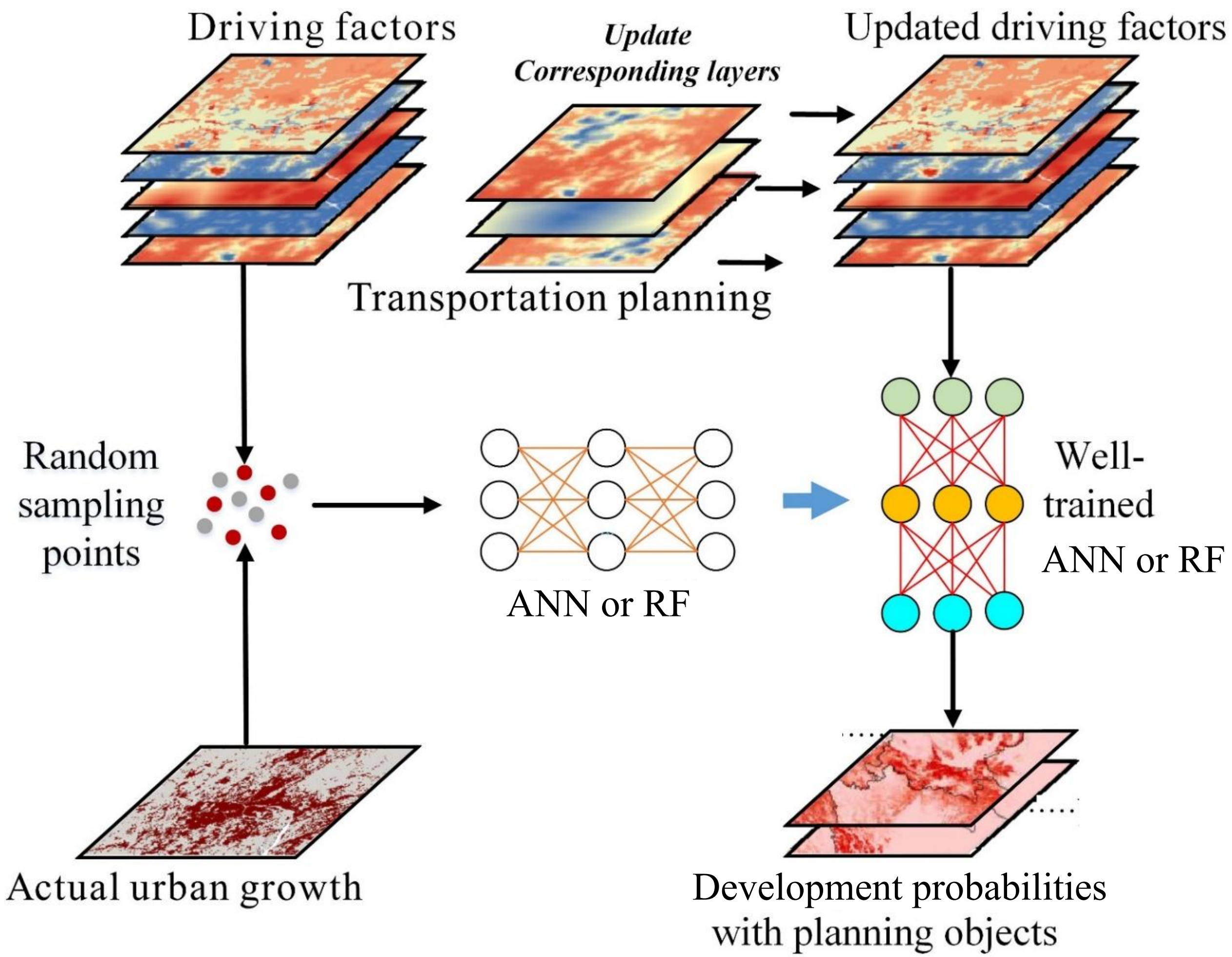
This study only considers planning policies in space, not macro-scale policies, including 1) planning traffic lines or sites and 2) planning development zone. Moreover, predicted variables exported by other models can also be imported to the PLUS model in the same way, for example, the future population, GDP, temperature, precipitation, etc.

References: Liang, X., Liu, X., Li, D., Zhao, H., Chen, G., 2018, Urban growth simulation by incorporating planning policies into a CA-based future land-use simulation model, International Journal of Geographical Information Science, 32(11): 2294-2316. . (ESI highly cited paper)

Liang X., Guan Q.*, Clarke KC, Liu S., Wang B., Yao Y., 2021. Understanding the drivers of sustainable land expansion using a patch-generating simulation (PLUS) model: A case study in Wuhan, China, Computers, Environment and Urban Systems, 85:101569

01

The updated mechanism



Flowchart

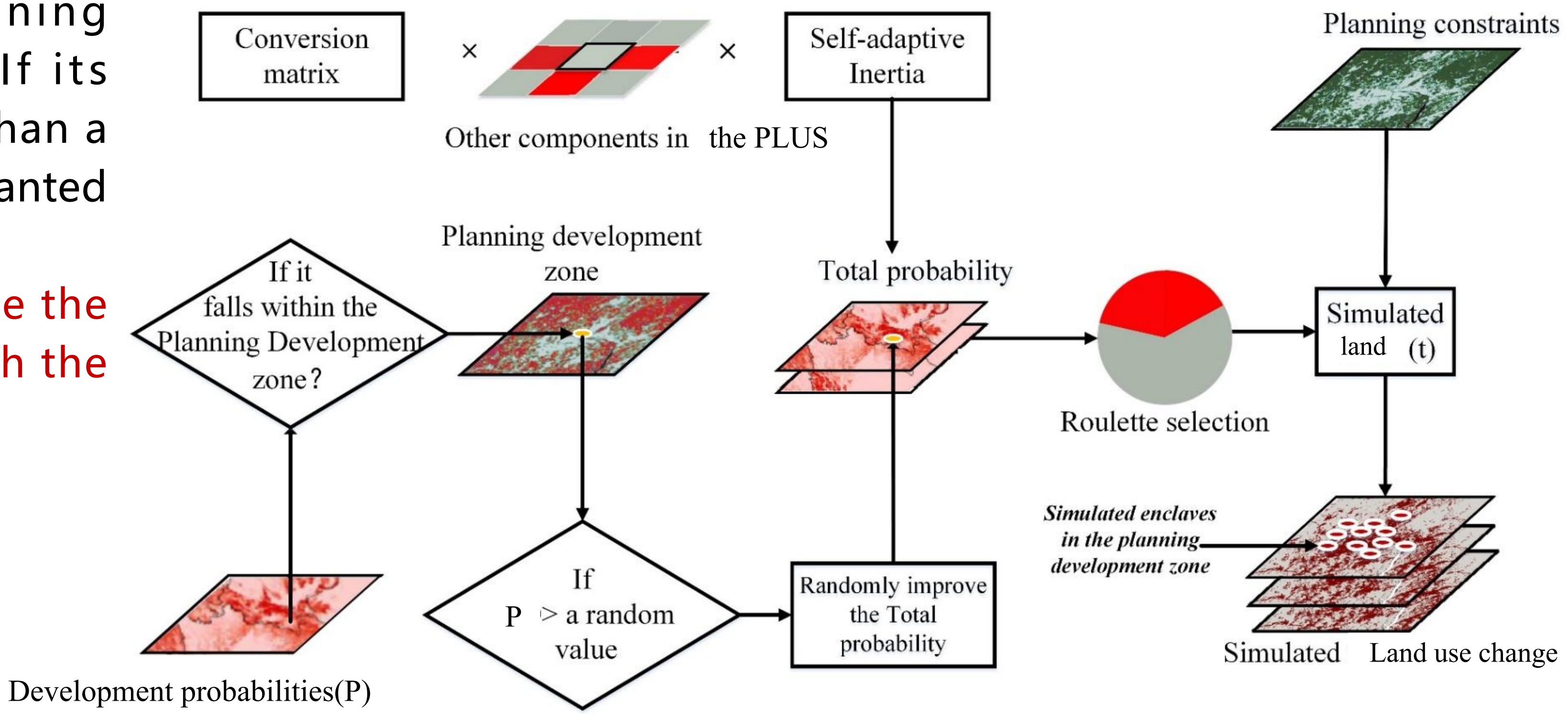
- First, sampled land-use map data and historical driving force data are employed to train the RF.
- The driving factors that will be updated are specified in this step (only driving factors with both historical and planning schemes (or future variables) can be updated).
- In the RF prediction process, the historical driving forces in the specified layers are replaced with data that include both historical and future driving forces and output the development probabilities.

01 The random seeding mechanism

Flow chart

- A cell that is located in the planning development zones is selected. If its development probability is greater than a random value within [0, 1], a seed is planted in the cell.
- A planted seed will randomly increase the total probability of an urban area with the following rule:

$$TP_k = \begin{cases} (r + TP_k) \times \mathrm{w} & if\ r + TP_k \le 1 \\ 1 \times \mathrm{w} & if\ r + TP_k > 1 \end{cases}$$



TP_k - denotes the total probability of specific land k

r - a random value within [0, 1]

w – the weight of the strength of planning development zones

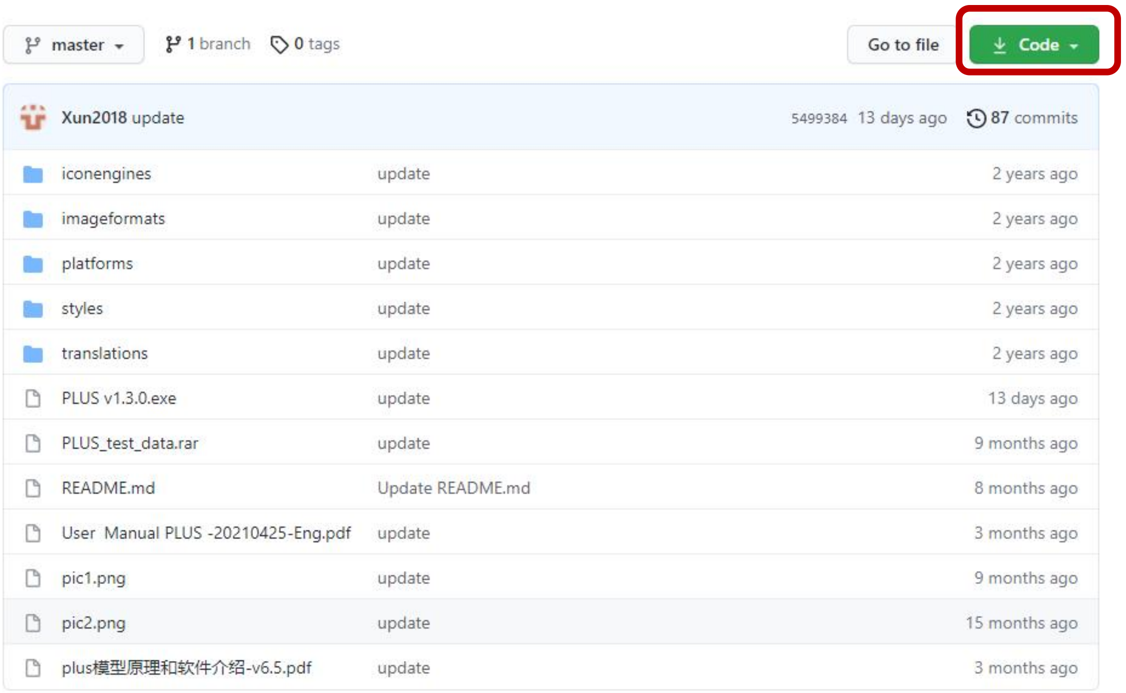
02

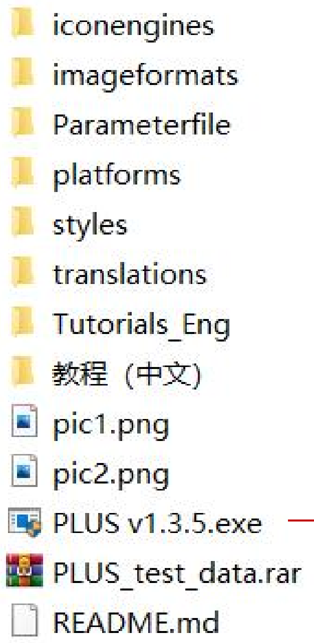
Consider the driving effects of planning transport infrastructure on LUCC

02 Download PLUS v1.3.5

Link： https://github.com/HPSCIL/Patch-generating_Land_Use_Simulation_Model

Click here to download

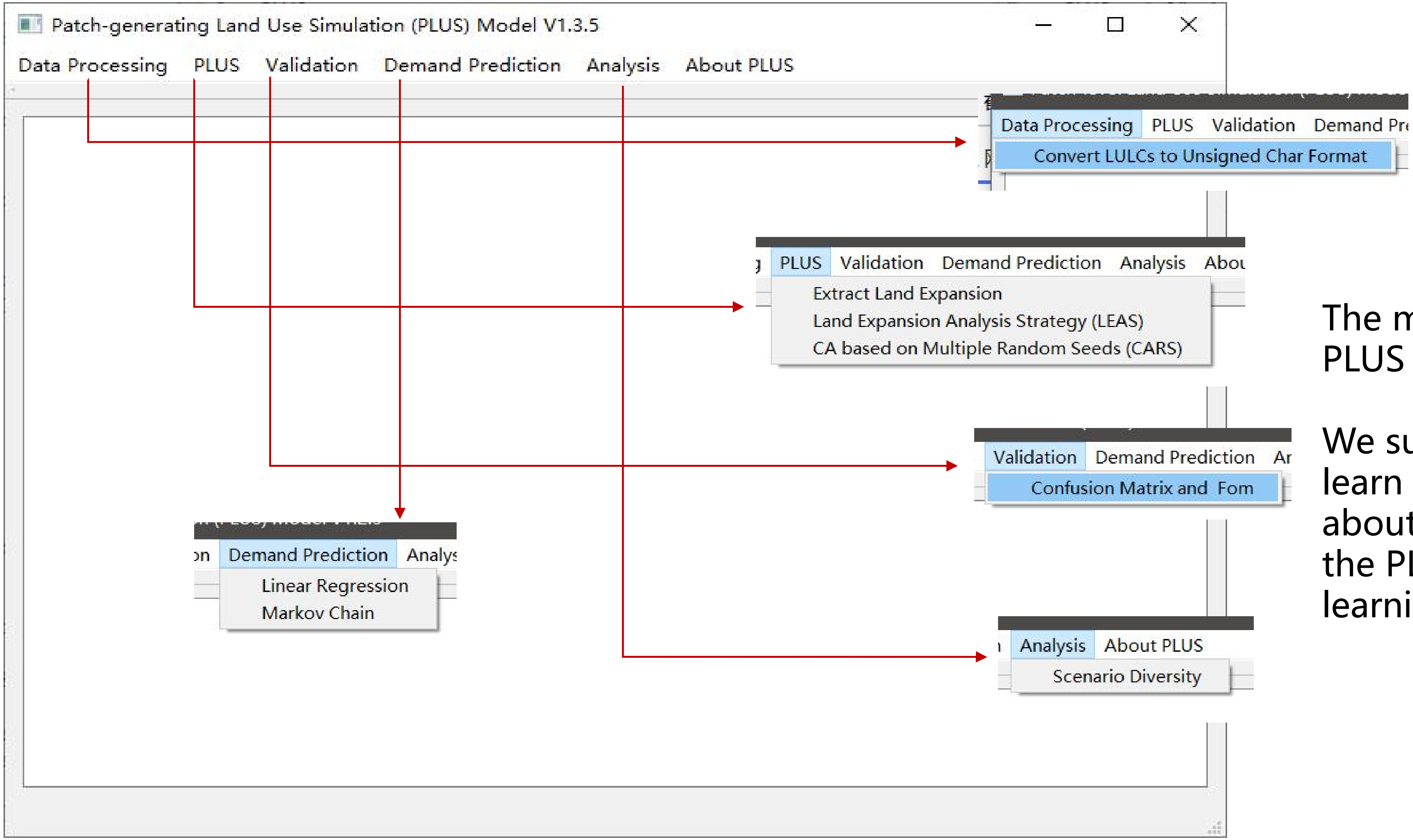




Click here to start

PLUS can run in the environment of Windows Vista/7/8/X64 without install process and the support of other software.

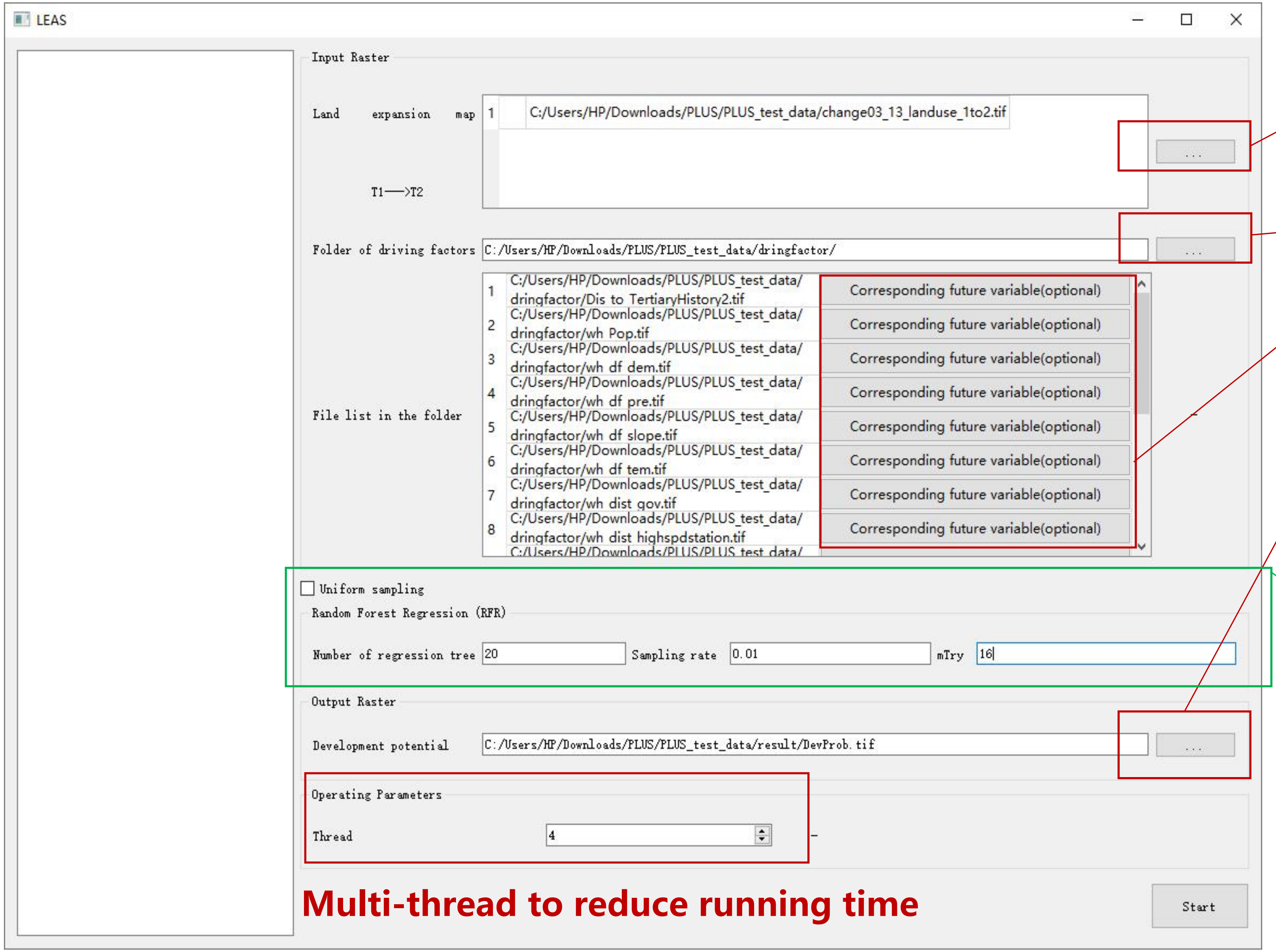
02 Main interface



The main interface of the PLUS software.

We suggest the users learn tutorial A to know about all the modules of the PLUS model before learning this tutorial.

02 Add in the planning traffic data



Multi-thread to reduce running time

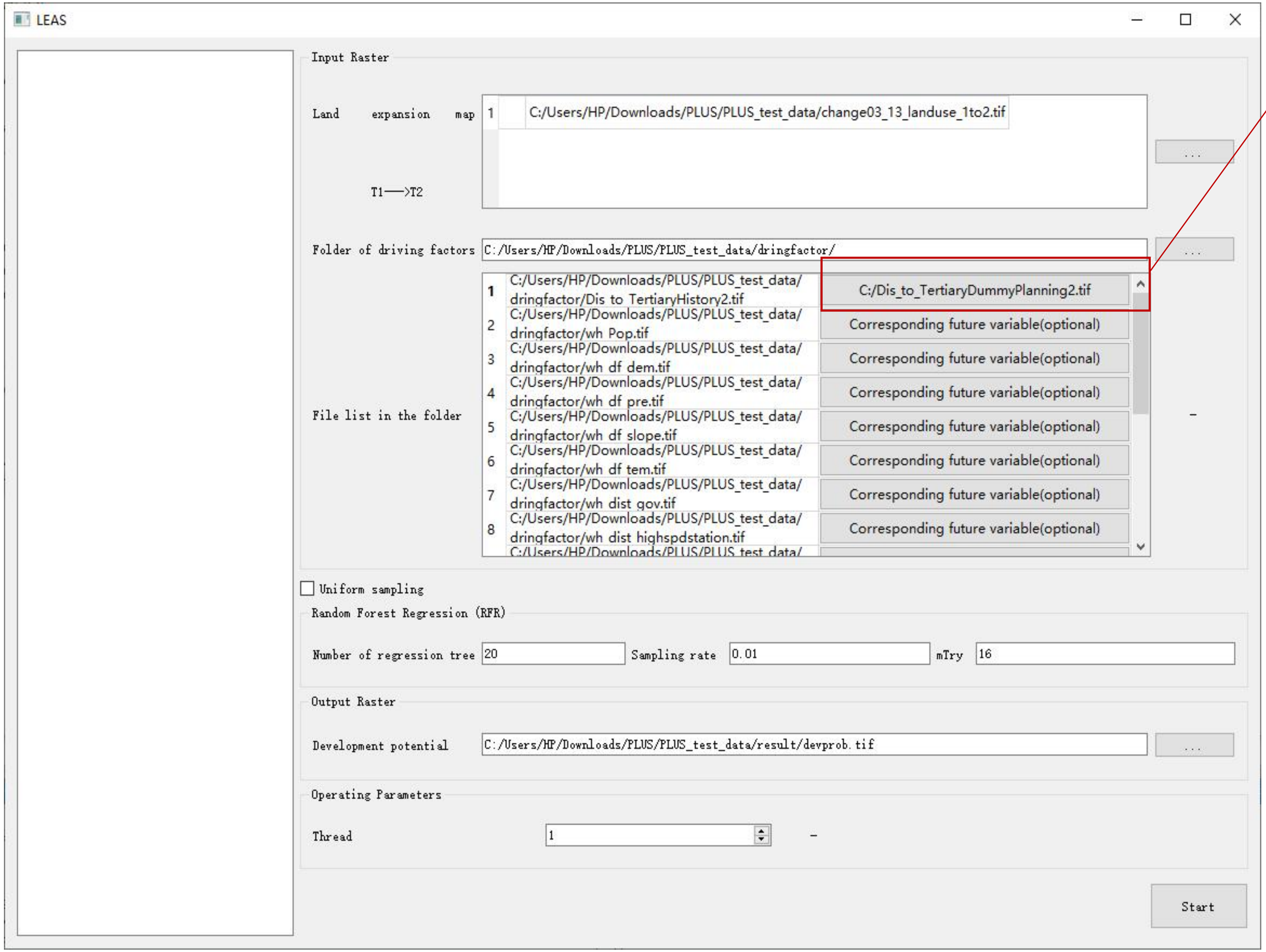
Input&Output

- ✓ Land expansion map (see tutorial A to know how to obtain this file)
- ✓ Driving factors of LUCC
- ✓ **Import the planning policies or future variables (optional)**
- ✓ Output path

Other parameters

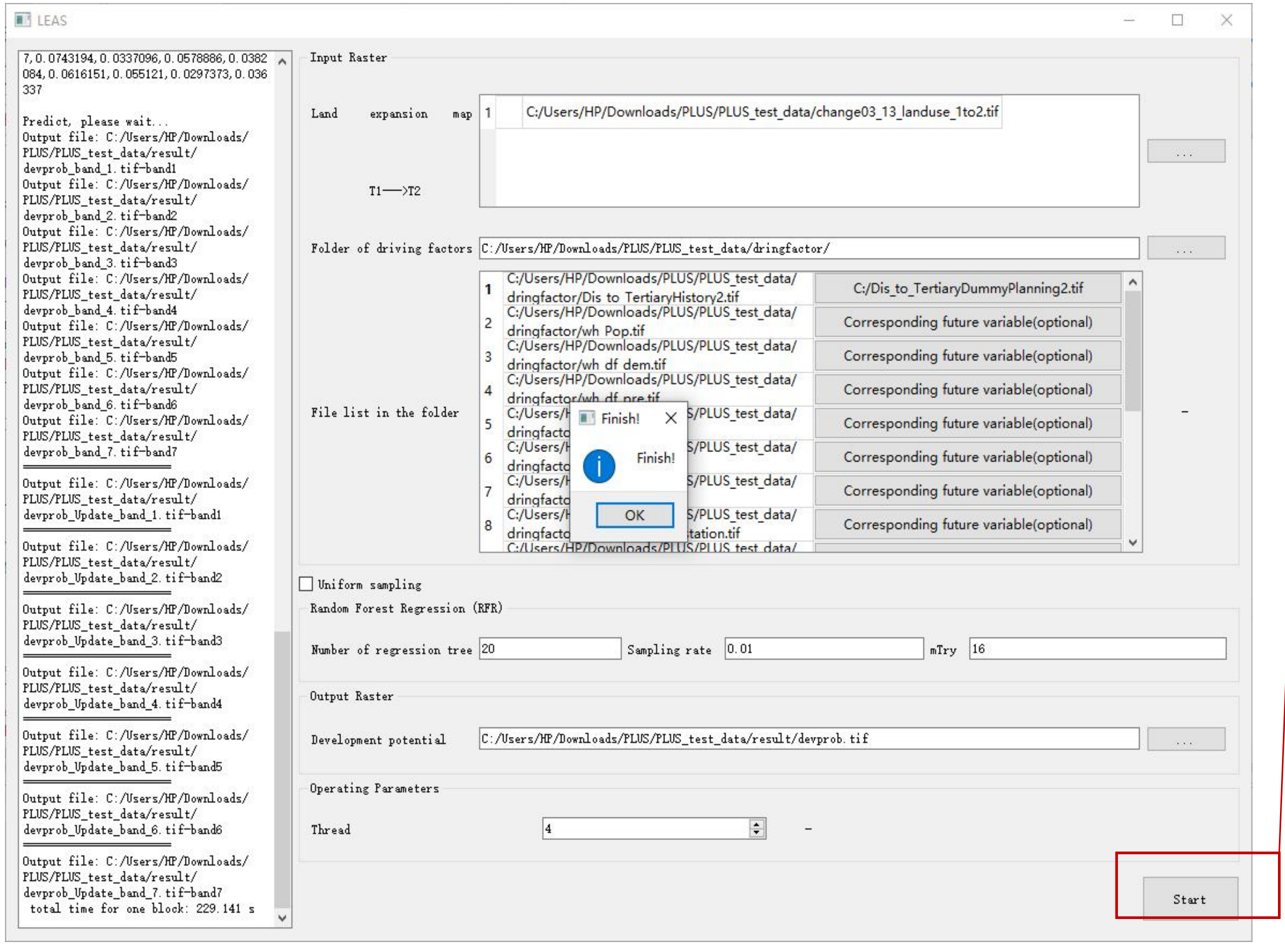
- ✓ *See tutorial A*

02 Add in the planning traffic data

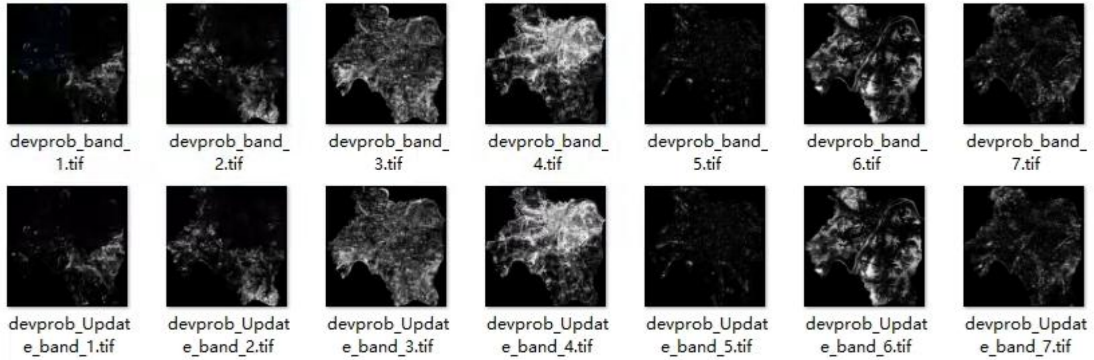


- ✓ Click the button that corresponding to the proximity to the historical tertiary road “Dis_toTertiaryHistory2.tif” to import the planning tertiary road data “Dis_to_TertiaryDummyPlanning.tif” (All the test data can be found in the zip file PLUS_test_data.rar)

02 Output results



✓ Click the "Start" button and wait for the output files.



✓ Output two groups of development probabilities:

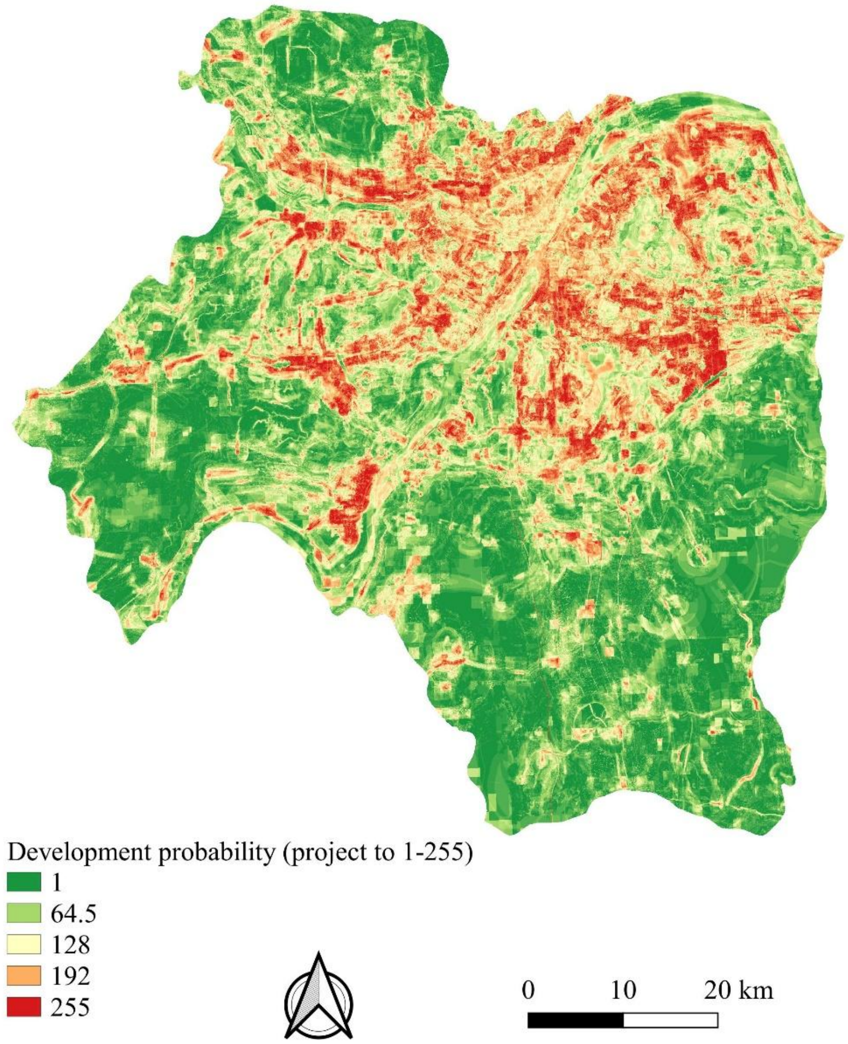
devprob_Update_band1-7.tif: development probabilities under the influences of planning policies, which is the input of the next step.

devprob_band1-7.tif: development probabilities without the influences of planning policies, which is used to compare with the one with the influences of planning policies.

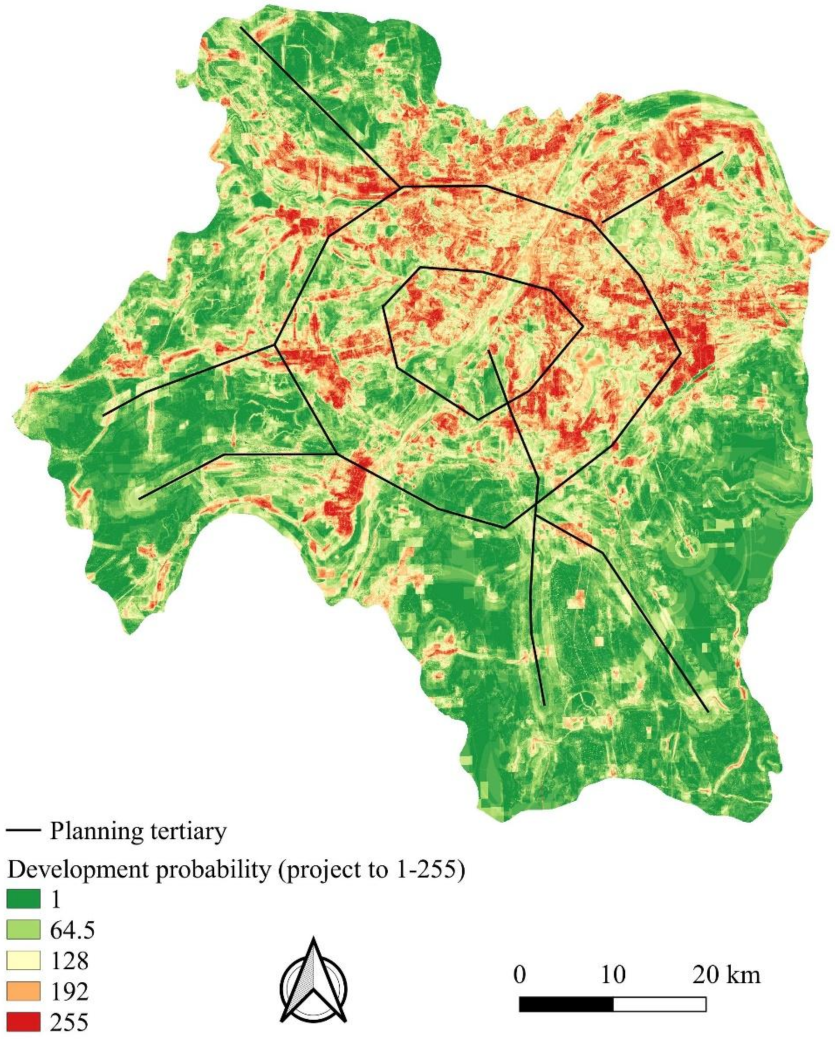
02

Output results

Urban development potential without the influences of planning policies



Urban development potential under the influences of planning policies

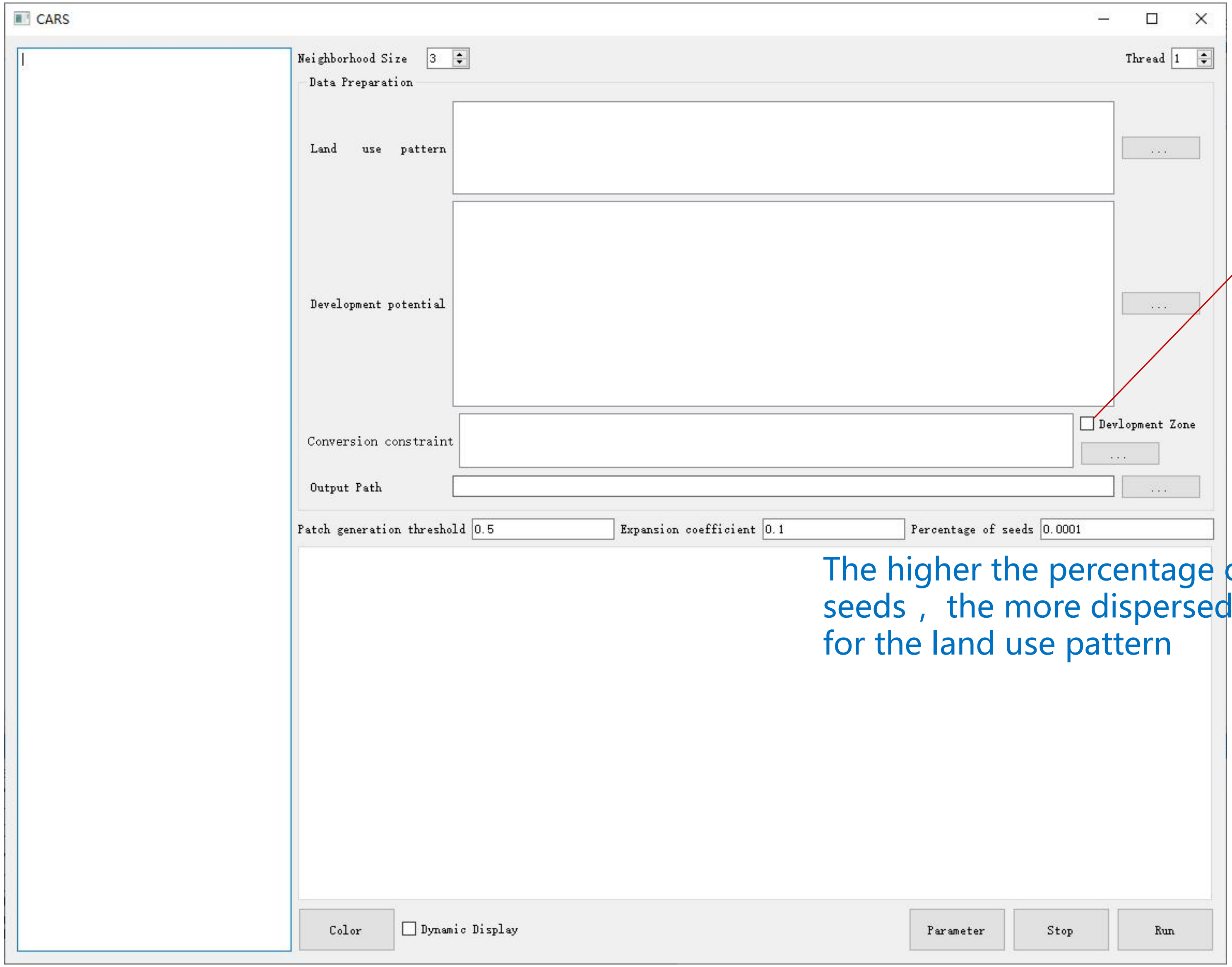


03

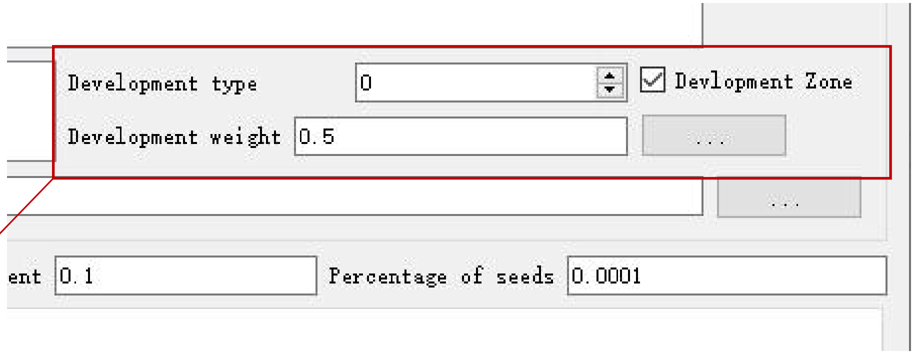
Consider the driving effects of development zones on LUCC

03

Add in the planning development zones



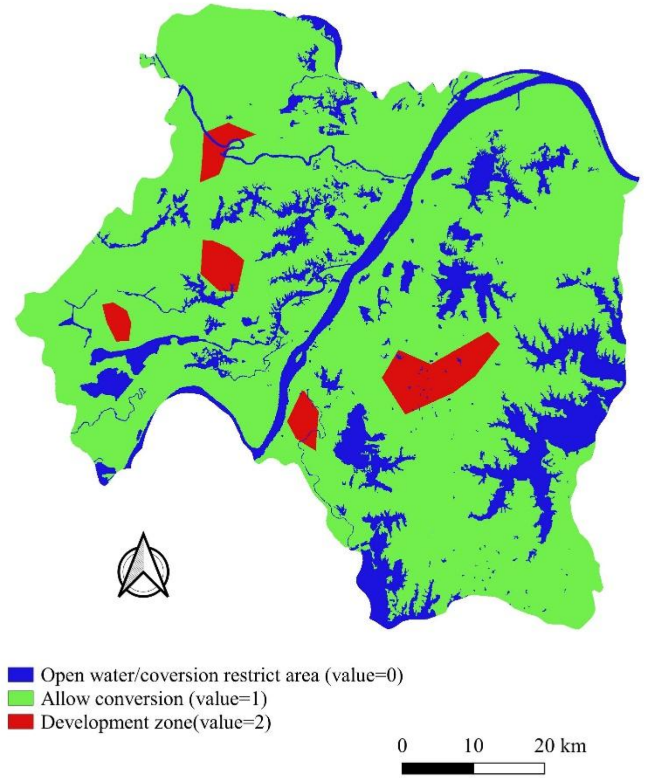
The higher the percentage of seeds， the more dispersed for the land use pattern



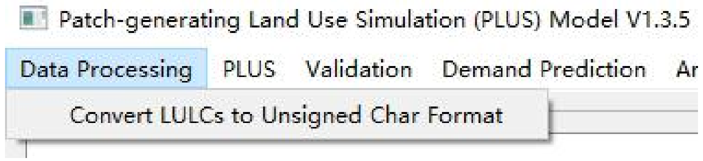
- ✓ Click the “Development Zone” checkbox to active the “Development type” and “Development weight” parameters
- ✓ “Development Zone” is used to defining the land use type that is influenced by the planning policies; “Development weight” ranges from 0-1, which is used to defining the strength of the planning policies.

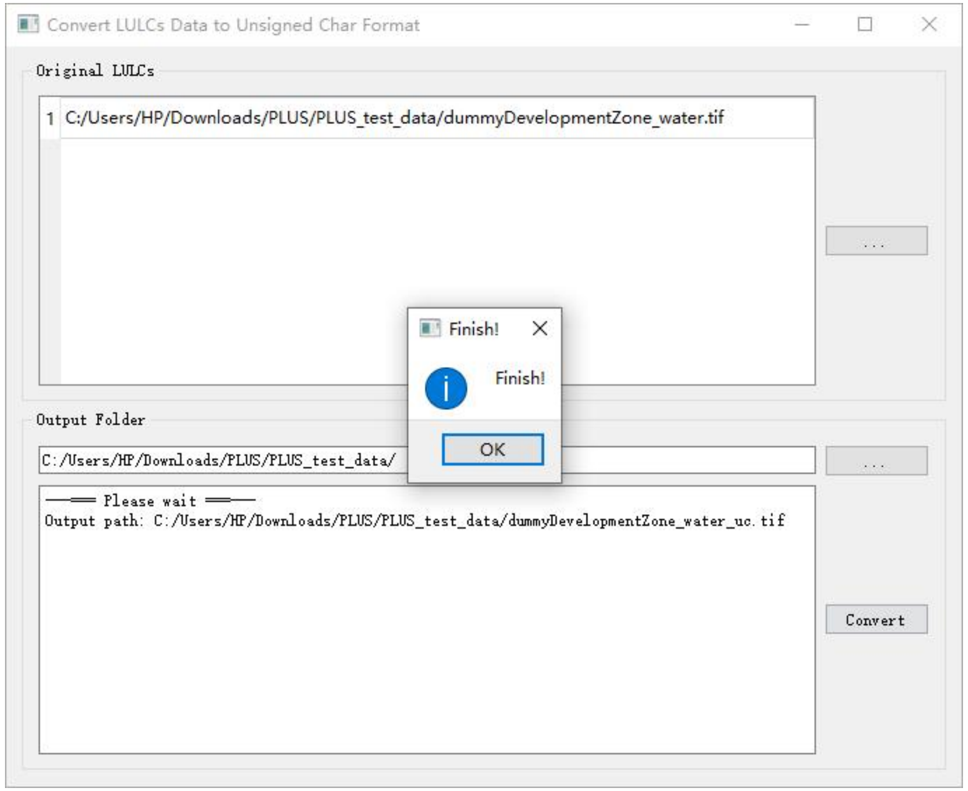
03 Make the planning development zones

Make the “conversion constraints area and planning development zones” , the value of conversion constraints area is 0 , and the value of planning development zone is 2. Value 1 means transitions are allowed.

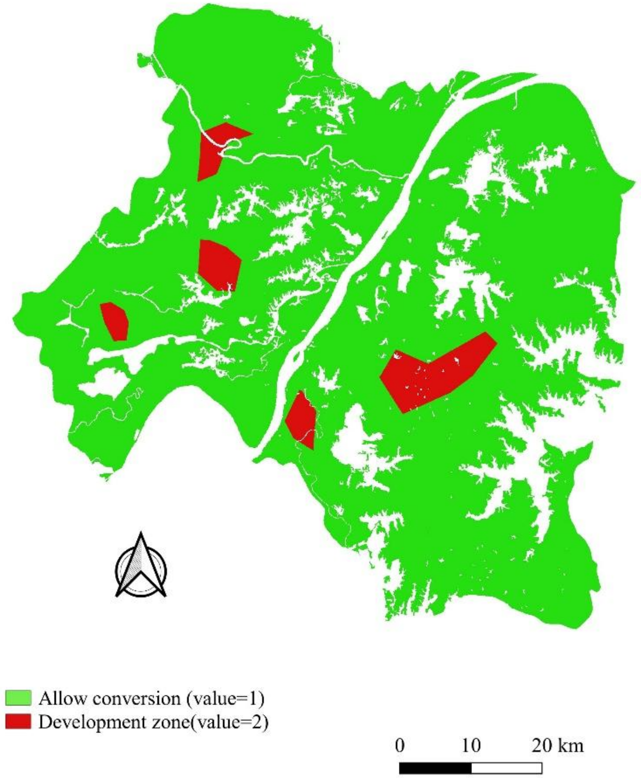


Import this file to the conversion tools of the PLUS model to convert it to ‘unsigned char’ format.

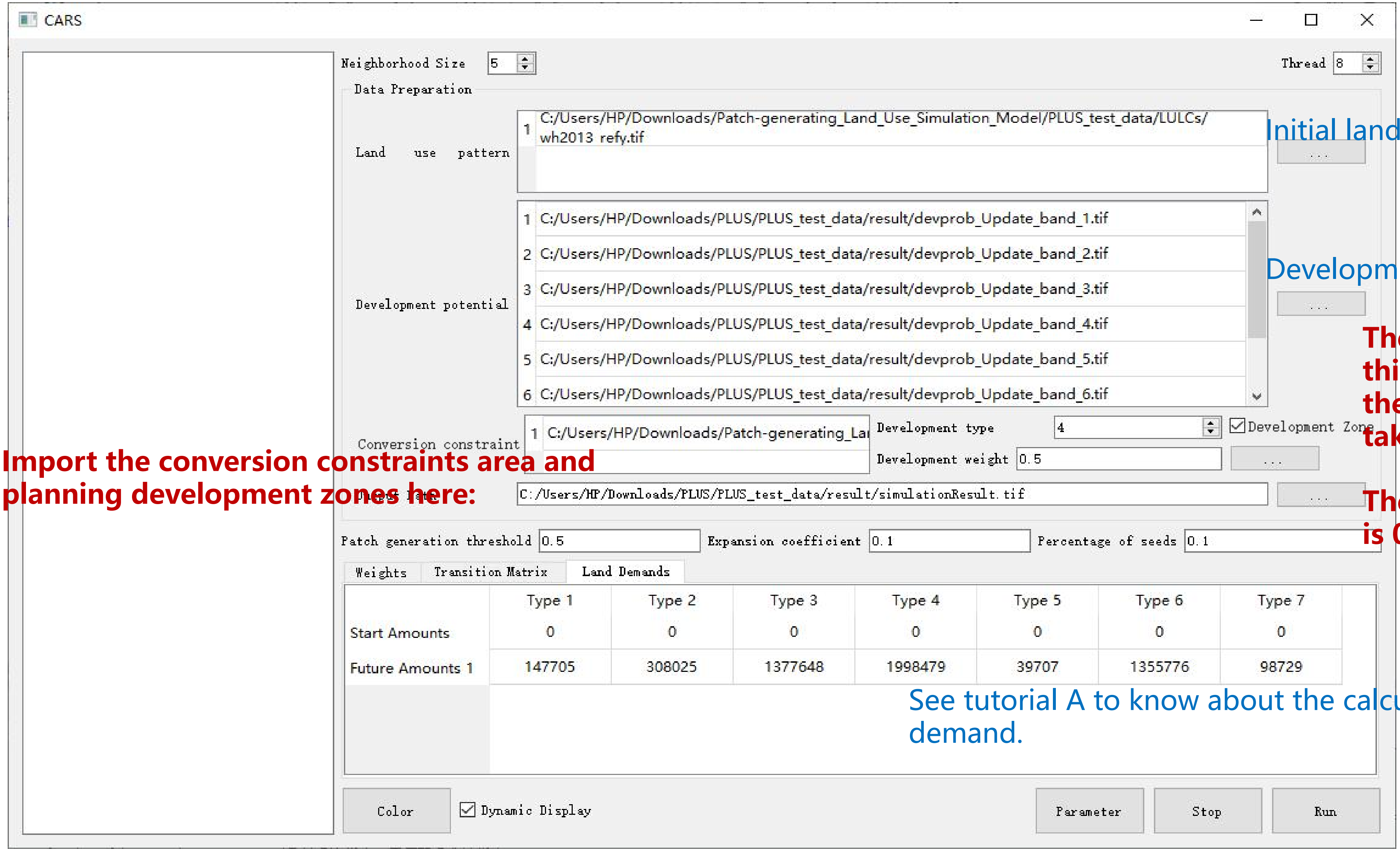




The value 0 will be converted to nodata value in this tool and will not be shown in the final results.



03 Parameter setting



Initial land use data

Development potential

The value of urban land is 4 in this data. When the value is 0, the development zone will not take effect.

Import the conversion constraints area and planning development zones here:

The default development weight is 0.5.

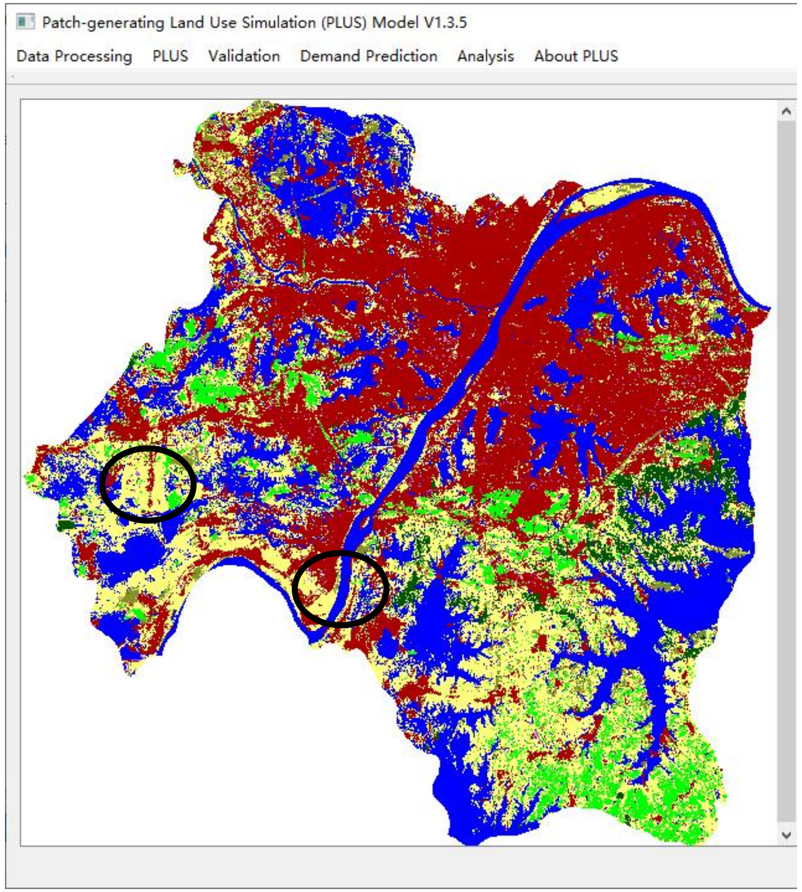
See tutorial A to know about the calculation of future Land use demand.

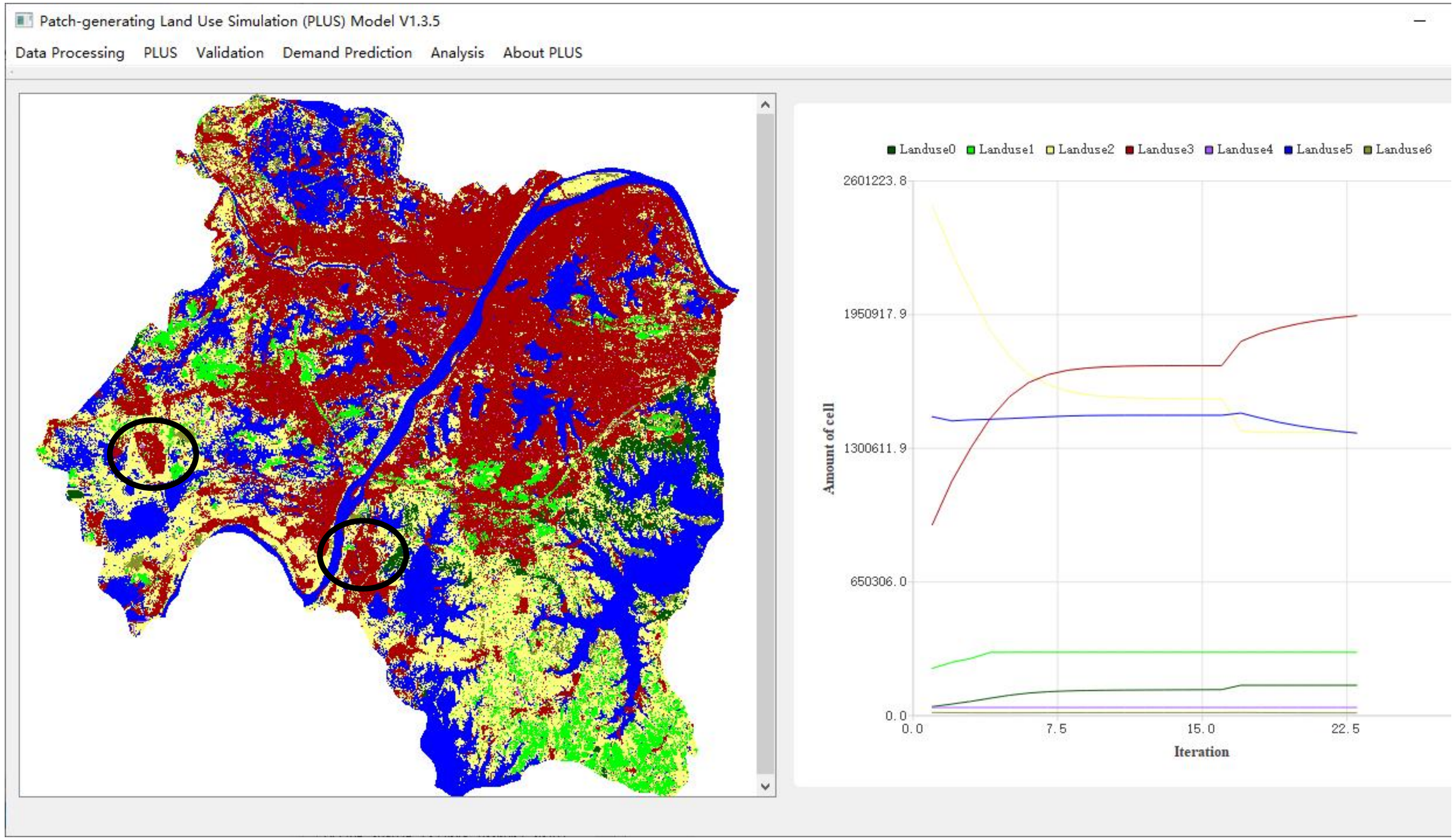
Click the 'Run' button to start simulation.

03 Results affected by the development zones

The picture on the left is the result without the influences of planning development zones

The picture on the right is the result under the influences of planning development zones .We can see the newly developed urban patches affected by the planning development zones.





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Thanks!